Circuit Theorems Assignment

ECE2107 Circuits & Systems · assignment marathon · built so far: Part A Superposition (5.1–5.14), Thevenin & Norton (5.15–5.34), maximum power (5.35–5.44); Part B Sadiku superposition (PP 4.3–4.5, 4.8–4.19), source transformation (PP 4.6–4.7, 4.20–4.32)

How to use this

Every question follows the same order:

1. **Question**, with the figure taken straight from the book.
2. **Try it first** (yellow). Cover the rest of the question and solve it.
3. **Hint: what's new** (orange). Only when the question brings an idea you haven't met earlier in this PDF. If you're stuck, read it and try again.
4. **Solution** in compact assignment form: Given → To find → each step with its intermediate values → boxed answer. Copy it into your submission, or compare it line by line with your attempt to find the step that went wrong.
5. **Side note** (grey) or **book error** (red), only when there is something worth knowing.

The tag next to each question number shows how much teaching it carries: `teach` full explanation · `new idea` one new thing · `practice` plain solution.

If you couldn't solve one, just tell me its number. Nothing else needs marking.

What's in it (full version)

Part	Book	Questions	Count
A	Ghosh & A K Chakraborty, *Circuit Theory and Networks*, ch. 5 (**solved**)	Superposition 5.1–5.14 · Thevenin/Norton 5.15–5.34 · Max power 5.35–5.44	44
B	Alexander & Sadiku, *Fundamentals of Electric Circuits* 5e (**unsolved**)	Practice problems + end-of-chapter: ch. 4 (superposition, source transformation, Thevenin/Norton, max power), ch. 10 (the same in AC), ch. 11.3 (AC max power)	107 + practice problems

Scope follows the official MUJ ECE2107 syllabus: superposition, Thevenin's and Norton's theorems, maximum power transfer, DC and AC, plus source transformation.

Checking. Every answer was worked out again and confirmed by a full circuit solve with all sources on (`src/verify.py`), including each case's partial answer. Matches with the book are marked "= book". Where the book is wrong, the correct answer is given and a red box explains the book's slip.

Part A · Chakraborty (solved problems)

Ghosh & A K Chakraborty, *Circuit Theory and Networks* 4e, ch. 5 "Network Theorems", Solved Problems

A · Superposition theorem

KVL, KCL, Ohm's law (you have these) → Nodal / mesh: all sources at once → **Superposition: one source at a time** → Thevenin / Norton (next)

Where this sits. You can already solve any circuit with nodal or mesh analysis, taking every source at once. Superposition breaks the same job into smaller circuits, **one source at a time**, which is often quicker by hand. It is also the reason Thevenin's theorem works, which comes next.

Three terms first.

- **Linear circuit:** made of R, L, C, linear dependent sources and independent sources. Doubling every source doubles every current and voltage.
- **Independent source:** a fixed voltage or current (a battery, a 2 A source, 10 cos 4t).
- **Dependent source** (diamond symbol): its value follows some other voltage or current in the circuit, for example $2V_x$.

Statement. In a linear circuit with several independent sources, any current or voltage is the **algebraic sum** of the values caused by **each independent source acting alone**.

Switching a source off means setting its value to zero:

Source	Switched off it becomes	Why
Ideal voltage source	**short circuit**	0 V across it
Ideal current source	**open circuit**	0 A through it
Dependent source	**never switched off**	it is not a source on its own; it follows the circuit

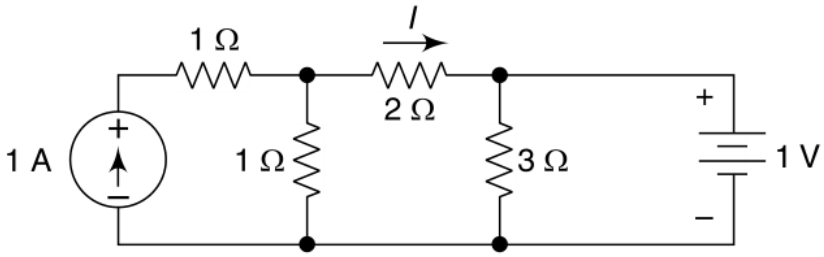
Superposition adds currents and voltages only, **not power** (P goes as I^2, and $(I' + I'')^2 \neq I'^2 + I''^2$).

A1 · Problem 5.1

Superposition · DC **teach**

QUESTION

Find the current I in the circuit shown, using the superposition theorem.



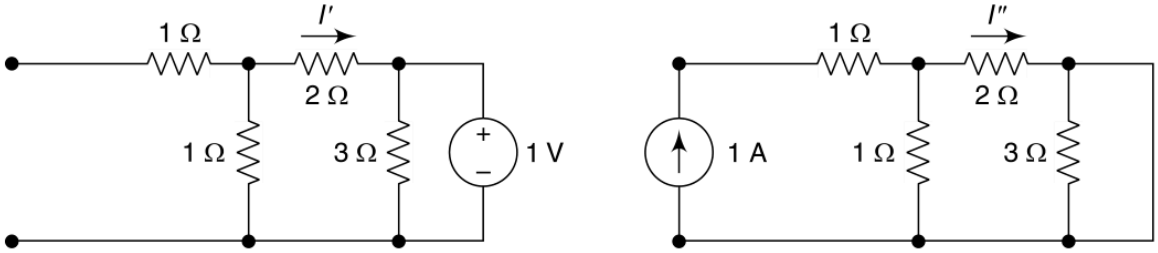
I flows left to right through the 2 Ω resistor.

TRY IT FIRST

Cover everything below. Draw two circuits, 1 V alone and 1 A alone, find the current in the 2 Ω in each, and decide each one's sign before adding.

HINT: WHAT'S NEW

- **The whole method:** two sources, so solve **two small circuits** and add.
- **Switching off:** the 1 V source becomes a short; the 1 A source becomes an open.
- **Signs:** measure each part against the arrow drawn for I. A part flowing the other way is negative.



(i) 1 V acting alone (1 A opened) · (ii) 1 A acting alone (1 V shorted)

Given: 1 A source, 1 V source; 1 Ω (series), 1 Ω (shunt), 2 Ω, 3 Ω.

To find: I through the 2 Ω (left → right).

Case (i): 1 V alone (1 A opened). The series 1 Ω carries nothing. The 3 Ω sits directly across the 1 V, so it changes nothing else. The 1 V drives current through 2 Ω + 1 Ω (shunt):

$$\text{current} = \frac{1}{2+1} = \frac{1}{3}\ \text{A (right} \rightarrow \text{left)} \qquad \Rightarrow \qquad I' = -\frac{1}{3}\ \text{A}$$

Case (ii): 1 A alone (1 V shorted, so the 3 Ω is shorted out). The 1 A divides between the 1 Ω shunt and the 2 Ω:

$$I'' = 1 \times \frac{1}{1+2} = \frac{1}{3}\ \text{A (left} \rightarrow \text{right)}$$

Superposition: $I = I' + I'' = -\frac{1}{3} + \frac{1}{3} = 0$

ANSWER

I = 0 A (= book)

WHY ZERO, AND TWO SHORTCUTS

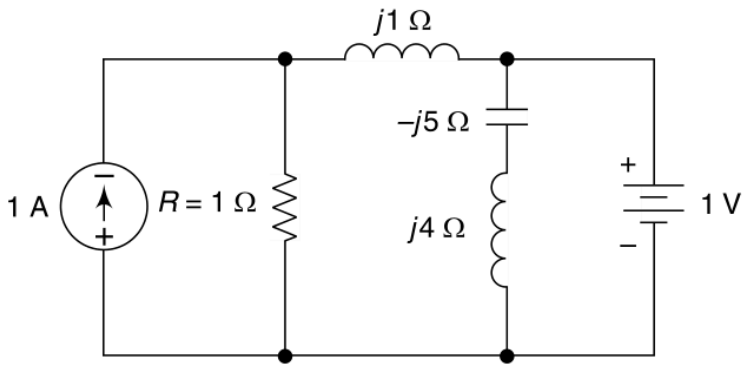
- Node check: $1 = \frac{V_A}{1} + \frac{V_A - 1}{2}$ gives $V_A = 1$ V, the same as the right-hand node, so no current flows in the 2 Ω.
- An element **in series with an ideal current source** never changes any other current (the left 1 Ω here).
- An element **in parallel with an ideal voltage source** never changes any other current (the 3 Ω here).

A2 · Problem 5.2

Superposition · AC **teach**

QUESTION

Calculate the voltage V across the resistor R by using the superposition theorem.



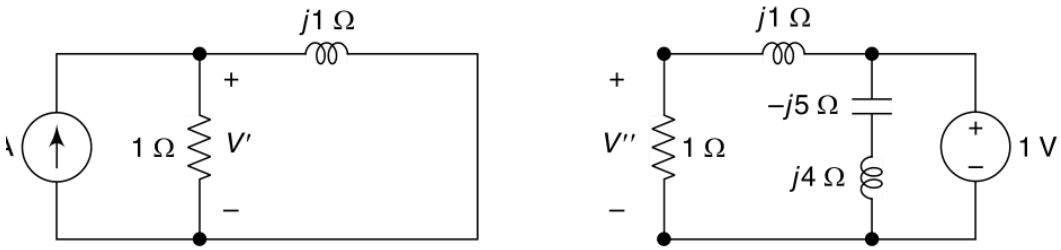
The book draws +/− signs inside the current source. Read it by its arrow: 1 A pushed up into the top node.

TRY IT FIRST

Which elements drop out in each case? Aim for V′ and V″ in the form $\frac{\text{something}}{1 + j}$.

HINT: WHAT'S NEW

- **AC (phasor) circuit.** j1 Ω is an inductor, −j5 Ω a capacitor. Superposition is unchanged; the arithmetic uses complex numbers.
- **Complex dividers:** the same formulas as for resistors, with Z in place of R.
- The −j5 + j4 branch is **across the 1 V source** (5.1's shortcut), so it never matters.



(i) 1 A acting alone · (ii) 1 V acting alone

Given: 1 A, 1 V; R = 1 Ω, j1 Ω, −j5 Ω, j4 Ω.

To find: V across R (+ at top).

Case (i): 1 A alone (1 V shorted; this also shorts out the −j5 + j4 branch). R ∥ j1 across the 1 A:

$$V' = 1 \times \frac{1 \times j1}{1 + j1} = \frac{j}{1 + j}\ \text{V}$$

Case (ii): 1 V alone (1 A opened). j1 and R in series across 1 V:

$$I'' = \frac{1}{1 + j} \quad \Rightarrow \quad V'' = 1 \times I'' = \frac{1}{1 + j}\ \text{V}$$

Superposition: $V = \frac{j}{1 + j} + \frac{1}{1 + j} = 1\ \text{V}$

ANSWER

V = 1∠0° V (= book)

CHECK

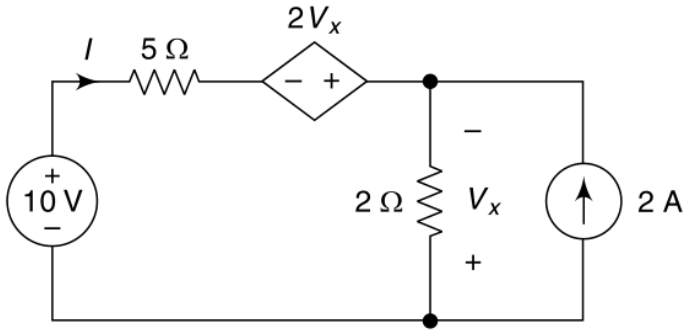
Node equation at the top node: $1 = \frac{V}{1} + \frac{V - 1}{j1}$ gives V(1 + j) = 1 + j, so V = 1 V ✓.

A3 · Problem 5.3

Superposition · dependent source **teach**

QUESTION

Use the superposition theorem on the circuit shown to find I.



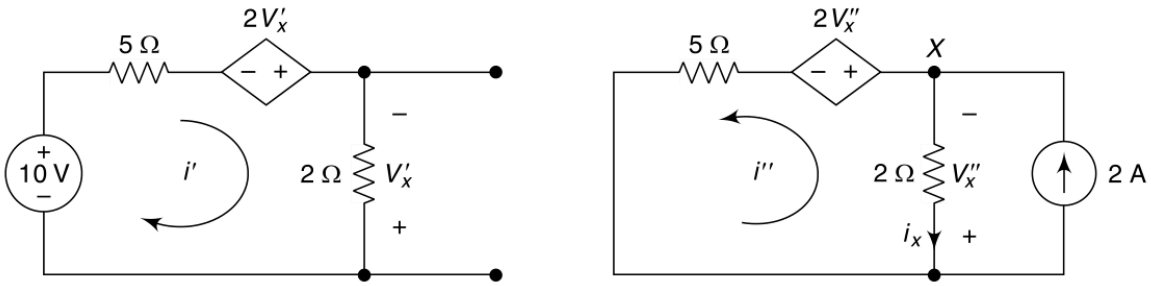
$2V_x$ is a voltage-controlled voltage source. V_x is across the 2 Ω with + at the bottom.

TRY IT FIRST

Write V_x' in terms of the loop current in case (i). For case (ii), write one KCL at the top of the 2 Ω and one KVL around the left loop. Expect elevenths.

HINT: WHAT'S NEW

- **Dependent source:** it **stays in every case**. Only the 10 V and the 2 A take turns.
- **Its value changes per case:** $2V_x'$ in case (i), $2V_x''$ in case (ii). Re-find V_x in each sub-circuit.
- **Polarity of V_x:** + is at the **bottom** of the 2 Ω. Most wrong answers to this question come from this sign.



(i) 10 V alone (2 A opened) · (ii) 2 A alone (10 V shorted); the dependent source stays in both

Given: 10 V, 2 A, 5 Ω, 2 Ω, VCVS $2V_x$.

To find: I (out of the 10 V source, through the 5 Ω).

Case (i): 10 V alone. One loop, current i′ clockwise (= direction of I), flowing down the 2 Ω, so $V_x' = -2i'$.

$$10 + 2V_x' = 5i' + 2i' \quad \Rightarrow \quad 10 - 4i' = 7i' \quad \Rightarrow \quad I' = i' = \frac{10}{11}\ \text{A}$$

Case (ii): 2 A alone. At node X: 2 A splits into i_x (down the 2 Ω, so $V_x'' = -2i_x$) and i″ (leftward through the dependent source and 5 Ω).

$$\text{KCL at X:} \qquad 2 = -\frac{V_x''}{2} + i'' \qquad \ldots(1)$$

$$\text{Left loop:} \qquad 5i'' + 3V_x'' = 0 \qquad \Rightarrow \qquad i'' = -\frac{3}{5}V_x'' \qquad \ldots(2)$$

(X sits at $-V_x''$; the dependent source drops another $2V_x''$, so the 5 Ω has $-3V_x''$ across it.)

$$\text{(2) in (1):} \qquad 2 = -\frac{11}{10}V_x'' \qquad \Rightarrow \qquad V_x'' = -\frac{20}{11}\text{ V}, \qquad i'' = \frac{12}{11}\text{ A}$$

i″ runs anticlockwise, against I, so **I″ = −12/11 A**.

Superposition: $I = \frac{10}{11} - \frac{12}{11} = -\frac{2}{11}$ A

ANSWER

I = −2/11 A ≈ −0.182 A (= book). The minus sign means the current actually flows into the + of the 10 V source.

NEVER SWITCH OFF A DEPENDENT SOURCE

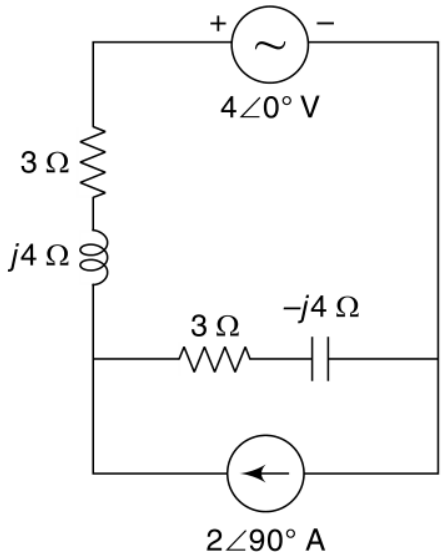
Shorting $2V_x$ as if it were independent gives a wrong answer. It stays in; only its value changes from case to case.

A4 · Problem 5.4

Superposition · AC **new idea**

QUESTION

Determine the current in the capacitor branch by the superposition theorem.



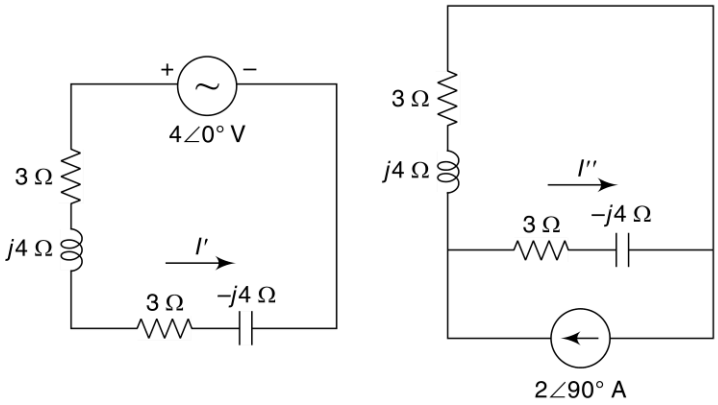
$Z_1 = 3 + j4\ \Omega$ (left branch); $Z_2 = 3 - j4\ \Omega$ (capacitor branch); current source $2\angle 90°$ A.

TRY IT FIRST

Which way does each source see Z_1 and Z_2: in series, or in parallel?

HINT: WHAT'S NEW

- **Source with an angle:** $2\angle 90°$ A = j2 A. Convert to rectangular form before multiplying.
- **Polar form with a negative real part:** a calculator's $\tan^{-1}$ lands in the wrong quadrant; add 180°.



(i) $4\angle 0°$ V alone · (ii) $2\angle 90°$ A alone

Given: $4\angle 0°$ V, $2\angle 90°$ A, $Z_1 = 3 + j4\ \Omega$, $Z_2 = 3 - j4\ \Omega$.

To find: I in the capacitor branch (left → right).

Case (i): $4\angle 0°$ V alone (current source opened). Z_1, Z_2 in series:

$$I' = \frac{4\angle 0°}{(3 + j4) + (3 - j4)} = \frac{2}{3}\angle 0°\ \text{A}$$

Case (ii): 2∠90° A alone (voltage source shorted). $Z_1 \parallel Z_2$ across the source:

$$I'' = j2 \times \frac{3 + j4}{6} = -\frac{4}{3} + j1 \text{ A}$$

Superposition: $I = \frac{2}{3} - \frac{4}{3} + j1 = -\frac{2}{3} + j1$ A

$$|I| = \sqrt{((2/3)^2 + 1)} = 1.20 \text{ A}, \qquad \theta = 180° - \tan^{-1}(1.5) = 123.7°$$

ANSWER

I = 1.2∠123.7° A (= book)

SIDE NOTE

$Z_1 + Z_2 = 6\ \Omega$ is purely real (+j4 and −j4 cancel), which is why both cases are clean. Equal and opposite reactances in series act like a short: **series resonance**, met later.

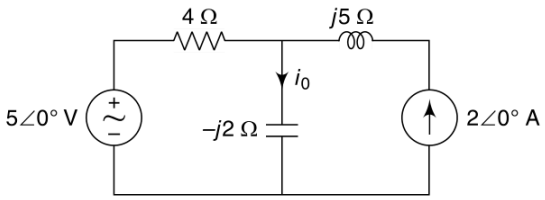
A5 · Problem 5.5

Superposition · AC, DC + AC **teach**

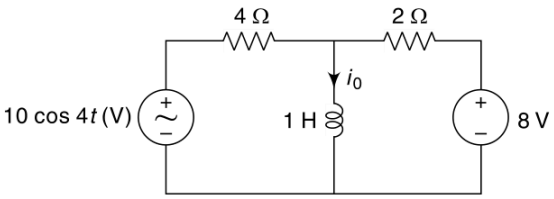
QUESTION

Find the current i_0 using the superposition theorem, in each of circuits (a), (b) and (c).

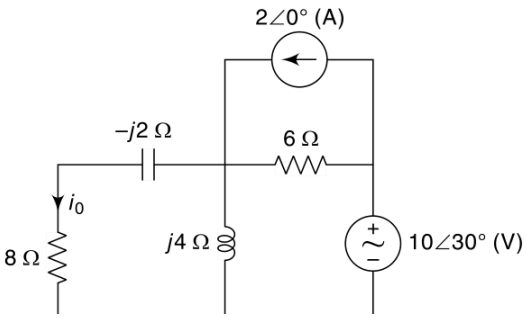
(a)



(b)



(c)



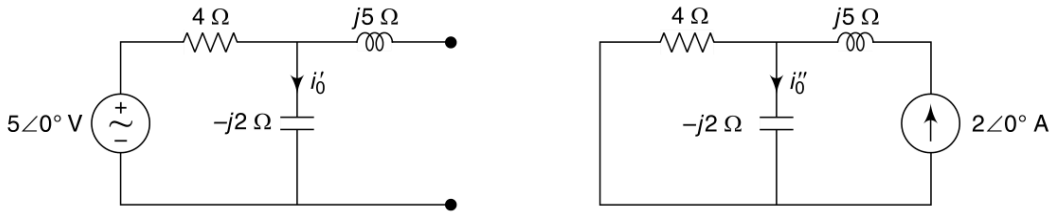
TRY IT FIRST

(a) is quick practice. In (b), stop and ask: what does an inductor look like to the 8 V **DC** source? In (c), redraw the circuit after switching off each source before you calculate anything.

HINT: WHAT'S NEW

- **(b) Two different frequencies (DC and 10 cos 4t).** Each case is solved **at its own frequency**: at DC (ω = 0), $X_L = \omega L = 0$, so the **inductor is a short**; at ω = 4, it is j4 Ω. The two answers are added **in the time domain**, never as phasors, because phasors only add when they share one frequency.
- **(c) Redraw after switching off.** Once the voltage source is shorted, the 6 Ω and the current source both run to ground and everything becomes parallel.

(a) Given: 5∠0° V, 2∠0° A, 4 Ω, −j2 Ω, j5 Ω. **To find:** i_0 (down the capacitor).



(i) voltage source alone · (ii) current source alone

$$5\angle 0° \text{ alone:} \qquad i_0' = \frac{5}{4 - j2} = 1 + j0.5 \text{ A}$$

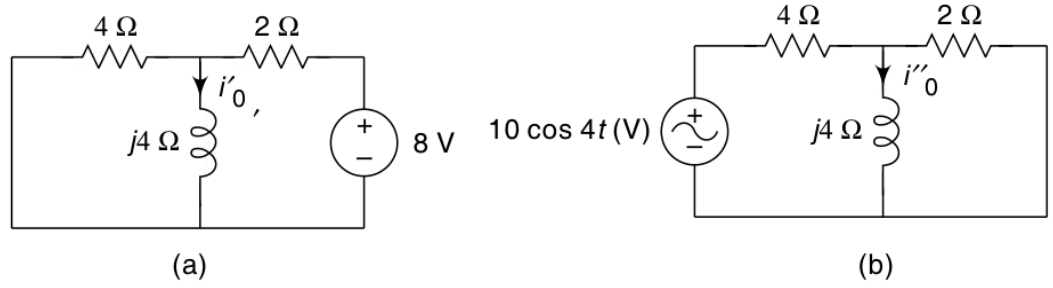
2∠0° alone (j5 is in series with the source, so irrelevant):

$$i_0'' = 2 \times \frac{4}{4 - j2} = 1.6 + j0.8 \text{ A}$$

ANSWER

(a) i_0 = 2.6 + j1.3 = 2.91∠26.57° A (= book, 2.9∠26.56° A)

(b) Given: 10 cos 4t V, 8 V DC, 4 Ω, 1 H, 2 Ω. **To find:** $i_0(t)$ (down the inductor).



The book's sub-circuits (a) DC alone, (b) AC alone. In (a) the inductor must be read as a short circuit.

8 V DC alone (AC source shorted; ω = 0, so the inductor is a short):

$$i_0' = \frac{8}{2} = 4\ \text{A} \qquad \text{(the 4 Ω has 0 V across it)}$$

10 cos 4t alone (8 V shorted; ω = 4 rad/s, Z_L = j4 Ω):

$$Z = 4 + \frac{j4 \times 2}{2 + j4} = 5.6 + j0.8\ \Omega, \qquad I = \frac{10\angle 0°}{Z} = 1.75 - j0.25\ \text{A}$$

$$i_0'' = I \times \frac{2}{2 + j4} = 0.25 - j0.75 = 0.791\angle -71.57°\ \text{A}$$

Superposition (time domain):

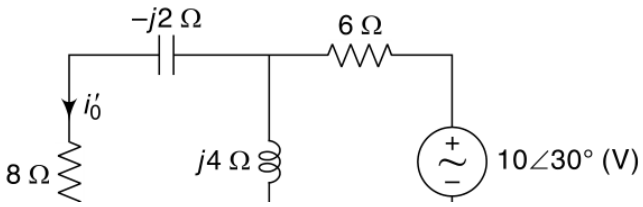
ANSWER

(b) $i_0(t) = 4 + 0.791 \cos(4t - 71.57°)$ A

BOOK ERROR IN 5.5(B)

The book treats the inductor as j4 Ω for the **DC** source too, and then adds the DC and AC results as if both were phasors, printing 2.63∠−63.58° A. Both steps are wrong: at DC an inductor is a short, and quantities at different frequencies cannot be added as phasors. The answer above was confirmed by a node solve of each case.

(c) Given: 10∠30° V, 2∠0° A, 8 Ω, −j2 Ω, j4 Ω, 6 Ω. **To find:** i_0 (down the 8 Ω).



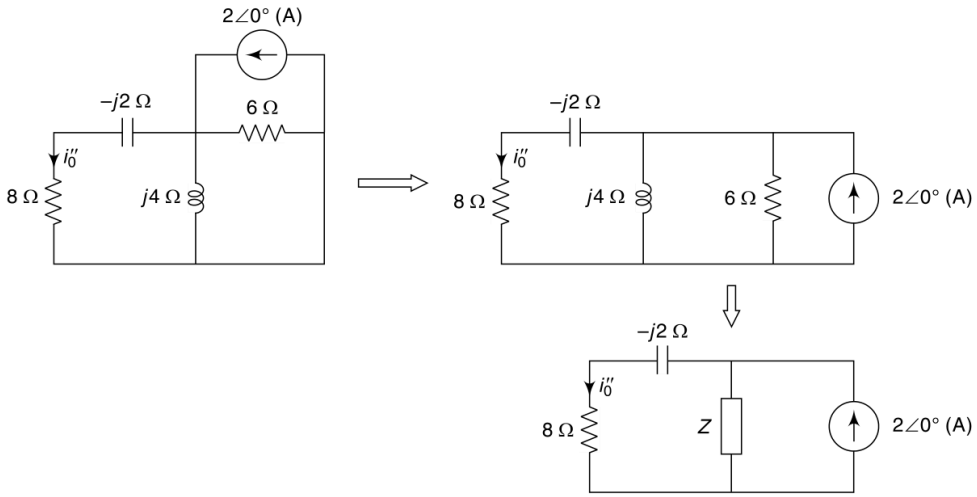
(i) voltage source alone

10∠30° V alone (current source opened):

$$Z = 6 + j4 \parallel (8 - j2) = 6 + (1.882 + j3.529) = 7.882 + j3.529\ \Omega,$$

$$I = \frac{10\angle 30°}{Z} = 1.158\angle 5.88°\ \text{A}$$

$$i_0' = I \times \frac{j4}{(8 - j2) + j4} = I \times \frac{j4}{8 + j2} = 0.562\angle 81.84°\ \text{A}$$



(ii) current source alone, redrawn: the 6 Ω and the source now both go to ground

2∠0° A alone (voltage source shorted). Z = j4 ∥ 6 = 1.846 + j2.769 Ω:

$$i_0'' = 2 \times \frac{Z}{(8 - j2) + Z} = \frac{j24}{28 + j22} = 0.674\angle 51.84^\circ \text{ A}$$

$$i_0 = (0.080 + j0.556) + (0.416 + j0.530) = 0.496 + j1.086 \text{ A}$$

ANSWER

(c) $i_0 = 1.19\angle 65.45^\circ$ A

BOOK ERROR IN 5.5(C)

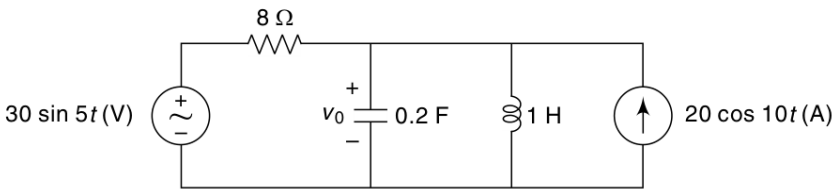
The book prints 0.81∠39.38° A. Two slips: in case (i) it divides the current with (8 − j2)/(8 + j2) instead of j4/(8 + j2) (the current into a branch is set by the **other** branch's impedance); in case (ii) it expands (8 − j2)(3 + j2) + j12 as 12 + j11 instead of 28 + j22. The answer above was confirmed by a full node solve.

A6 · Problem 5.6

Superposition · two frequencies **new idea**

QUESTION

Find v_0 using the superposition theorem.

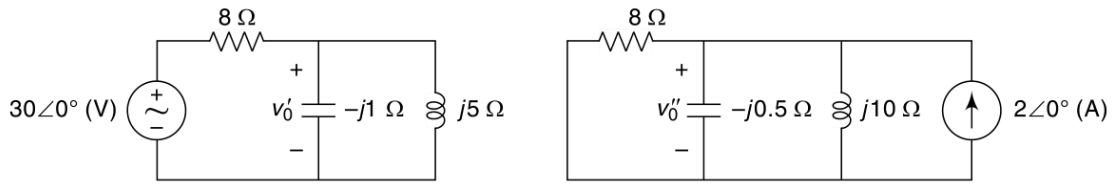


TRY IT FIRST

Work out X_C and X_L separately for ω = 5 and ω = 10 before drawing anything.

HINT: WHAT'S NEW

- **Reactances change with frequency.** $X_C = \frac{1}{\omega C}$ and $X_L = \omega L$ must be recomputed for each source's own ω.
- **Mixed sin and cos:** keep each term in the function it started with (sin for the voltage source, cos for the current source).



(i) voltage source alone, ω = 5 · (ii) current source alone, ω = 10

Given: 30 sin 5t V with 8 Ω; 0.2 F, 1 H, 20 cos 10t A. **To find:** $v_0(t)$.

30 sin 5t alone (ω = 5; current source opened): $X_C = \frac{1}{5 \times 0.2} = 1\ \Omega \to -j1\ \Omega$; $X_L = 5 \times 1 = 5\ \Omega \to j5\ \Omega$.

$$\frac{30 - v'}{8} = \frac{v'}{-j1} + \frac{v'}{j5} \quad \Rightarrow \quad v' = \frac{30}{8(0.125 + j0.8)} = 4.631\angle -81.12°\ \text{V}$$

20 cos 10t alone (ω = 10; voltage source shorted): $X_C = \frac{1}{10 \times 0.2} = 0.5\ \Omega \to -j0.5\ \Omega$; $X_L = 10\ \Omega \to j10\ \Omega$.

$$20 = v''\left(\frac{1}{8} + \frac{1}{-j0.5} + \frac{1}{j10}\right) = v''(0.125 + j1.9) \quad \Rightarrow \quad v'' = 10.50\angle -86.24°\ \text{V}$$

ANSWER

$v_0(t) = 4.631 \sin(5t - 81.12°) + 10.50 \cos(10t - 86.24°)$ V

BOOK INCONSISTENCY IN 5.6

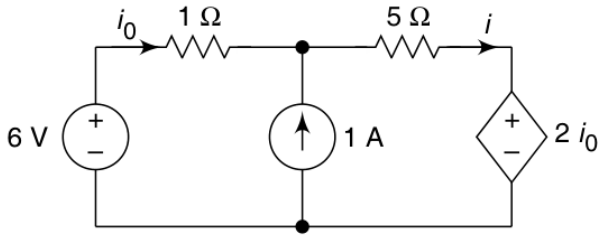
The figure gives the current source as **20** cos 10t A, but the book's solution uses **2**∠0° A and prints 1.051 cos(10t − 86.24°) for the second term. The answer above follows the figure. If your source is 2 A, the second term is 1.051 cos(10t − 86.24°) V; the first term is unchanged.

A7 · Problem 5.7

Superposition · current-controlled source **new idea**

QUESTION

Find i_0 and i from the circuit of the figure using the superposition theorem.



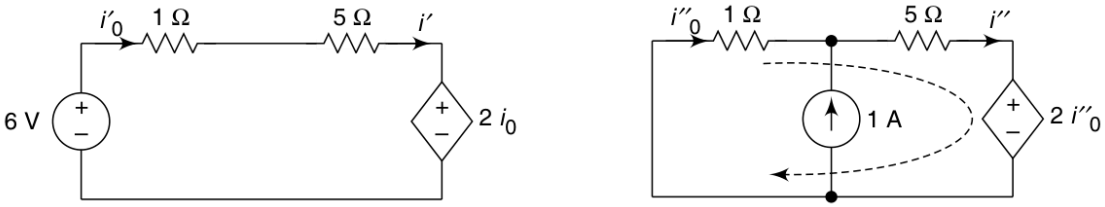
$2i_0$ is a current-controlled voltage source.

TRY IT FIRST

In each case, the dependent source's value is 2 × (that case's i_0). Write KCL at the middle node and one KVL around the outer loop.

HINT: WHAT'S NEW

- **Controlled by a current, not a voltage** (5.3 was voltage-controlled). The rule is the same: keep the source, and re-find its controlling current i_0 in each case.



(i) 6 V alone · (ii) 1 A alone

Given: 6 V, 1 A, 1 Ω, 5 Ω, CCVS $2i_0$. **To find:** i_0, i.

6 V alone (1 A opened): one loop, so $i_0' = i'$.

$$6 = 1\cdot i' + 5\cdot i' + 2i_0' = 8i' \quad \Rightarrow \quad i' = i_0' = 0.75 \text{ A}$$

1 A alone (6 V shorted):

$$\text{KCL:} \quad i'' = 1 + i_0'' \qquad \text{outer KVL:} \quad 1\cdot i_0'' + 5i'' + 2i_0'' = 0$$

$$\Rightarrow \quad 3i_0'' + 5(1 + i_0'') = 0$$

$$8i_0'' = -5 \quad \Rightarrow \quad i_0'' = -0.625 \text{ A}, \quad i'' = 0.375 \text{ A}$$

ANSWER

$i = 0.75 + 0.375 = 1.125$ A, $i_0 = 0.75 - 0.625 = 0.125$ A

BOOK ERROR IN 5.7

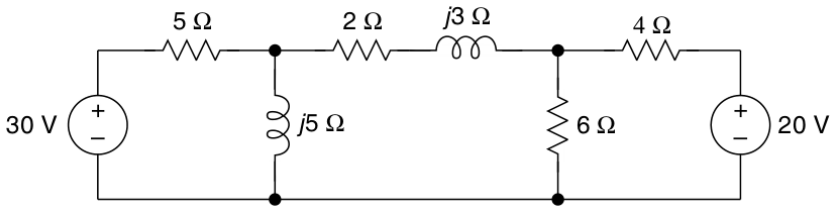
The book sets up the same equation, then solves $8i_0'' = -5$ as $i_0'' = -5/4 = -1.25$ A, and prints $i = 0.5$ A, $i_0 = -0.5$ A. The correct value is $-5/8 = -0.625$ A. Direct check: KCL $i = i_0 + 1$ and KVL $6 - i_0 - 5i = 2i_0$ give $i_0 = 1/8$ A ✓.

A8 · Problem 5.8

Superposition · AC **practice**

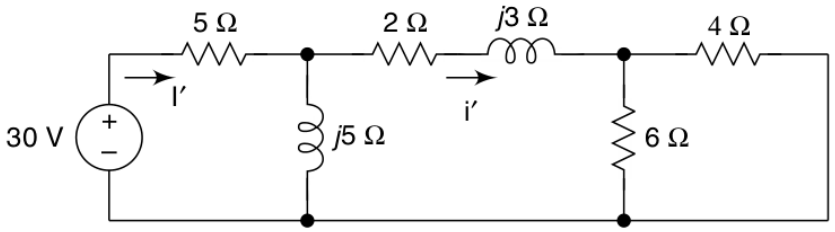
QUESTION

Using the superposition theorem, calculate the current through the (2 + j3) Ω impedance branch.



TRY IT FIRST

Series-parallel reduction, twice. Watch which way the branch current flows in case II.

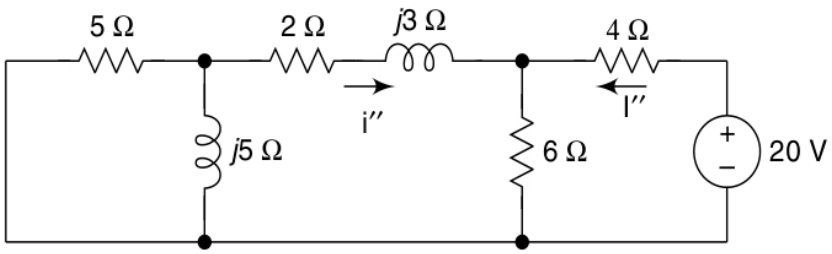


Case I: 30 V alone

Case I: 30 V alone (20 V shorted). Branch: $Z_b = (2 + j3) + (6 \parallel 4) = 4.4 + j3\ \Omega$.

$$Z = 5 + \frac{Z_b \times j5}{Z_b + j5} = 6.32 + j2.60\ \Omega, \qquad I' = \frac{30}{Z} = 4.06 - j1.67\ \text{A},$$

$$i' = I' \times \frac{j5}{Z_b + j5} = 2.389 + j0.270\ \text{A}$$



Case II: 20 V alone

Case II: 20 V alone (30 V shorted). Branch: $Z_a = (5 \parallel j5) + 2 + j3 = 4.5 + j5.5\ \Omega$.

$$Z = 4 + \frac{6Z_a}{Z_a + 6} = 7.31 + j1.41\ \Omega, \qquad I'' = \frac{20}{Z} = 2.638 - j0.509\ \text{A}$$

$$i'' = -I'' \times \frac{6}{Z_a + 6} = -1.063 + j0.848\ \text{A} \qquad \text{(it flows right → left, against i)}$$

ANSWER

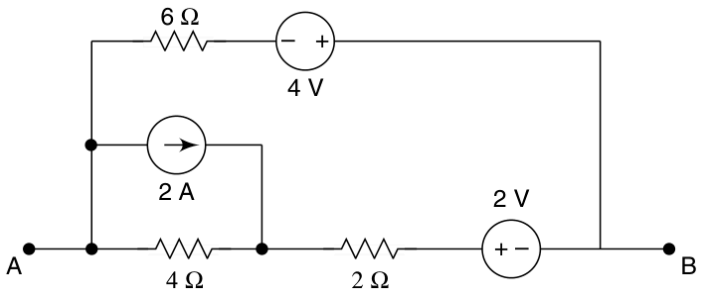
$i = i' + i'' = 1.326 + j1.117 = 1.73\angle 40.13°$ A (= book)

A9 · Problem 5.9

Superposition · three sources, open terminals **new idea**

QUESTION

Using the superposition theorem, find V_{AB}.

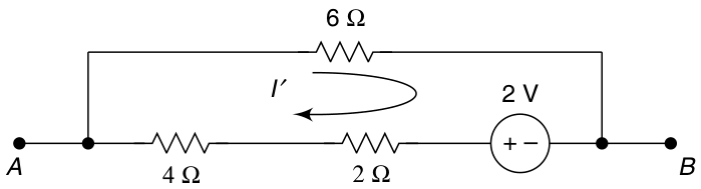


TRY IT FIRST

Three sources, so three cases. Nothing is connected across A–B, so no current leaves at A or B.

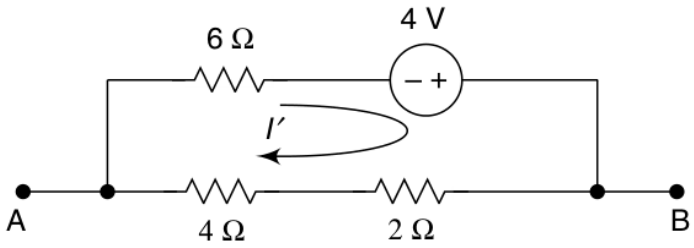
HINT: WHAT'S NEW

- **Three sources:** same method, one more case.
- **Open terminals:** V_{AB} is the voltage across a gap. With no load, the only path is the loop 4 Ω + 2 Ω + 6 Ω, and V_{AB} is the drop across the 6 Ω.
- **Source transformation inside a case:** 2 A in parallel with 4 Ω equals an 8 V source in series with 4 Ω, which makes case III a single loop.



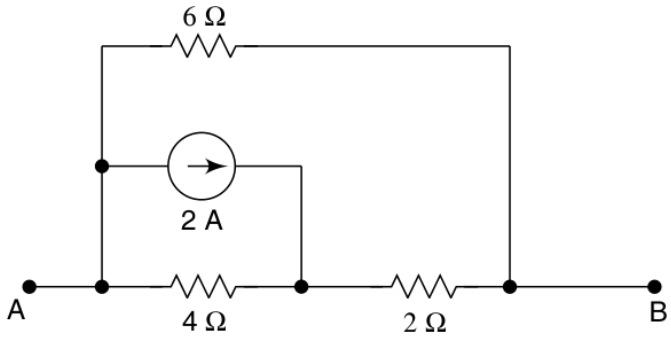
Case I: 2 V alone

Case I (2 V): $I' = \dfrac{2}{4+2+6} = \dfrac{1}{6}$ A, $V_{AB}' = 6I' = 1$ V

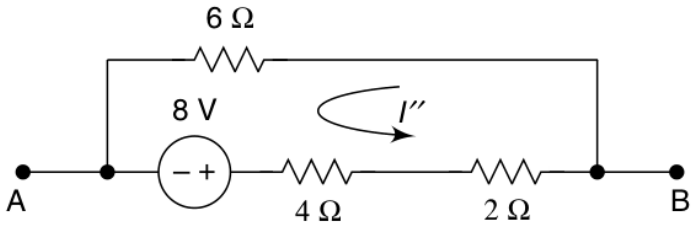


Case II: 4 V alone

Case II (4 V): $I'' = \dfrac{4}{12} = \dfrac{1}{3}$ A, $V_{AB}'' = -6I'' = -2$ V



Case III: 2 A alone



Case III after source transformation: 2 A ∥ 4 Ω → 8 V in series with 4 Ω

Case III (2 A): $I''' = \dfrac{8}{12} = \dfrac{2}{3}$ A, $V_{AB}''' = -6I''' = -4$ V

(In case I the current runs through the 6 Ω from A's side to B's side, so A is higher; in cases II and III it runs the other way, hence the minus signs.)

ANSWER

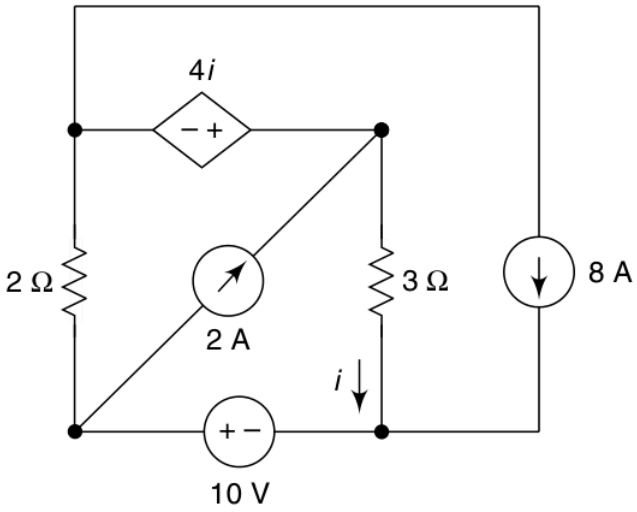
$V_{AB} = 1 - 2 - 4 = -5$ V (= book)

A10 · Problem 5.10

Superposition · dependent source, three sources **new idea**

QUESTION

Find the current i in the circuit shown using the superposition theorem.



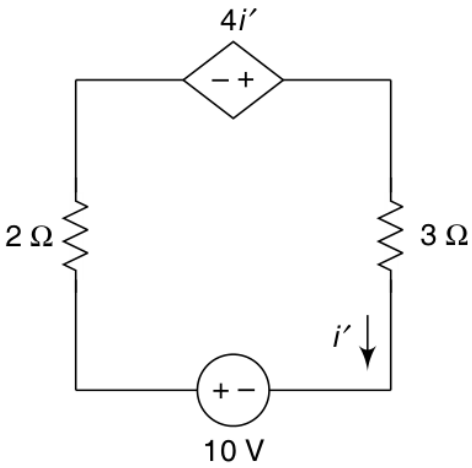
4i is a current-controlled voltage source; i flows down the 3 Ω.

TRY IT FIRST

Case I is one loop. In case II, look for a resistor in series with the dependent source.

HINT: WHAT'S NEW

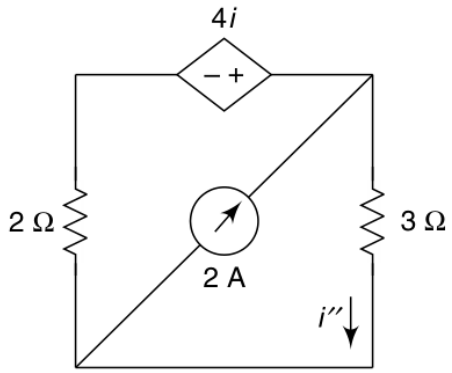
- **Source transformation works on dependent sources too.** In case II, the 4i source in series with the 2 Ω becomes a dependent current source 4i/2 = **2i** in parallel with 2 Ω.
- **A current source inside a mesh fixes that mesh current** (case III: the 8 A source sets the mesh difference I − 8).



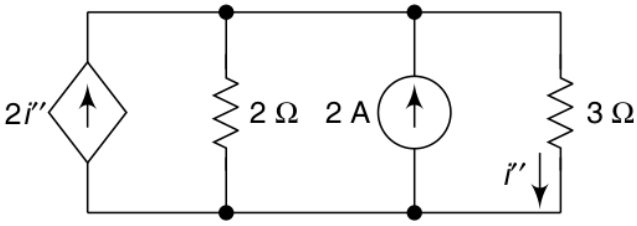
Case I: 10 V alone

Case I: 10 V alone (2 A and 8 A opened). Loop current i′ clockwise:

$$-4i' + 3i' - 10 + 2i' = 0 \qquad \Rightarrow \qquad i' = 10\ \text{A}$$



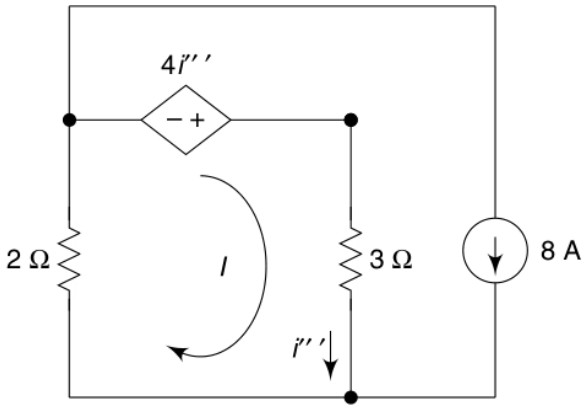
Case II: 2 A alone



Case II after converting 4i″ + 2 Ω into 2i″ ‖ 2 Ω

Case II: 2 A alone (10 V shorted, 8 A opened). Total (2 + 2i″) splits between 2 Ω and 3 Ω:

$$i'' = \frac{2}{2+3}(2 + 2i'') \qquad \Rightarrow \qquad 5i'' = 4 + 4i'' \qquad \Rightarrow \qquad i'' = 4\ \text{A}$$



Case III: 8 A alone

Case III: 8 A alone (10 V shorted, 2 A opened). Mesh current I; the 3 Ω carries i‴ = I − 8:

$$-4i''' + 3(I - 8) + 2I = 0, \qquad I = i''' + 8 \qquad \Rightarrow \qquad -4i''' + 3i''' + 2i''' + 16 = 0 \qquad \Rightarrow$$
$$i''' = -16\ \text{A}$$

ANSWER

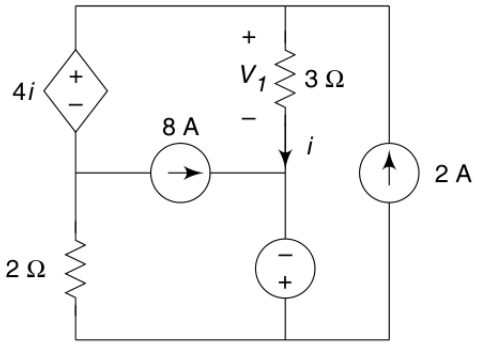
i = 10 + 4 − 16 = −2 A (= book)

A11 · Problem 5.11

Superposition · dependent source, three sources **practice**

QUESTION

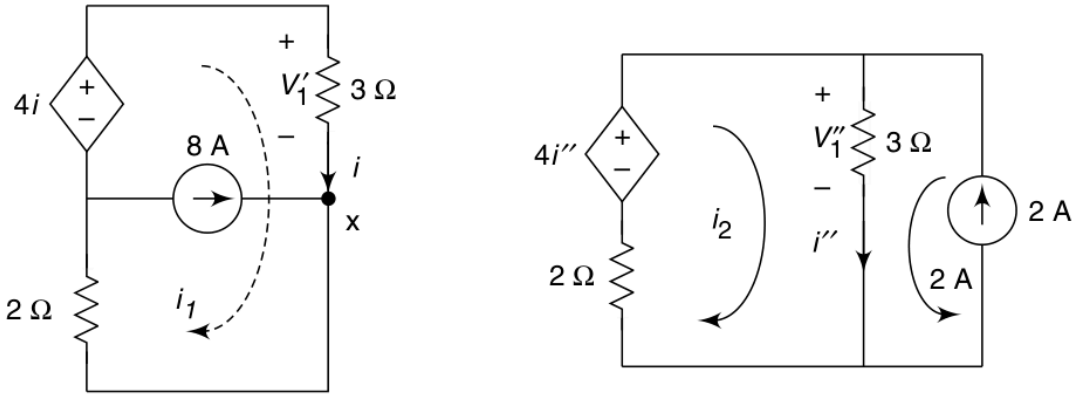
Using the superposition theorem, determine V_1, the voltage across the 3 Ω resistor.



The figure prints no value on the bottom-middle voltage source; the book's solution uses 10 V.

TRY IT FIRST

Three cases again; the dependent source 4i stays in all three. Case I has a supermesh around the 8 A source.



Case I: 8 A alone · Case II: 2 A alone

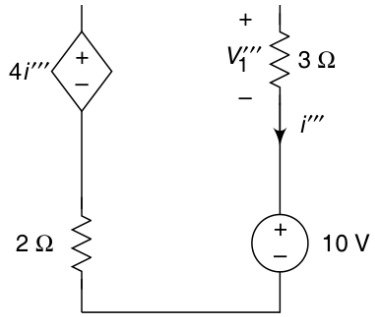
Case I: 8 A alone. Supermesh KVL and KCL at node x:

$$3i' + 2i_1 - 4i' = 0 \Rightarrow i_1 = \frac{i'}{2}; \qquad i_1 = 8 + i' \Rightarrow \frac{i'}{2} = 8 + i' \Rightarrow i' = -16\ \text{A},$$

$$V_1' = 3i' = -48\ \text{V}$$

Case II: 2 A alone. Mesh current i_2 on the left; the 3 Ω carries $i'' = i_2 + 2$:

$$3(i_2 + 2) + 2i_2 - 4i'' = 0 \Rightarrow 5i_2 + 6 - 4(i_2 + 2) = 0 \Rightarrow i_2 = 2\ \text{A}, \qquad i'' = 4\ \text{A}, \qquad V_1'' = 12\ \text{V}$$



Case III: 10 V alone

Case III: 10 V alone. One loop:

$$3i''' - 10 + 2i''' - 4i''' = 0 \Rightarrow i''' = 10\ \text{A}, \qquad V_1''' = 30\ \text{V}$$

ANSWER

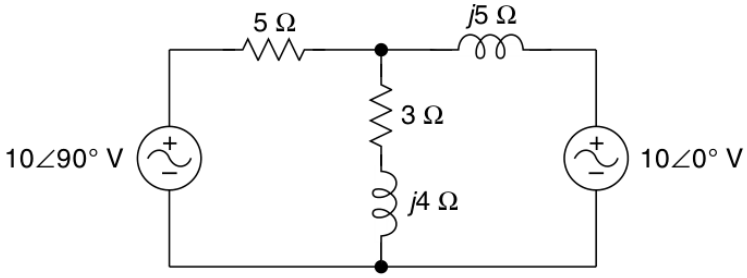
$V_1 = -48 + 12 + 30 = -6$ V (= book)

A12 · Problem 5.12

Superposition · AC **new idea**

QUESTION

Calculate the current through the impedance (3 + j4) Ω using the superposition theorem.

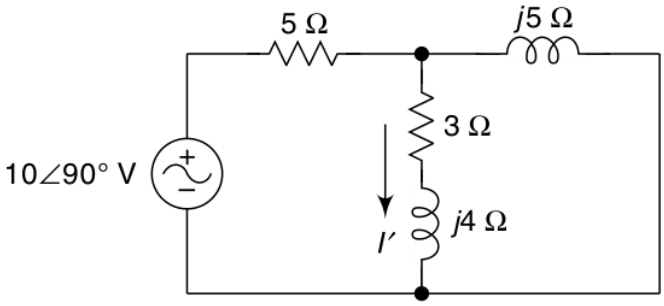


TRY IT FIRST

Two voltage sources, same frequency. Get each case's current as a fraction before simplifying.

HINT: WHAT'S NEW

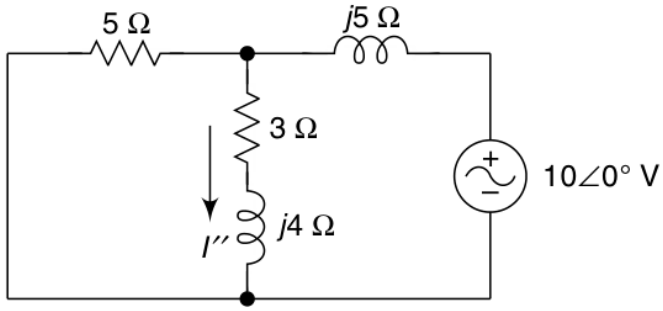
- **The two contributions can cancel.** Here they come out equal and opposite, so the branch current is zero. A zero answer is not an error; check it with one node equation.



10∠90° V alone

$$10\angle 90^\circ \text{ alone:} \qquad I = \frac{j10}{5 + \dfrac{(3 + j4)j5}{3 + j9}} = \frac{j10(3 + j9)}{-5 + j60},$$

$$I' = I \times \frac{j5}{3 + j9} = \frac{-10}{-1 + j12} = 0.069 + j0.828 \text{ A}$$



10∠0° V alone

$$10\angle 0^\circ \text{ alone:} \qquad I = \frac{10}{j5 + \dfrac{5(3 + j4)}{8 + j4}} = \frac{10(8 + j4)}{-5 + j60},$$

$$I'' = I \times \frac{5}{8 + j4} = \frac{10}{-1 + j12} = -0.069 - j0.828 \text{ A}$$

ANSWER

I = I′ + I″ = 0 A (= book)

CHECK

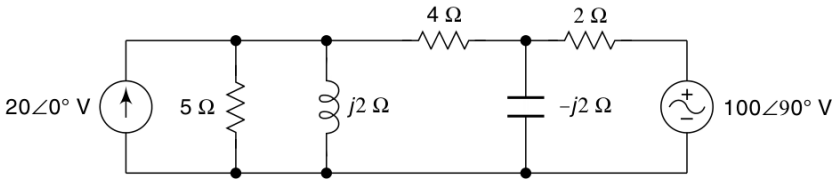
Node equation at the top node: $V(\frac{1}{5} + \frac{1}{3 + j4} + \frac{1}{j5}) = \frac{j10}{5} + \frac{10}{j5} = j2 - j2 = 0$, so V = 0 and no current flows in 3 + j4 ✓.

A13 · Problem 5.13

Superposition · AC **practice**

QUESTION

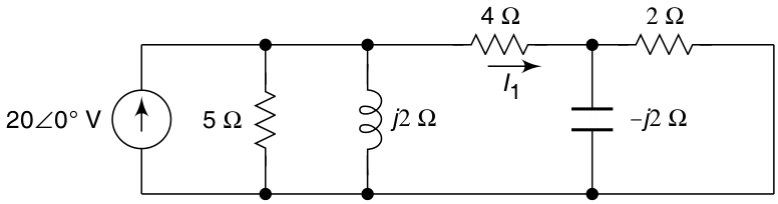
Using the superposition theorem, determine the current in the 4 Ω resistor.



The book labels the current source "20∠0° V"; it is a current source (arrow), 20∠0° A.

TRY IT FIRST

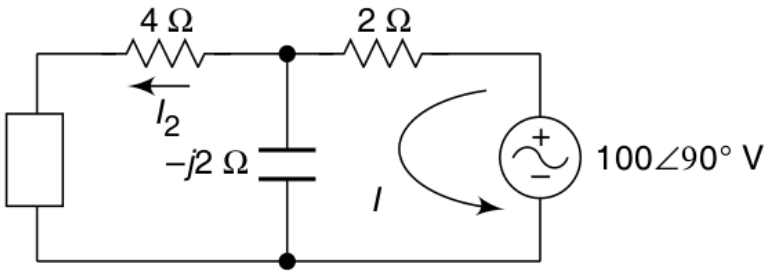
Combine the parallel pairs first: 5 Ω ‖ j2 Ω and 2 Ω ‖ −j2 Ω.



Case I: 20∠0° A alone

Case I: 20∠0° A alone (voltage source shorted). $Z_1 = 5 \parallel j2 = 0.690 + j1.724\ \Omega$; $Z_2 = 2 \parallel (-j2) = 1 - j1\ \Omega$.

$$I_1 = 20 \times \frac{Z_1}{Z_1 + 4 + Z_2} = 3.145 + j5.660\ \text{A} \qquad \text{(left} \rightarrow \text{right)}$$



Case II: 100∠90° V alone

Case II: 100∠90° V alone (current source opened). $(Z_1 + 4) \parallel (-j2) = 0.850 - j1.950\ \Omega$:

$$I_s = \frac{j100}{2 + (0.850 - j1.950)} = -16.35 + j23.90\ \text{A},$$

$$V_B = I_s(0.850 - j1.950) = 32.70 + j52.20\ \text{V}$$

$$I_2 = \frac{V_B}{Z_1 + 4} = 9.748 + j7.547 \text{ A} \qquad \text{(right} \rightarrow \text{left)}$$

ANSWER

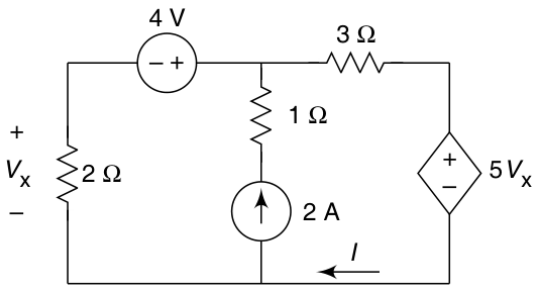
$I = I_1 - I_2 = -6.604 - j1.887 = 6.87\angle -164.05°$ A (left → right), i.e. 6.87 A flowing right → left (= book, $6.89\angle -163.67°$ A, to rounding)

A14 · Problem 5.14

Superposition · dependent source **practice**

QUESTION

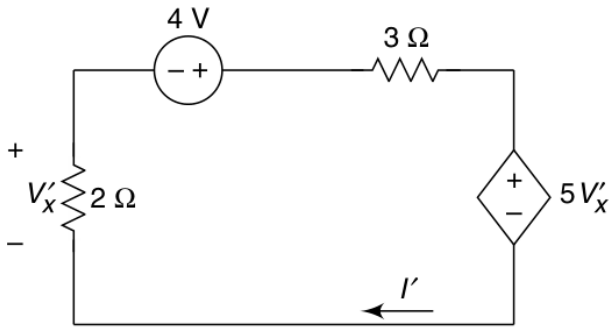
Find I in the figure using the superposition theorem.



$5V_x$ is a voltage-controlled voltage source; V_x is across the 2 Ω (+ at top).

TRY IT FIRST

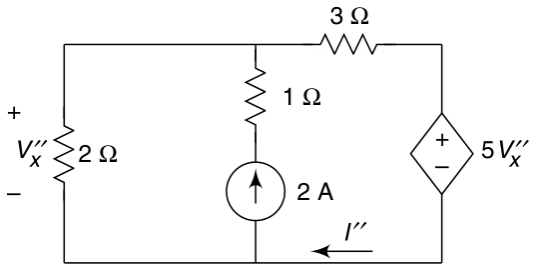
Same idea as 5.3. In case II, the 4 V short joins the two top-left nodes; write one KCL there.



Case I: 4 V alone

Case I: 4 V alone (2 A opened). One loop; I′ flows up the 2 Ω, so $V_x' = -2I'$:

$$-4 + 3I' + 5V_x' - V_x' = 0 \Rightarrow 3I' + 4(-2I') = 4 \Rightarrow I' = -0.8 \text{ A}$$



Case II: 2 A alone

Case II: 2 A alone (4 V shorted). KCL at the top node (voltage V_x''):

$$2 = \frac{V_x''}{2} + \frac{V_x'' - 5V_x''}{3} \Rightarrow V_x'' = -2.4\ \text{V}, \qquad I'' = \frac{V_x'' - 5V_x''}{3} = 3.2\ \text{A}$$

ANSWER

I = −0.8 + 3.2 = 2.4 A (= book)

B · Thevenin's and Norton's theorems

Superposition: one source at a time (done) → **Thevenin: whole network → one source + one impedance** → Norton: the same as a current source → Maximum power transfer (next)

Where this sits. Superposition showed that a linear circuit's response is a sum of simple parts. Thevenin's theorem uses that fact to shrink everything a load is connected to into **two numbers**, so changing the load never means re-solving the whole circuit.

Three terms first.

- **Terminals a–b:** the two points where the load connects. Everything on the other side of them is the **network**.
- **Open-circuit voltage V_{oc}:** the voltage across a–b with the load removed (nothing connected).
- **Short-circuit current I_{sc}:** the current through a wire placed across a–b, counted **from a to b through the wire** (out of the + terminal).

Statement. Seen from a–b, any linear network behaves exactly like **one voltage source $V_{th} = V_{oc}$ in series with one impedance Z_{th}**. Norton's form is the same network as **one current source $I_N = I_{sc}$ in parallel with $Z_N = Z_{th}$**. The two forms are a source transformation of each other: $V_{th} = I_N Z_{th}$.

Finding Z_{th}: three cases.

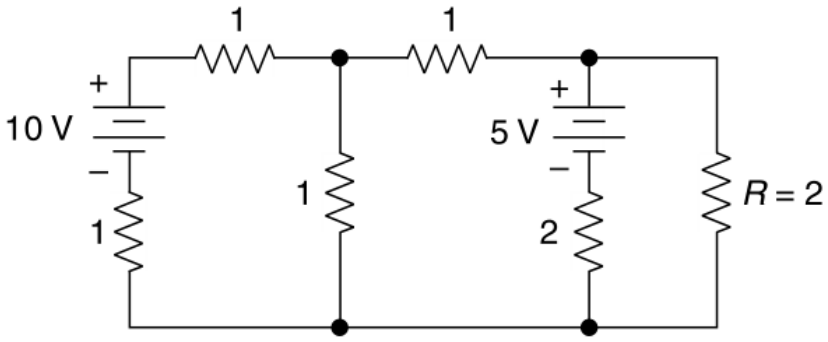
The network contains	Z_{th} is found by
Independent sources only	switch every source off (V → short, I → open), then combine impedances seen from a–b
Independent **and** dependent sources	$Z_{th} = V_{oc} / I_{sc}$, or switch off the independent sources only and apply a test source
Dependent sources only	$V_{th} = 0$; apply a test source (1 A in, measure V): $Z_{th} = V / I$

With dependent sources Z_{th} can come out **negative** (the network supplies energy). Keep the sign of I_{sc}: it decides the sign of Z_{th}.

A15 · Problem 5.15

Thevenin · DC **teach**

> **QUESTION**
> Draw the Thevenin's equivalent of the circuit in the figure and find the load current i. All values are in ohm.



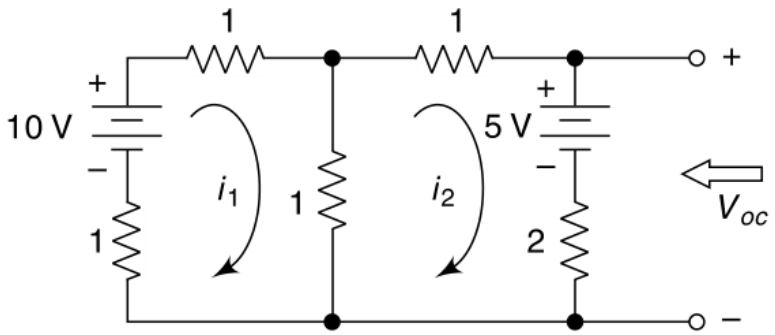
i flows down through R = 2 Ω.

TRY IT FIRST

Remove R. Find the voltage across its terminals (two meshes), then short both sources and find the resistance seen from there. Last, put R back on the equivalent.

HINT: WHAT'S NEW

- **The Thevenin recipe:** (1) remove the load; (2) V_{th} = open-circuit voltage at its terminals; (3) R_{th} = resistance seen into the terminals with every independent source switched off; (4) load current = $V_{th} / (R_{th} + R_L)$.
- **Switching off for R_{th}** uses the same rule as superposition: voltage sources become shorts.



Load removed: two mesh currents, V_{oc} across the open terminals

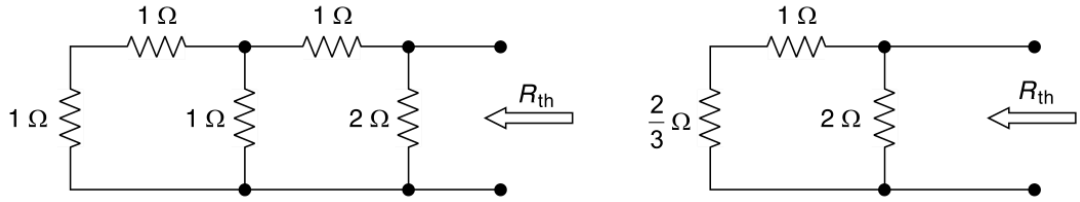
Given: 10 V, 5 V; 1 Ω (series with 10 V), 1 Ω, 1 Ω, 2 Ω; R = 2 Ω. **To find:** V_{th}, R_{th}, i.

Step 1: V_{th} (R removed). Mesh currents clockwise:

$$\text{Mesh 1:} \quad 3i_1 - i_2 = 10 \qquad \text{Mesh 2:} \quad -i_1 + 4i_2 = -5$$

$$\Rightarrow \quad i_1 = \frac{35}{11}\ \text{A}, \qquad i_2 = -\frac{5}{11}\ \text{A}$$

$$V_{th} = V_{oc} = 5 + 2i_2 = 5 - \frac{10}{11} = \frac{45}{11} = 4.091\ \text{V}$$



Both sources shorted: R_{th} seen from the load terminals

Step 2: R_{th} (both sources shorted):

$$(1 + 1) \parallel 1 = \frac{2}{3}\ \Omega, \qquad \frac{2}{3} + 1 = \frac{5}{3}\ \Omega, \qquad R_{th} = \frac{5}{3} \parallel 2 = \frac{10}{11} = 0.909\ \Omega$$

Step 3: load current:

$$i = \frac{V_{th}}{R_{th} + R} = \frac{45/11}{10/11 + 2} = \frac{45}{32}\ \text{A}$$

ANSWER

Thevenin equivalent: 45/11 V in series with 10/11 Ω. i = 1.406 A (= book)

SIDE NOTE

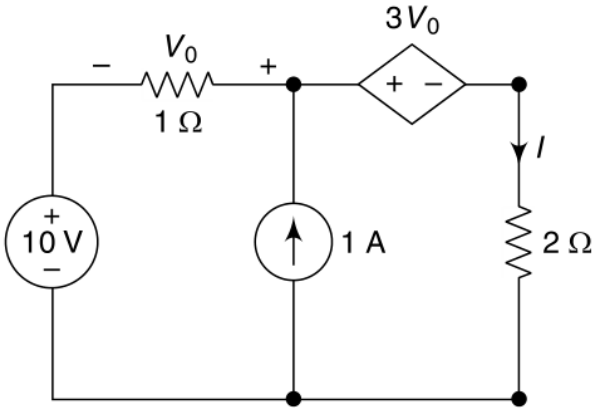
The book prints $i_1 = 5/11$; it is 35/11 (only i_2 is used, so the answer is unaffected). Full node solve with R in place: i = 1.40625 A ✓.

A16 · Problem 5.16

Thevenin · dependent source **teach**

QUESTION

Find I in the given figure, using Thevenin's theorem.



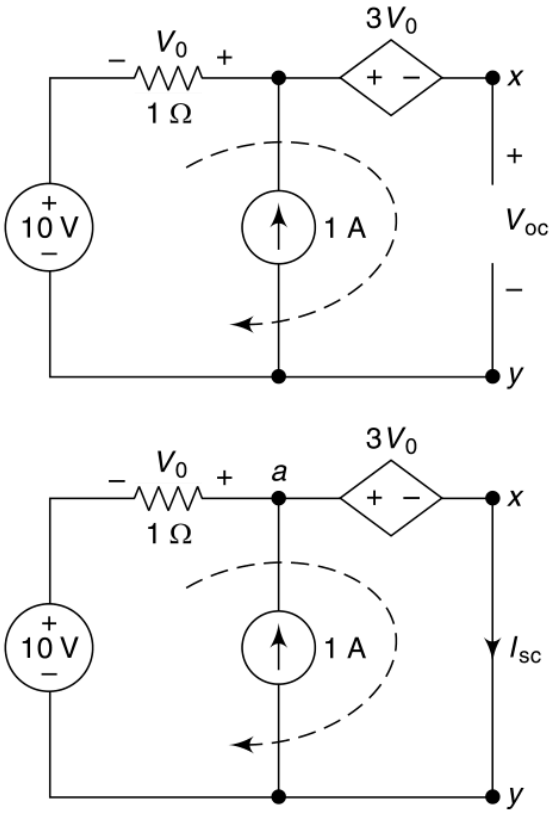
$3V_0$ is a voltage-controlled voltage source; V_0 is across the 1 Ω (+ on the right).

TRY IT FIRST

The dependent source cannot be switched off, so R_{th} must come from V_{oc} / I_{sc}. Find both, and keep the sign of I_{sc}.

HINT: WHAT'S NEW

- **Dependent source ⇒ $R_{th} = V_{oc} / I_{sc}$** (or a test source). Shorting the independent sources and combining resistors would ignore the dependent source.
- **I_{sc} is measured from x to y through the short.** A negative I_{sc} gives a **negative R_{th}**, which is allowed: the dependent source can supply energy.



Top: terminals x–y open (V_{oc}). Bottom: x–y shorted (I_{sc}).

Given: 10 V, 1 A, 1 Ω, VCVS $3V_0$, load 2 Ω. **To find:** I.

V_{oc} (2 Ω removed). Nothing leaves at x, so the 1 A must flow left through the 1 Ω: $V_0 = 1 \times 1 = 1$ V.

$$V_a = 10 + V_0 = 11 \text{ V}, \qquad V_{oc} = V_a - 3V_0 = 11 - 3 = 8 \text{ V}$$

I_{sc} (x shorted to y, so $V_x = 0$):

$$0 = V_a - 3V_0 = (10 + V_0) - 3V_0 \qquad \Rightarrow \qquad V_0 = 5 \text{ V}$$

$$\text{KCL at a:} \qquad 1 = \frac{V_0}{1} + I_{sc} \qquad \Rightarrow \qquad I_{sc} = 1 - 5 = -4\ \text{A}$$

$$R_{th} = \frac{V_{oc}}{I_{sc}} = \frac{8}{-4} = -2\ \Omega$$

Load: $R_{th} + 2\ \Omega = -2 + 2 = 0$, so

$$I = \frac{V_{th}}{R_{th} + 2} = \frac{8}{0} \qquad \Rightarrow \qquad \text{no finite solution}$$

ANSWER

$V_{th} = 8$ V, $R_{th} = -2\ \Omega$. With the 2 Ω load, $R_{th} + R_L = 0$: the circuit as drawn has no solution. (Book: 2 A.)

BOOK ERROR IN 5.16

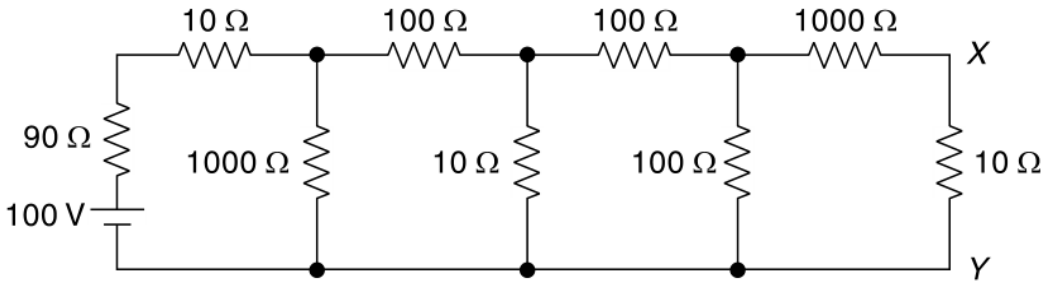
The book finds $I_{sc} = -4$ A, then writes $R_{th} = 8/4 = 2\ \Omega$, dropping the minus sign, and prints $I = 2$ A. Direct check with the 2 Ω in place: KCL at a gives $V_0 = 1 - I$; the right-hand path gives $V_x = 10 - 2V_0 = 8 + 2I$; but $V_x = 2I$, so $8 = 0$, a contradiction. A test source (10 V shorted, 1 A opened, 1 A pushed into x) confirms $R_{th} = -2\ \Omega$. For any other load, $I = \frac{8}{R_L - 2}$.

A17 · Problem 5.17

Thevenin · repeated reduction **new idea**

QUESTION

By the iterative use of Thevenin's theorem, reduce the circuit shown in the figure to a single emf acting in series with a single resistor. Hence, calculate the current in the 10 Ω resistor connected across XY.



Ladder network; the load is the 10 Ω across XY.

TRY IT FIRST

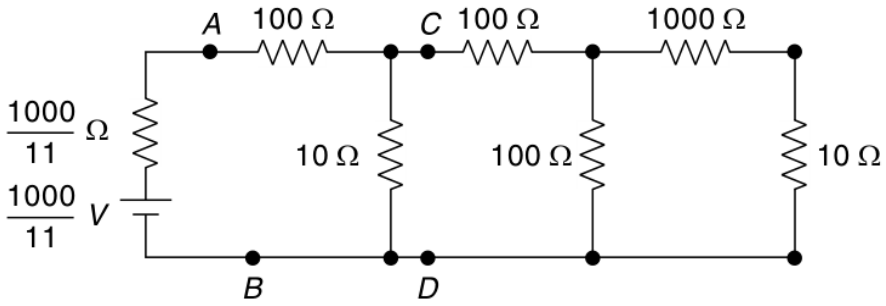
Cut the ladder just right of the first shunt resistor and replace the left part by its Thevenin equivalent. Repeat, one rung at a time.

HINT: WHAT'S NEW

- **Iterative Thevenin:** a ladder is reduced rung by rung. Each step is a voltage divider for V_{th} and a parallel combination for R_{th}: $V_{new} = V_{old} \times \dfrac{R_{sh}}{R_{old} + R_{series} + R_{sh}}$, $R_{new} = (R_{old} + R_{series}) \parallel R_{sh}$.

Step 1 (left of AB: 100 V behind 90 + 10 = 100 Ω, shunt 1000 Ω):

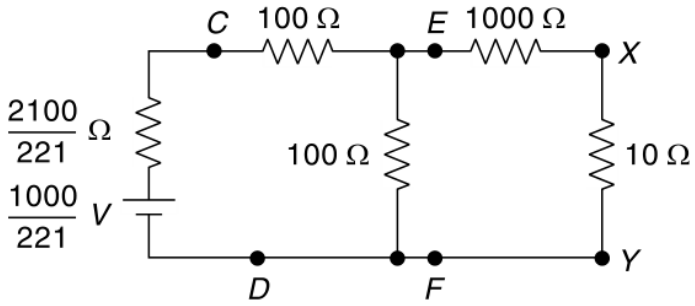
$$V_1 = 100 \times \frac{1000}{1100} = \frac{1000}{11}\ \text{V}, \qquad R_1 = 100 \parallel 1000 = \frac{1000}{11}\ \Omega$$



After step 1

Step 2 (add 100 Ω series, shunt 10 Ω):

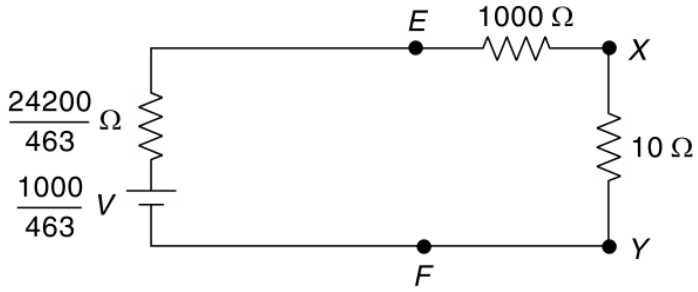
$$V_2 = \frac{1000}{11} \times \frac{10}{2100/11 + 10} = \frac{1000}{221}\ \text{V}, \qquad R_2 = \frac{2100}{11} \parallel 10 = \frac{2100}{221}\ \Omega$$



After step 2

Step 3 (add 100 Ω series, shunt 100 Ω):

$$V_3 = \frac{1000}{221} \times \frac{100}{24200/221 + 100} = \frac{1000}{463}\ \text{V}, \qquad R_3 = \frac{24200}{221} \parallel 100 = \frac{24200}{463}\ \Omega$$



After step 3, with the 1000 Ω still in series

Step 4 (add 1000 Ω series):

$$V_{th} = \frac{1000}{463} = 2.160\ \text{V}, \qquad R_{th} = \frac{24200}{463} + 1000 = \frac{487200}{463} = 1052.3\ \Omega$$

$$I = \frac{1000/463}{487200/463 + 10} = \frac{1000}{491830}\ \text{A}$$

ANSWER

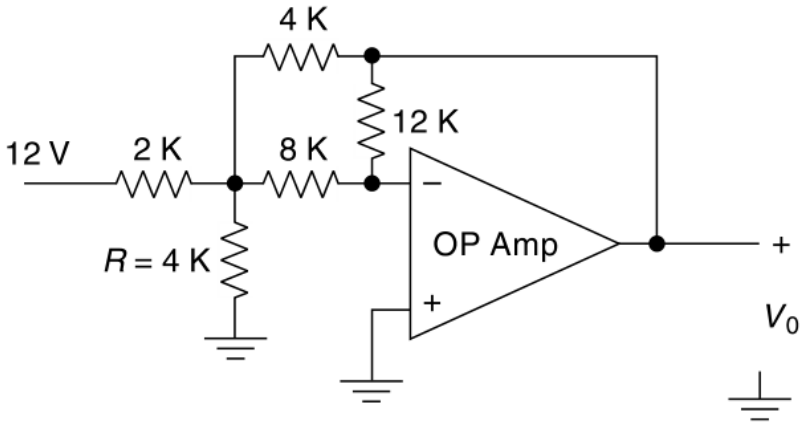
Single emf 2.160 V in series with 1052.3 Ω; I = 2.03 mA ≈ 0.002 A (= book)

A18 · Problem 5.18

Thevenin · op-amp, test source **new idea**

QUESTION

In the operational-amplifier circuit shown in the figure, find I in the R = 4 kΩ resistor, using Thevenin's theorem.



Ideal op-amp; the load is R = 4 kΩ from node e_1 to ground.

TRY IT FIRST

With R removed, write KCL at e_1 and at the inverting input. For R_{th}, short the 12 V and push 1 A into e_1.

HINT: WHAT'S NEW

- **Ideal op-amp rules:** no current into its inputs, and (with negative feedback) $e_2 = e_+ = 0$ V. The output can supply any current, so **no KCL is written at the output node**.
- **Test source for R_{th}:** the op-amp acts like a dependent source, so switch off the 12 V only, inject 1 A at the terminals and read the voltage: $R_{th} = e_1 / 1$ A.

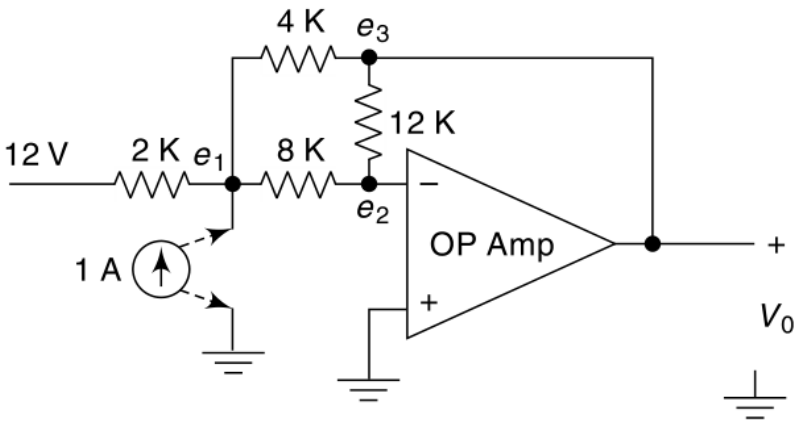
Given: 12 V, 2 kΩ, 4 kΩ (feedback $e_1 \to V_0$), 8 kΩ, 12 kΩ, ideal op-amp. **To find:** I in R = 4 kΩ.

V_{th} (R removed; $e_2 = 0$, $e_3 = V_0$):

$$\text{KCL at } e_2: \quad \frac{0 - e_1}{8k} + \frac{0 - V_0}{12k} = 0 \quad \Rightarrow \quad V_0 = -\frac{3}{2}e_1$$

$$\text{KCL at } e_1: \quad \frac{e_1 - 12}{2k} + \frac{e_1 - V_0}{4k} + \frac{e_1}{8k} = 0 \quad \Rightarrow \quad 7e_1 = 48 + 2V_0$$

$$7e_1 = 48 - 3e_1 \quad \Rightarrow \quad V_{th} = e_1 = 4.8 \text{ V}$$



12 V shorted, 1 A test source at the load terminals

R_{th} (12 V shorted, 1 A injected at e_1):

$$\frac{e_1}{2k} + \frac{e_1 - V_0}{4k} + \frac{e_1}{8k} = 1 \quad \Rightarrow \quad 7e_1 - 2V_0 = 8000$$

$$10e_1 = 8000 \quad \Rightarrow \quad e_1 = 800 \text{ V}$$

$$R_{th} = \frac{800 \text{ V}}{1 \text{ A}} = 800 \ \Omega, \qquad I = \frac{4.8}{800 + 4000} = 1 \times 10^{-3} \text{ A}$$

ANSWER

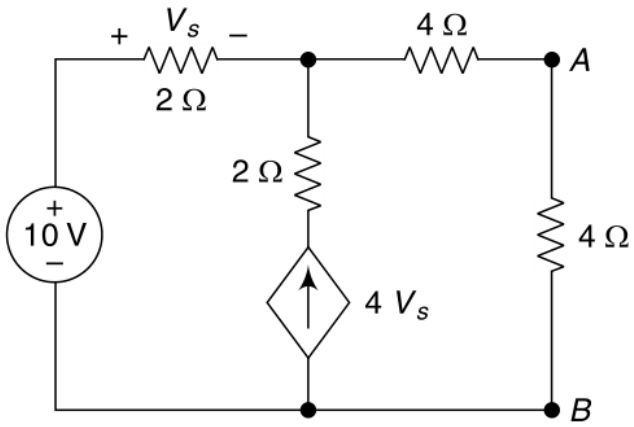
V_{th} = 4.8 V, R_{th} = 800 Ω, I = 1 mA (= book)

A19 · Problem 5.19

Thevenin · dependent current source **practice**

QUESTION

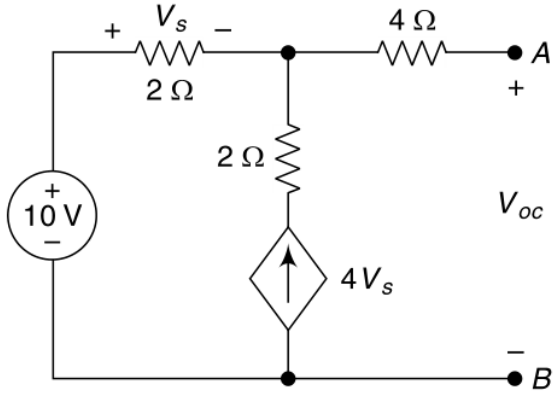
Find Thevenin's equivalent about AB for the circuit shown in the figure.



The 4 Ω across AB is the load (the book removes it).

TRY IT FIRST

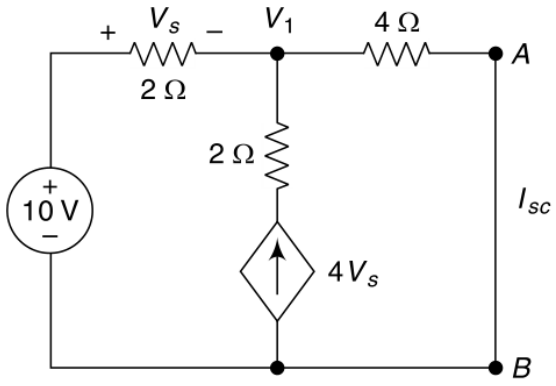
V_{oc}: with AB open, what does KCL at the middle node say about V_s? Then short AB for I_{sc}.



AB open

V_{oc} (AB open, so the series 4 Ω carries nothing; $V_s = 10 - V_M$). KCL at the middle node M:

$$\frac{V_s}{2} + 4V_s = 0 \quad \Rightarrow \quad V_s = 0 \quad \Rightarrow \quad V_{oc} = V_M = 10\ \text{V}$$



AB shorted

I_{sc} (AB shorted; $V_1 = V_M$, $V_s = 10 - V_1$):

$$\frac{V_1 - 10}{2} + \frac{V_1}{4} = 4(10 - V_1) \quad \Rightarrow \quad 19V_1 = 180 \quad \Rightarrow \quad V_1 = 9.474 \text{ V}$$

$$I_{sc} = \frac{V_1}{4} = 2.368 \text{ A}$$

$$R_{th} = \frac{10}{2.368} = 4.22 \ \Omega$$

ANSWER

V_{th} = 10 V, R_{th} = 38/9 = 4.22 Ω (= book)

SIDE NOTE

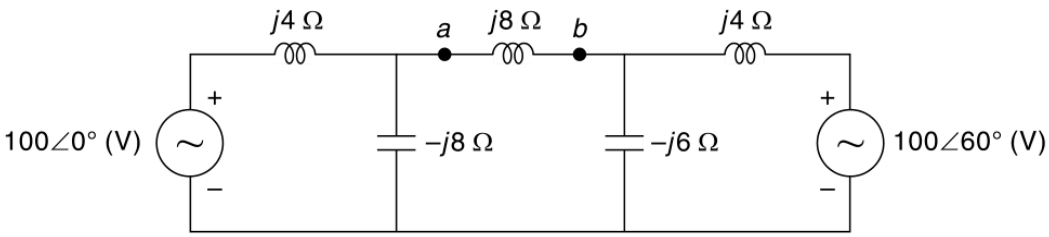
The 2 Ω in series with the dependent current source never matters (5.1's shortcut). Test-source check (10 V shorted, 1 A into A): $V_A = 4 + \frac{1}{4.5} = 4.22 \text{ V} \Rightarrow R_{th} = 4.22 \ \Omega$ ✓.

A20 · Problem 5.20

Thevenin · AC, floating terminals **new idea**

QUESTION

In the network, determine the steady current in the 8 Ω inductor using Thevenin's theorem.



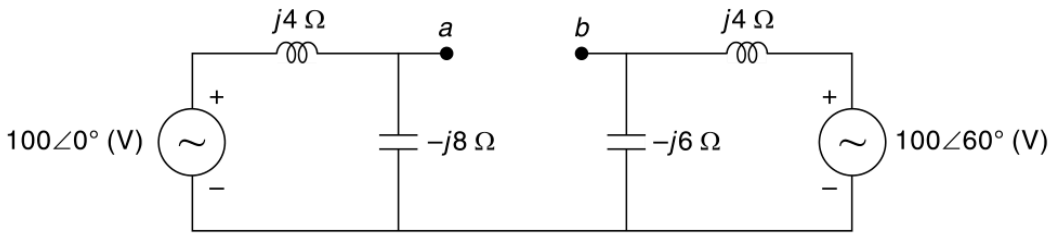
The load is the j8 Ω between a and b.

TRY IT FIRST

With j8 removed, the circuit splits into two separate halves. Find V_a and V_b with a divider each.

HINT: WHAT'S NEW

- **Neither terminal is ground.** $V_{th} = V_a - V_b$, each found from its own half.
- With the sources shorted, each half is a parallel pair, and the two pairs are **in series** between a and b.



j8 Ω removed: two independent halves

$$V_a = 100\angle 0° \times \frac{-j8}{j4 - j8} = 200\angle 0° \text{ V}, \qquad V_b = 100\angle 60° \times \frac{-j6}{j4 - j6} = 300\angle 60° \text{ V}$$

$$V_{th} = V_a - V_b = 200 - (150 + j259.81) = 50 - j259.81 = 264.58\angle -79.11° \text{ V}$$

$$Z_{th} = \frac{(j4)(-j8)}{j4 - j8} + \frac{(j4)(-j6)}{j4 - j6} = j8 + j12 = j20\ \Omega$$

$$i = \frac{V_{th}}{Z_{th} + j8} = \frac{264.58\angle -79.11°}{28\angle 90°}$$

ANSWER

i = 9.45∠−169.11° A (a → b) (= book)

A21 · Problem 5.21

Thevenin / Norton · AC, s-domain, dependent source **new idea**

QUESTION

Obtain Thevenin's equivalent circuit with respect to terminals A–B in the networks shown below.

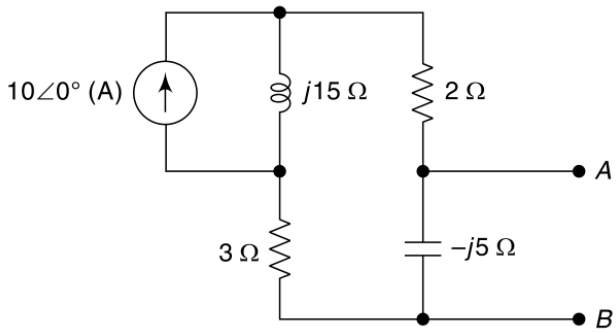
TRY IT FIRST

(a)–(c) are practice. In (d), treat s like jω. In (e) the dependent source stays, so use V_{oc} / I_{sc}.

HINT: WHAT'S NEW

- **(d) s-domain:** impedances are R, sL and $\frac{1}{sC}$; the source becomes its Laplace transform, 2 cos 2t $\rightarrow \frac{2s}{s^2 + 4}$. Everything else is unchanged.
- **(e) AC with a dependent source:** find V_{th} and I_N separately; $Z_{th} = V_{th} / I_N$.

(a)



Source transformation: $10\angle 0°$ A ‖ j15 Ω → j150 V in series with j15 Ω. With AB open, one loop:

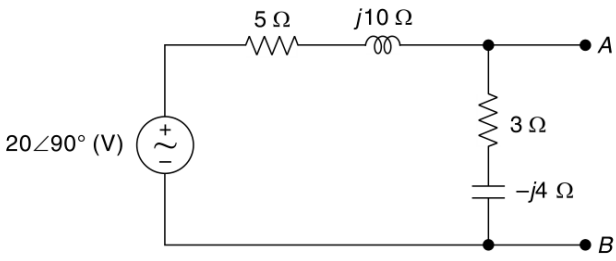
$$I = \frac{j150}{j15 + 2 - j5 + 3} = \frac{j150}{5 + j10} = 13.42\angle 26.57° \text{ A}, \qquad V_{th} = I \times (-j5) = 30 - j60 \text{ V}$$

$$Z_{th} = (-j5) \parallel (5 + j15) = \frac{-j5(5 + j15)}{5 + j10} = 1 - j7\ \Omega$$

ANSWER

(a) $V_{th} = 67.08\angle -63.43°$ V, $Z_{th} = 7.07\angle -81.87°\ \Omega$ (= book)

(b)



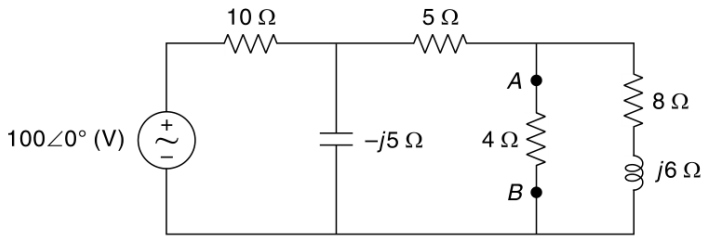
$$V_{th} = 20\angle 90° \times \frac{3 - j4}{(5 + j10) + (3 - j4)} = \frac{j20(3 - j4)}{8 + j6} = \frac{100\angle 36.87°}{10\angle 36.87°} = 10\angle 0° \text{ V}$$

$$Z_{th} = \frac{(5 + j10)(3 - j4)}{8 + j6} = \frac{55 + j10}{8 + j6} = 5 - j2.5\ \Omega$$

ANSWER

(b) $V_{th} = 10\angle 0°$ V, $Z_{th} = 5.59\angle -26.57°\ \Omega$ (= book; its first line prints j120, a typo for j20)

(c) The 4 Ω across AB is the load.



$$Z = 10 + (-j5) \parallel (13 + j6) = 11.91 - j5.15 = 12.98\angle -23.37°\ \Omega$$

$$I = \frac{100}{Z} = 7.706\angle 23.37°\ \text{A}$$

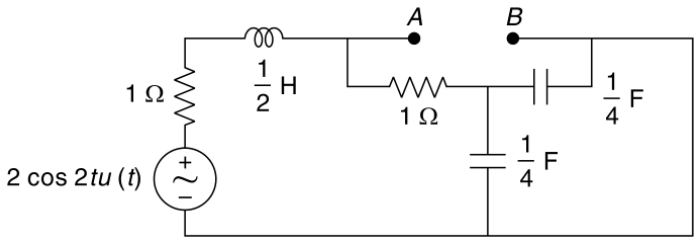
$$I_{right} = I \times \frac{-j5}{13 + j1} = 2.955\angle -71.03°\ \text{A}, \qquad V_{th} = I_{right}(8 + j6) = 29.55\angle -34.16°\ \text{V}$$

$$Z_{th} = [(10 \parallel -j5) + 5] \parallel (8 + j6) = (7 - j4) \parallel (8 + j6) = 5.328 - j0.044\ \Omega$$

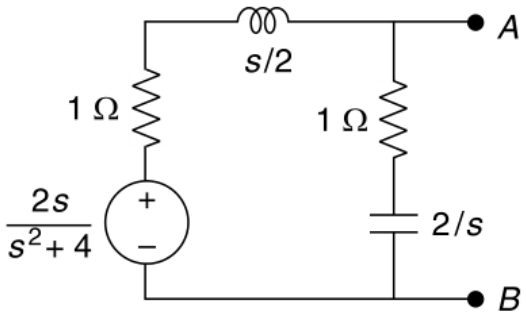
ANSWER

(c) $V_{th} = 29.55\angle -34.16°$ V, $Z_{th} = 5.33\angle -0.47°\ \Omega$ (= book)

(d)



The two $\frac{1}{4}$ F capacitors both run from the same node to ground, so they add: $\frac{1}{2}$ F → $\frac{2}{s}$. A is the node after the inductor; B is ground.



Redrawn in the s-domain

$$V_{th}(s) = \frac{2s}{s^2+4} \times \frac{1 + 2/s}{(1 + 2/s) + (1 + s/2)} = \frac{2s}{s^2+4} \times \frac{2}{s+2} = \frac{4s}{(s^2+4)(s+2)}$$

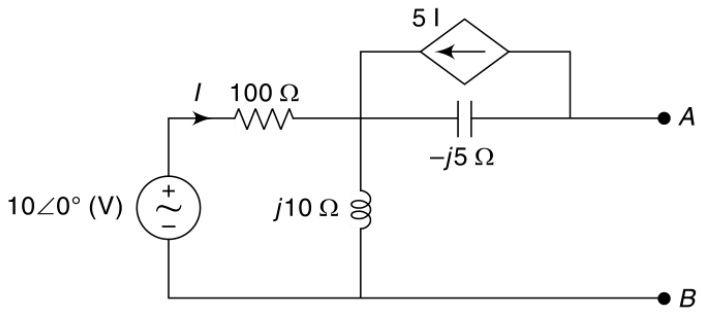
(Simplify: $1 + 2/s = \frac{s+2}{s}$ and $2 + s/2 + 2/s = \frac{(s+2)^2}{2s}$, so the ratio is $\frac{2}{s+2}$.)

$$Z_{th}(s) = (1 + 2/s) \parallel (1 + s/2) = \frac{(s+2)^2/(2s)}{(s+2)^2/(2s)} = 1\ \Omega$$

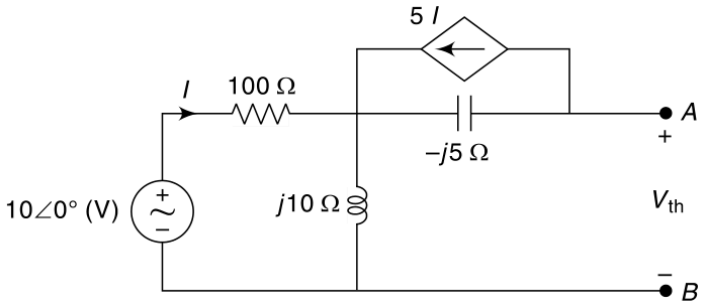
ANSWER

(d) $V_{th}(s) = \frac{4s}{(s^2+4)(s+2)}$ V, $Z_{th} = 1\ \Omega$ (= book)

(e)



I flows through the 100 Ω; the dependent source 5I points from A towards the middle node.

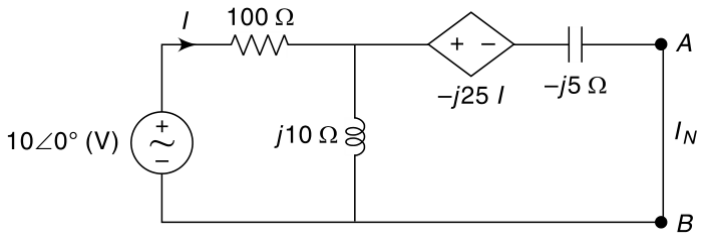


AB open

V_{th}. With AB open, the 5I circulates only through the −j5 Ω, so the source current is set by the left loop alone:

$$I = \frac{10\angle 0°}{100 + j10} = 0.0995\angle -5.71° \text{ A}$$

$$V_{th} = V_{AB} = I(j10) - 5I(-j5) = j35I = 3.483\angle 84.29° \text{ V}$$



AB shorted, with 5I ‖ −j5 transformed to −j25I in series with −j5

I_N. AB shorted. Right loop: $j10(I - I_N) + j25I + j5I_N = 0 \Rightarrow I_N = 7I$. Left loop:

$$10 = (100 + j10)I - j10I_N = (100 - j60)I \qquad \Rightarrow \qquad I = 0.0857\angle 30.96° \text{ A}$$

$$I_N = 7I = 0.600\angle 30.96° \text{ A}$$

$$Z_{th} = \frac{V_{th}}{I_N} = \frac{3.483\angle 84.29°}{0.600\angle 30.96°} = 5.80\angle 53.33° \ \Omega$$

ANSWER

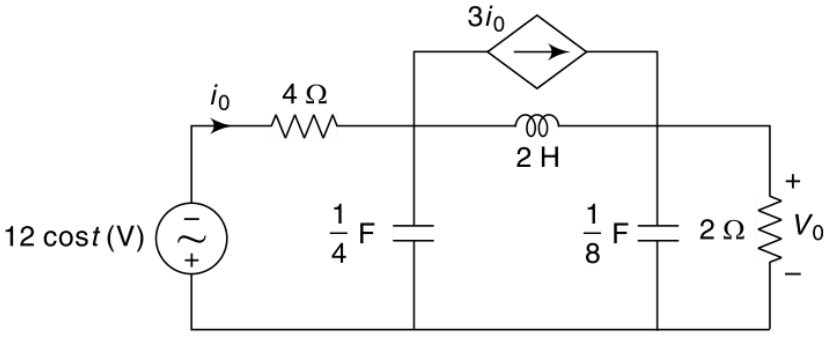
(e) $V_{th} = 3.48\angle 84.3°$ V, $I_N = 0.600\angle 31.0°$ A, $Z_{th} = 5.80\angle 53.3°$ Ω (= book)

A22 · Problem 5.22

Thevenin · AC, dependent source **practice**

QUESTION

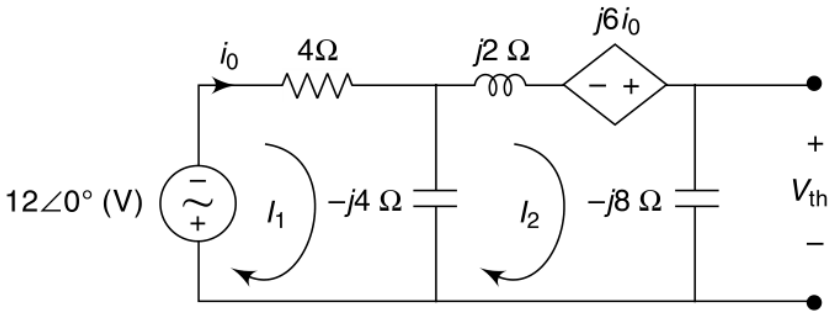
Find V_0 using Thevenin's theorem.



ω = 1 rad/s. The source's + is at the bottom; i_0 flows right through the 4 Ω.

TRY IT FIRST

ω = 1: work out the three reactances first. Transform $3i_0 \parallel j2$ into a voltage source, then write two mesh equations for each of V_{th} and I_N.



2 Ω removed; $3i_0 \parallel j2\ \Omega \rightarrow j6i_0$ in series with j2 Ω (+ on the right)

Reactances (ω = 1): 2 H → j2 Ω, $\frac{1}{4}$ F → −j4 Ω, $\frac{1}{8}$ F → −j8 Ω.

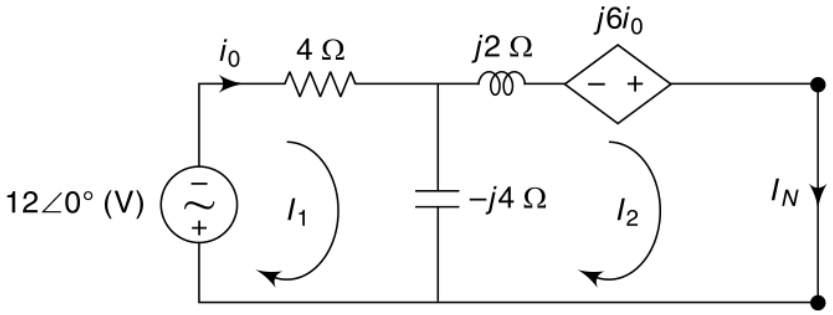
V_{th} (2 Ω removed; clockwise mesh currents, $i_0 = I_1$):

$$\text{Mesh 1:} \quad (4 - j4)I_1 + j4I_2 = -12$$

$$\text{Mesh 2:} \quad -j4(I_2 - I_1) + j2I_2 - j6I_1 - j8I_2 = 0 \quad \Rightarrow \quad -j2I_1 - j10I_2 = 0 \quad \Rightarrow \quad I_2 = -0.2I_1$$

$$(4 - j4.8)I_1 = -12 \quad \Rightarrow \quad I_1 = -1.230 - j1.475\ \text{A}, \quad I_2 = 0.246 + j0.295\ \text{A}$$

$$V_{th} = I_2(-j8) = 2.361 - j1.967 = 3.073\angle -39.81°\ \text{V}$$



Terminals shorted (the −j8 is shorted out)

I_N (terminals shorted):

$$\text{Mesh 2:} \quad -j4(I_2 - I_1) + j2I_2 - j6I_1 = 0 \quad \Rightarrow \quad I_2 = -I_1; \quad (4 - j8)I_1 = -12$$

$$I_N = I_2 = \frac{12}{4 - j8} = 0.6 + j1.2 = 1.342\angle 63.43^\circ \text{ A}$$

$$Z_{th} = \frac{V_{th}}{I_N} = 2.290\angle -103.24^\circ = -0.525 - j2.230\ \Omega$$

$$V_0 = V_{th} \times \frac{2}{Z_{th} + 2} = \frac{3.073\angle -39.81^\circ \times 2}{2.674\angle -56.50^\circ} = 2.299\angle 16.70^\circ \text{ V}$$

ANSWER

$V_0 = 2.30\angle 16.70^\circ$ V, i.e. $v_0(t) = 2.30\cos(t + 16.70^\circ)$ V

BOOK ERROR IN 5.22

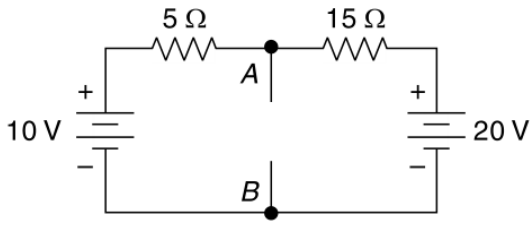
For V_{th}, the book writes mesh 2 as $-j2I_1 - j6I_2 = 0$: it keeps the shared $-j4$ in the I_1 term but drops it from I_2's own impedance ($-j4 + j2 - j8 = -j10$, not $-j6$). That gives $V_{th} = 4.8\angle -36.87^\circ$ V, $Z_{th} = 3.58\angle -100.3^\circ\ \Omega$ and $V_0 = 1.27\angle 32^\circ$ V. Its $I_N = 1.342\angle 63.43^\circ$ A is right. The answer above was confirmed by a full node solve with the 2 Ω in place ($2.2988\angle 16.70^\circ$ V).

A23 · Problem 5.23

Norton · DC **new idea**

QUESTION

Obtain the Norton's equivalent circuit with respect to the terminals AB for the network shown in the figure.

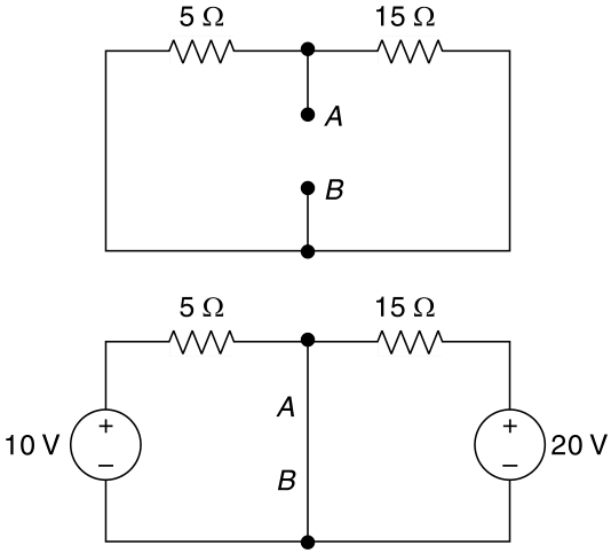


TRY IT FIRST

Short A to B. What current does each source push through the short?

HINT: WHAT'S NEW

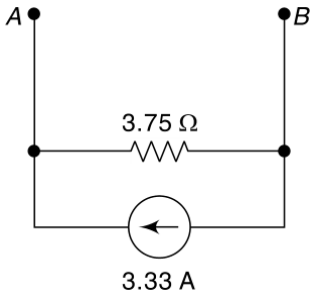
- **Norton's form:** $I_N = I_{sc}$ (current through the A–B short), $R_N = R_{th}$, drawn as a current source in parallel with R_N, arrow pointing towards A.
- With A–B shorted, each source sees only its own resistor, so I_{sc} is a sum of two Ohm's-law currents.



Top: sources removed (R_N). Bottom: A–B shorted (I_N).

$$R_N = 5 \parallel 15 = \frac{75}{20} = 3.75\ \Omega$$

$$I_N = I_{sc} = \frac{10}{5} + \frac{20}{15} = 2 + 1.333 = 3.33\ \text{A}$$



Norton equivalent (arrow towards A)

ANSWER

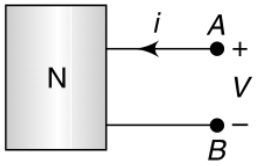
I_N = 3.33 A (towards A) in parallel with R_N = 3.75 Ω (= book). Check: $V_{th} = I_N R_N$ = 12.5 V = node voltage at A ✓.

A24 · Problem 5.24

Norton · from a v–i characteristic **new idea**

QUESTION

For the one-port shown in the figure, determine the Norton's equivalent at the terminals AB, if the v–i characteristic is given by 16v = 80 − 2i.



TRY IT FIRST

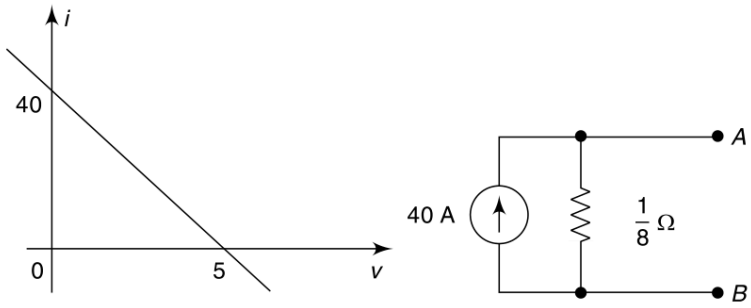
Which two points on the v–i line are V_{oc} and I_{sc}?

HINT: WHAT'S NEW

- **The equivalent is read off the terminal line.** i = 0 gives V_{oc}; v = 0 gives I_{sc}; $R_N = V_{oc} / I_{sc}$ (the size of the line's slope, $\Delta v/\Delta i$).

$$16v = 80 - 2i \quad \Rightarrow \quad \frac{v}{5} + \frac{i}{40} = 1$$

$$i = 0: \quad V_{oc} = 5\ V \qquad v = 0: \quad I_{sc} = 40\ A \qquad R_N = \frac{5}{40} = \frac{1}{8}\ \Omega$$



The v–i line and the Norton equivalent

ANSWER

I_N = 40 A (towards A) in parallel with R_N = 1/8 Ω (= book)

SIGN CONVENTION

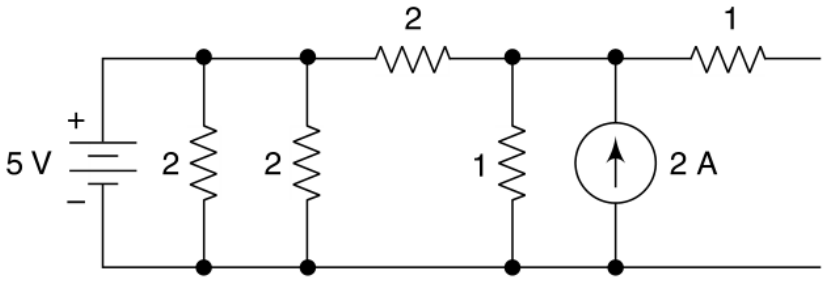
$v = 5 - \frac{i}{8}$ describes an ordinary source only if i is the current **delivered out of A** (v falls as the load draws more). The figure's arrow points into N; read literally, that would make $R_N = -1/8\ \Omega$ with the 40 A reversed. The intended answer is the book's.

A25 · Problem 5.25

Thevenin + Norton · superposition inside **practice**

QUESTION

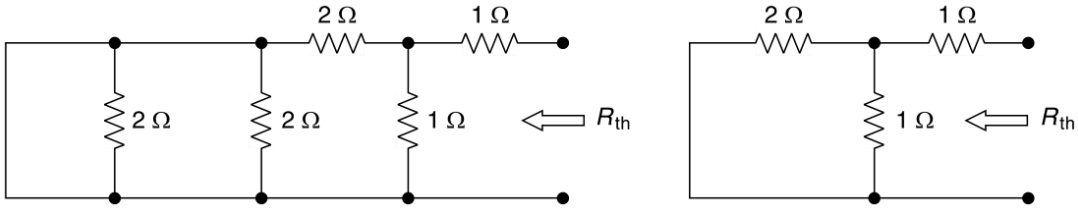
Find both Thevenin's and Norton's equivalent circuits for the network shown in the figure. All values are in ohm.



Terminals at the right-hand end of the 1 Ω series resistor and the bottom rail.

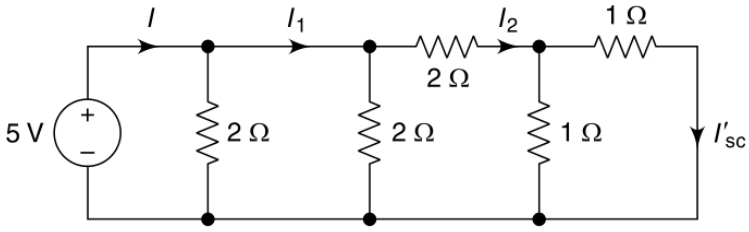
TRY IT FIRST

Which resistors does the shorted 5 V source remove from R_{th}? Then find I_{sc} one source at a time.



Sources switched off: the two 2 Ω across the 5 V are shorted out

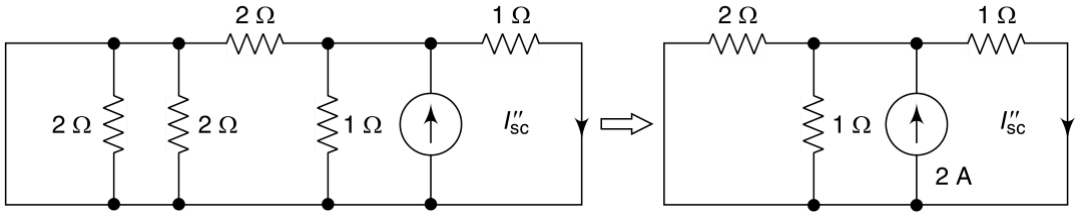
$$R_N = R_{th} = (2 \parallel 1) + 1 = \frac{2}{3} + 1 = \frac{5}{3}\ \Omega$$



5 V alone, terminals shorted

I_{sc}, 5 V alone (2 A opened): the middle node sees 2 Ω to the 5 V, and 1 Ω ∥ 1 Ω to ground:

$$\frac{V_m - 5}{2} + \frac{V_m}{1} + \frac{V_m}{1} = 0 \quad \Rightarrow \quad V_m = 1\ \text{V}, \qquad I_{sc}' = \frac{1}{1} = 1\ \text{A}$$

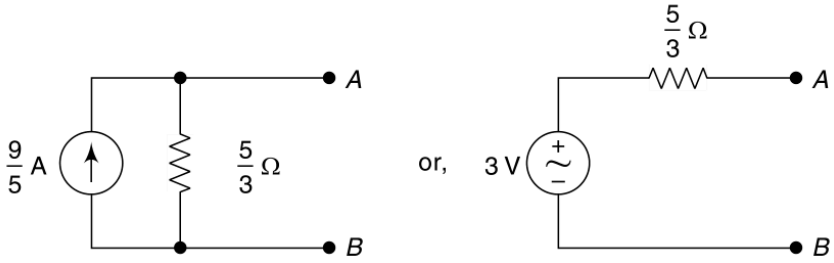


2 A alone, terminals shorted

I_{sc}, 2 A alone (5 V shorted): the 2 A divides between 2 Ω ‖ 1 Ω = 2/3 Ω and the 1 Ω to the short:

$$I_{sc}'' = 2 \times \frac{2/3}{2/3 + 1} = \frac{4}{5}\ \text{A}$$

$$I_N = 1 + \frac{4}{5} = \frac{9}{5}\ \text{A}, \qquad V_{th} = I_N R_{th} = \frac{9}{5} \times \frac{5}{3} = 3\ \text{V}$$



Norton and Thevenin forms

ANSWER

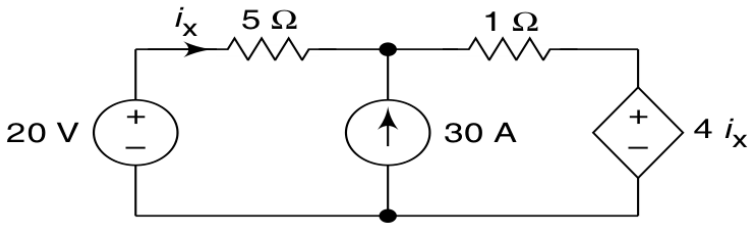
Norton: 1.8 A ‖ 5/3 Ω. Thevenin: 3 V in series with 5/3 Ω (= book)

A26 · Problem 5.26

Thevenin · dependent source **practice**

QUESTION

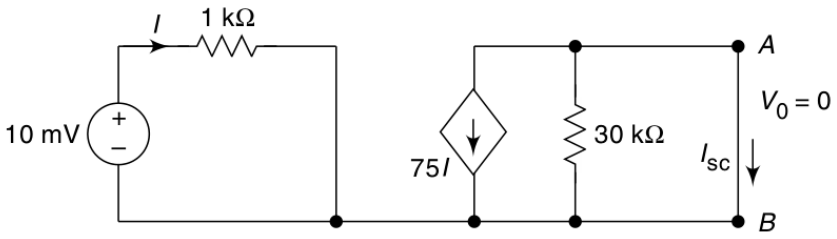
Replace the circuit in the figure with the Thevenin's equivalent circuit across A and B.



The question's figure, as printed: it marks no terminals A and B.

BOOK ERROR IN 5.26: FIGURE AND SOLUTION BELONG TO DIFFERENT CIRCUITS

The figure shows no A and B, so the question cannot be answered as printed. The book's solution solves a different circuit (a transistor model): 10 mV source, 1 kΩ carrying I, a controlled voltage $10^{-4}V_0$ in the input loop, and a controlled current source 75I in parallel with 30 kΩ across A–B. That solution is checked below. Ask your professor which circuit is meant before submitting.



The book's solution circuit, A–B shorted

V_{oc} (A–B open). The 75I pulls current down out of node A, and it can only return through the 30 kΩ, so $V_0 = 30k \times (-75I) = -2.25 \times 10^6 I$. Input loop:

$$10^3 I + 10^{-4}V_0 = 10 \times 10^{-3} \quad \Rightarrow \quad -4.444 \times 10^{-4}V_0 + 1 \times 10^{-4}V_0 = 10^{-2} \quad \Rightarrow$$

$$V_0 = -29.03 \text{ V}$$

I_{sc}. A–B shorted $\Rightarrow V_0 = 0$, so $I = \dfrac{10 \text{ mV}}{1 \text{ k}\Omega} = 10^{-5}$ A and $I_{sc} = -75I = -7.5 \times 10^{-4}$ A.

$$R_{th} = \frac{-29.03}{-7.5 \times 10^{-4}} = 38.71 \text{ k}\Omega$$

ANSWER

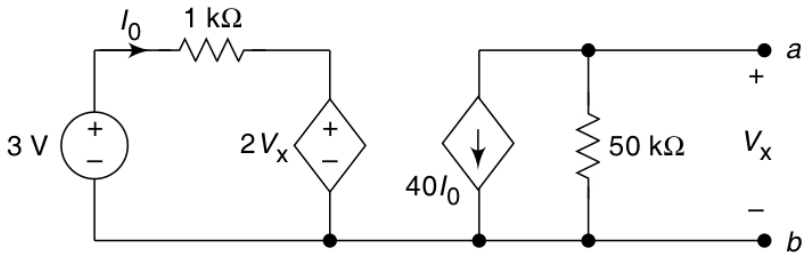
(solution circuit) $V_{th} = -29.0$ V, $R_{th} = 38.7$ kΩ (≈ book: it rounds V_0 to −29 V and prints 38.67 kΩ)

A27 · Problem 5.27

Thevenin · dependent sources, negative R_{th} **practice**

QUESTION

Find the Thevenin's equivalent between terminals a and b of the circuit shown in the figure.



The question figure gives 50 kΩ; the book's solution figure uses 50 Ω.

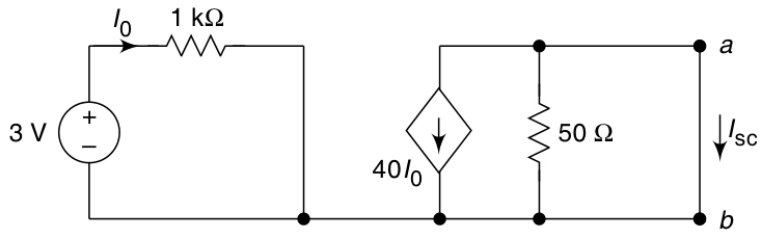
TRY IT FIRST

$V_x = V_{oc}$ feeds back into the left loop. For I_{sc}, $V_x = 0$ removes the $2V_x$ source.

V_{oc} ($R = 50$ kΩ, as in the question figure). Left loop and right node:

$$I_0 = \frac{3 - 2V_x}{1000}, \qquad V_{oc} = V_x = -40I_0 \times 50000 = -2000(3 - 2V_{oc})$$

$$3999V_{oc} = 6000 \qquad \Rightarrow \qquad V_{oc} = 1.5004 \text{ V}$$



a–b shorted

I_{sc} ($V_x = 0 \Rightarrow 2V_x = 0$): $I_0 = \frac{3}{1000} = 3$ mA. The 50 kΩ is shorted, and the dependent source pulls $40I_0$ from a down to b, so the short carries it from b to a:

$$I_{sc}\ (a \to b) = -40I_0 = -0.12 \text{ A}, \qquad R_{th} = \frac{1.5004}{-0.12} = -12.50\ \Omega$$

ANSWER

$V_{th} = 1.500$ V, $R_{th} = -12.50\ \Omega$ (figure value 50 kΩ)

BOOK ERRORS IN 5.27

1. The solution uses **50 Ω** ($V_x = -40I_0 \times 50 = -2000I_0$), not the figure's 50 kΩ. With 50 Ω: $V_{oc} = -2(3 - 2V_{oc}) \Rightarrow V_{oc} = 2$ V.
2. It then writes $R_{th} = 2/0.12 = 16.67\ \Omega$, dropping the sign of $I_{sc} = -0.12$ A. With 50 Ω the correct value is **$R_{th} = -16.67\ \Omega$**.

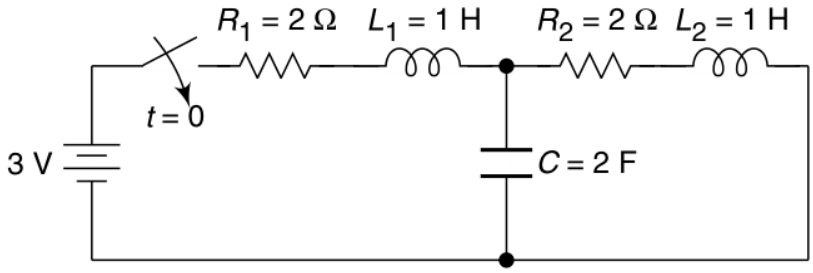
Test-source check (3 V shorted, 1 V applied at a): $I_0 = -\frac{2}{1000}$ A, current drawn $= \frac{1}{R} - 0.08$, giving $R_{th} = -12.50\ \Omega$ (50 kΩ) or $-16.67\ \Omega$ (50 Ω) ✓.

A28 · Problem 5.28

Norton · s-domain transient **new idea**

QUESTION

In the network shown in the figure, the switch is closed at time t = 0. Assuming all the initial currents and voltages as zero, find the current through the inductor L_2 by the use of Norton's theorem.

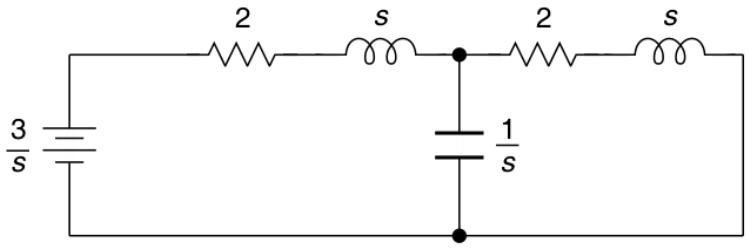


TRY IT FIRST

Draw the s-domain circuit (zero initial conditions). Make (R_2 + sL_2) the load and find the Norton equivalent of the rest.

HINT: WHAT'S NEW

- **Transient = theorem in the s-domain.** A DC source switched on at t = 0 becomes $\frac{V}{s}$; R → R, L → sL, C → $\frac{1}{sC}$. Find $I_L(s)$ exactly as in AC, then invert with partial fractions.
- With A–B shorted the capacitor is shorted out, so I_{sc} is one Ohm's-law step.



The book's s-domain network: note it draws the capacitor as 1/s

s-domain: source $\frac{3}{s}$; $R_1 + sL_1 = s + 2$; C = 2 F → $\frac{1}{2s}$; load $R_2 + sL_2 = s + 2$.

$$Z_N = (s+2) \parallel \frac{1}{2s} = \frac{s+2}{2s^2 + 4s + 1}, \qquad I_N = I_{sc} = \frac{3/s}{s+2} = \frac{3}{s(s+2)}$$

$$I_L(s) = I_N \times \frac{Z_N}{Z_N + (s+2)} = \frac{3}{s(s+2)} \times \frac{1}{1 + (2s^2 + 4s + 1)} = \frac{3}{2s(s+2)(s+1)^2}$$

Partial fractions: $I_L(s) = \frac{3/4}{s} - \frac{3/4}{s+2} + \frac{0}{s+1} - \frac{3/2}{(s+1)^2}$

(Coefficients: at $s = 0$, $\frac{3}{2\cdot2\cdot1} = \frac{3}{4}$; at $s = -2$, $\frac{3}{2(-2)(1)} = -\frac{3}{4}$; on $(s+1)^2$, $\frac{3}{2(-1)(1)} = -\frac{3}{2}$; on $1/(s+1)$, 0, because the s^3 terms must cancel.)

ANSWER

$\mathbf{i_{L2}(t) = \frac{3}{4} - \frac{3}{4}e^{-2t} - \frac{3}{2}te^{-t}\ A}$ $(t \geq 0)$

BOOK ERROR IN 5.28

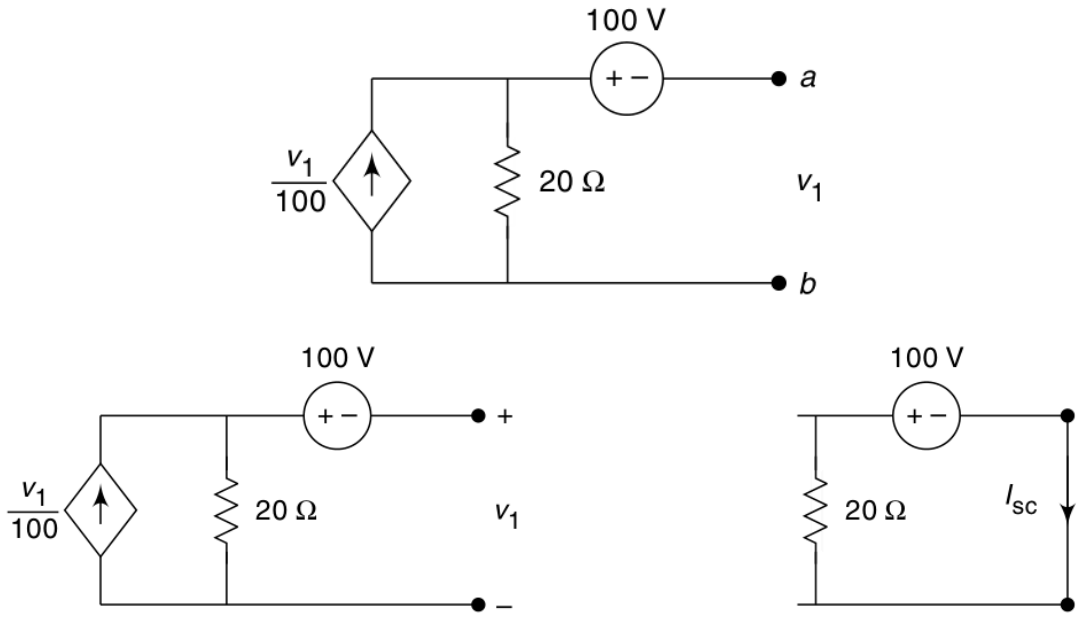
The figure gives C = 2 F, but the book's s-domain network uses $\frac{1}{s}$ (that is C = 1 F) and prints $i(t) = \frac{3}{4} - \frac{3}{4}e^{-2t} - \frac{3}{2}e^{-t}\sin t$ A. That answer is right for a 1 F capacitor only. A direct simulation of the circuit equations with C = 2 F matches the answer above (e.g. i(1 s) = 0.0967 A; the book's formula gives 0.184 A). Checks: $i(0) = 0$ and $i(\infty) = \frac{3}{2+2} = 0.75$ A.

A29 · Problem 5.29

Thevenin · dependent source **practice**

QUESTION

The following circuit has a dependent current source and an independent voltage source. Find the Thevenin equivalent network of the circuit across the terminals a and b.



Left: a–b open ($v_1 = v_{oc}$). Right: a–b shorted ($v_1 = 0$, so the dependent source is zero).

V_{oc}: the top node sits at $v_1 + 100$. KCL there:

$$\frac{v_1}{100} = \frac{v_1 + 100}{20} \quad \Rightarrow \quad v_1 = 5v_1 + 500 \quad \Rightarrow \quad V_{oc} = -125\ \text{V}$$

I_{sc}: $v_1 = 0$, so the dependent source is off; the top node is at 100 V and the 20 Ω draws 5 A from it, which must come from the short: I_{sc} (a → b) = −5 A.

$$R_{th} = \frac{-125}{-5} = 25\ \Omega$$

ANSWER

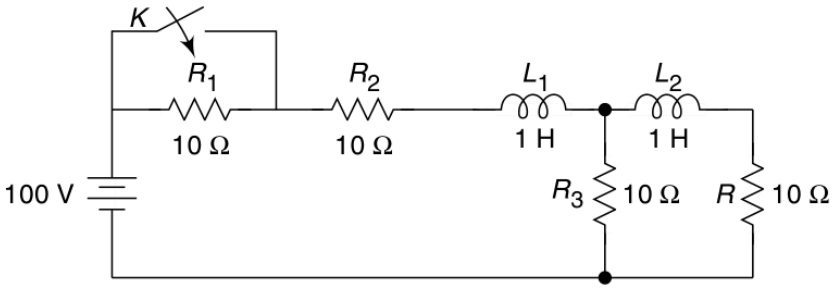
V_{th} = −125 V (i.e. 125 V with + at b), R_{th} = 25 Ω (= book)

A30 · Problem 5.30

Thevenin · transient with initial conditions **new idea**

QUESTION

In the network of the figure, the switch K is closed at time t = 0, a steady state having previously existed. Obtain the current in the resistor R using Thevenin's theorem.



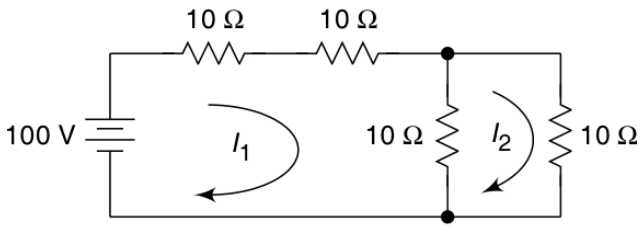
Closing K shorts R_1.

TRY IT FIRST

Find both inductor currents just before t = 0 (inductors are shorts in DC steady state). Then build the s-domain circuit with those currents as sources.

HINT: WHAT'S NEW

- **Non-zero initial current:** an inductor carrying i(0) becomes sL **in series with a voltage source L·i(0)**, polarity aiding the current's direction.



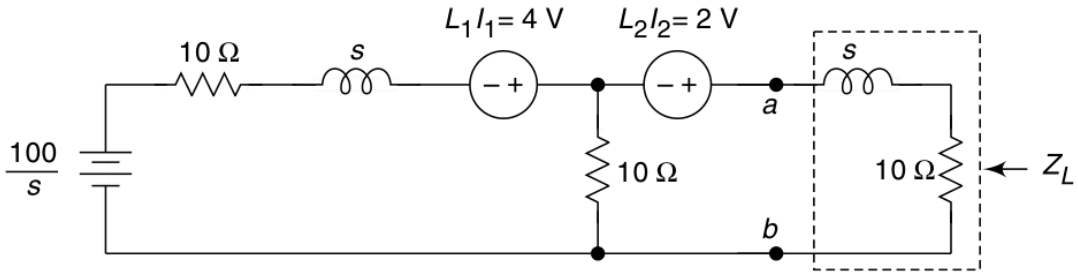
(a) Circuit at $t = 0-$

t = 0−: inductors as shorts

t = 0− (K open, R_1 in):

$$30I_1 - 10I_2 = 100, \qquad -10I_1 + 20I_2 = 0$$

$$i_{L1}(0) = I_1 = 4\ \text{A}, \qquad i_{L2}(0) = I_2 = 2\ \text{A}$$



(b) Transform network for $t > 0$

s-domain network for t > 0 (R_1 shorted); the load $Z_L = s + 10$ is in the dashed box

Thevenin at a–b (load = $sL_2 + R = s + 10$; the 2 V of L_2 stays on the source side):

$$Z_{th} = (s + 10) \parallel 10 = \frac{10(s + 10)}{s + 20}$$

$$I(s) = \frac{100/s + 4}{s + 20} = \frac{4s + 100}{s(s + 20)}, \qquad V_{th} = 10I(s) + 2 = \frac{2s^2 + 80s + 1000}{s(s + 20)}$$

$$I_R(s) = \frac{V_{th}}{Z_{th} + s + 10} = \frac{2s^2 + 80s + 1000}{s(s + 10)(s + 30)} = \frac{10/3}{s} - \frac{2}{s + 10} + \frac{2/3}{s + 30}$$

ANSWER

$$\mathbf{i_R(t) = \frac{10}{3} - 2e^{-10t} + \frac{2}{3}e^{-30t}\ A = 3.33 - 2e^{-10t} + 0.67e^{-30t}\ A}\ (= \text{book})$$

CHECK

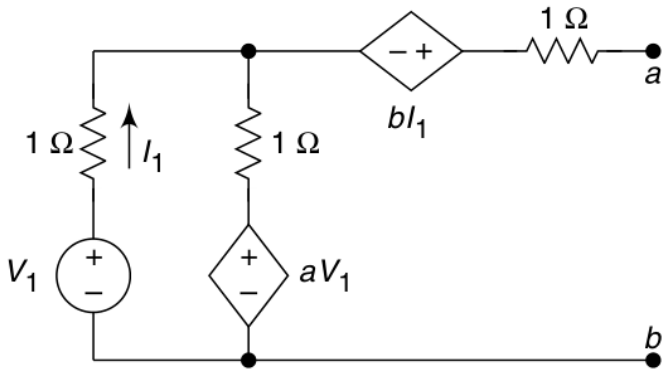
$i(0) = 10/3 - 2 + 2/3 = 2\text{ A} = i_{L2}(0)$ ✓. $i(\infty) = \dfrac{100}{10+5} \times \dfrac{1}{2} = 3.33\text{ A}$ ✓. A direct simulation of the circuit equations agrees at every time tested.

A31 · Problem 5.31

Thevenin · symbolic, two dependent sources **new idea**

QUESTION

For the network shown in the figure, show that the Thevenin equivalent at the terminals a-b is represented by $V_{Th} = \dfrac{V_1}{2}(1 + a + b - ab)$ and $Z_{Th} = \dfrac{3 - b}{2}$.



TRY IT FIRST

Keep a, b and V_1 as letters. Open circuit: no current in the right 1 Ω. Short circuit: two mesh equations.

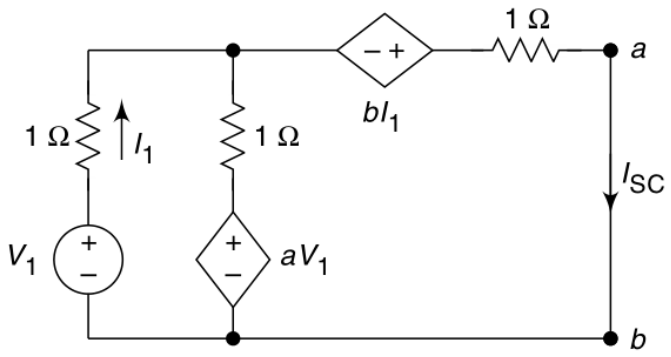
HINT: WHAT'S NEW

- **Symbolic answer:** the method is unchanged; carry the letters and simplify at the end. Substituting numbers (e.g. a = b = 0) checks the algebra.

V_{th} (no current in the right 1 Ω). Left mesh:

$$2I_1 + aV_1 = V_1 \qquad \Rightarrow \qquad I_1 = \frac{V_1}{2}(1 - a)$$

$$V_{th} = aV_1 + 1\cdot I_1 + bI_1 = aV_1 + (1 + b)\frac{V_1}{2}(1 - a) = \frac{V_1}{2}(1 + a + b - ab)$$



a–b shorted

I_{sc}. Two meshes:

$$2I_1 - I_{SC} = V_1(1 - a) \quad \text{...(i)} \qquad -(1 + b)I_1 + 2I_{SC} = aV_1 \quad \text{...(ii)}$$

$$I_{SC} = \frac{2aV_1 + (1 + b)V_1(1 - a)}{4 - (1 + b)} = \frac{V_1(1 + a + b - ab)}{3 - b}$$

$$Z_{th} = \frac{V_{th}}{I_{SC}} = \frac{3 - b}{2} \qquad \text{(proved)}$$

CHECK

Test source ($V_1 = 0$): the top node sits on $1 \parallel 1 = 0.5\ \Omega$, $I_1 = -0.5I_t$, so $V = I_t + 0.5I_t - 0.5bI_t = \frac{3 - b}{2} I_t$ ✓.

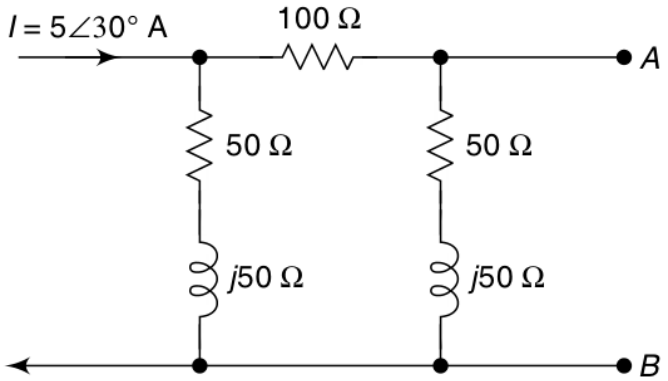
Also confirmed numerically for two sets of (a, b, V_1).

A32 · Problem 5.32

Thevenin · AC, current source **practice**

QUESTION

Find the Thevenin equivalent circuit for the network shown in the figure at terminals A-B.



The 5∠30° A is fed in at the left node and returns on the bottom rail.

$$I_{right} = 5\angle 30° \times \frac{50 + j50}{(100 + 50 + j50) + (50 + j50)} = 5\angle 30° \times \frac{1 + j}{4 + j2} = 1.581\angle 48.43° \text{ A}$$

$$V_{th} = I_{right}(50 + j50) = 111.8\angle 93.43° \text{ V}$$

$$Z_{th} = (150 + j50) \parallel (50 + j50) = \frac{5000 + j10000}{200 + j100} = 40 + j30\ \Omega$$

ANSWER

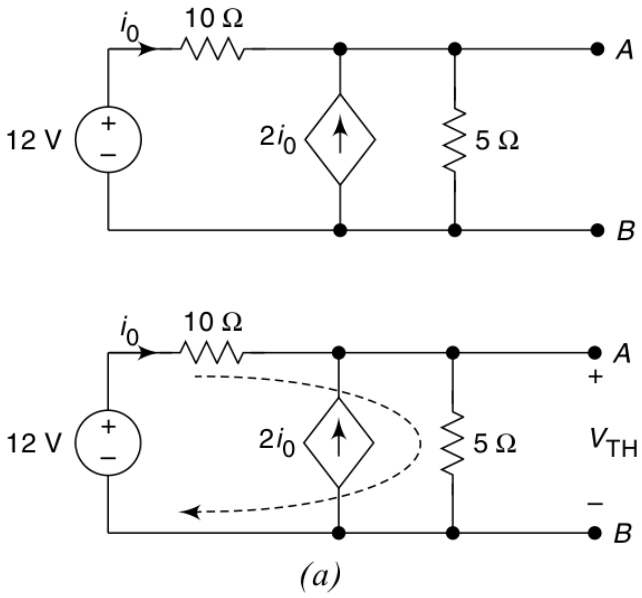
$V_{th} = 111.8\angle 93.43°$ V, $Z_{th} = 50\angle 36.87° = 40 + j30\ \Omega$ (= book)

A33 · Problem 5.33

Thevenin · dependent source **practice**

QUESTION

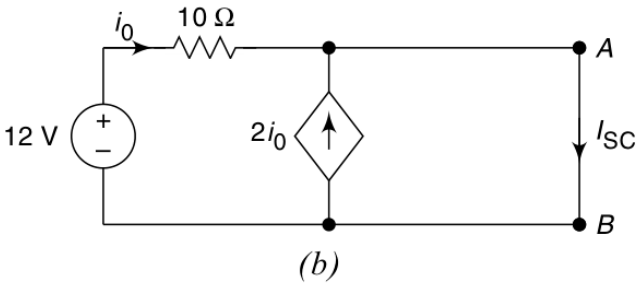
Find Thevenin's equivalent circuit across the terminals A and B for the network shown in the figure.



A–B open

KVL: $10i_0 + V_{th} = 12$ KCL at A: $i_0 + 2i_0 = \frac{V_{th}}{5}$ $\Rightarrow$ $V_{th} = 15i_0$

$25i_0 = 12$ $\Rightarrow$ $i_0 = 0.48$ A, $V_{th} = 7.2$ V



A–B shorted (the 5 Ω is shorted out)

$$i_0 = \frac{12}{10} = 1.2\ \text{A}, \qquad I_{sc} = i_0 + 2i_0 = 3.6\ \text{A}, \qquad R_{th} = \frac{7.2}{3.6} = 2\ \Omega$$

ANSWER

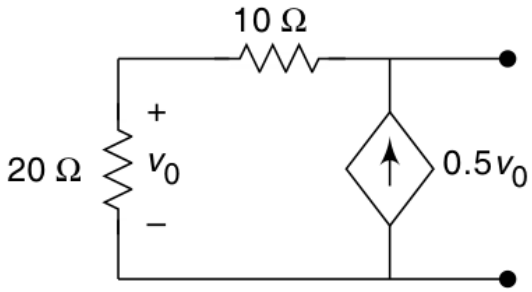
V_{th} = 7.2 V, R_{th} = 2 Ω (= book)

A34 · Problem 5.34

Thevenin · no independent source **new idea**

QUESTION

Find Thevenin's equivalent circuit for the network shown in the figure.

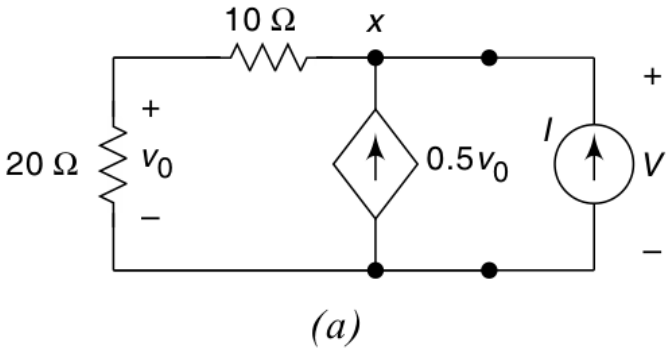


TRY IT FIRST

There is nothing to drive the circuit. What is V_{th}? For Z_{th}, connect a 1 A source and find the voltage it produces.

HINT: WHAT'S NEW

- **Only dependent sources:** V_{th} = 0 (no independent source, no response), and I_{sc} = 0 too, so V_{oc}/I_{sc} fails. Use a **test source**: Z_{th} = V / I.



(a)

Test current I applied; V measured

KCL at x: $\dfrac{V - v_0}{10} - 0.5v_0 - I = 0 \quad \Rightarrow \quad V = 10I + 6v_0$

Current down the 20 Ω = $\dfrac{V - v_0}{10} = 0.5v_0 + I$

$v_0 = 20(0.5v_0 + I) \quad \Rightarrow \quad v_0 = -\dfrac{20}{9}I$

$V = 10I - \dfrac{120}{9}I = -\dfrac{10}{3}I \quad \Rightarrow \quad Z_{th} = -3.33\ \Omega$

ANSWER

$V_{th} = 0$, $Z_{th} = -10/3 = -3.33\ \Omega$ (= book; its final sketch omits the minus sign)

C · Maximum power transfer theorem

Thevenin / Norton (done) → Load sees only V_{th} and Z_{th} → **Maximum power: choose the load to match Z_{th}**

Where this sits. Thevenin's theorem reduced any network to V_{th} in series with Z_{th}. The load current is then $V_{th} / (Z_{th} + Z_L)$, so the only question left is which load draws the most power from that one source.

Three terms first.

- **Load:** the element connected at the terminals, R_L (DC) or $Z_L = R_L + jX_L$ (AC).
- **Average power in the load:** $P = |I|^2 R_L$. Only the resistive part of the load absorbs power.
- **Complex conjugate:** for $Z_{th} = R_{th} + jX_{th}$, the conjugate is $Z_{th}^* = R_{th} - jX_{th}$ (same resistance, opposite reactance).

Statement.

Load allowed to vary	Maximum power when	P_{max}
R_L only, DC	$R_L = R_{th}$	$\frac{V_{th}^2}{4R_{th}}$
$Z_L = R_L + jX_L$, both free (AC)	$Z_L = Z_{th}^*$ (reactances cancel)	$\frac{\lvert V_{th}\rvert^2}{4R_{th}}$
Pure resistance R_L, AC source	$R_L = \lvert Z_{th}\rvert$	$\lvert I\rvert^2 R_L$ (calculate)

The recipe is always: **find V_{th} and Z_{th}, then match.** At maximum power the load gets exactly half the power, so efficiency is only 50%.

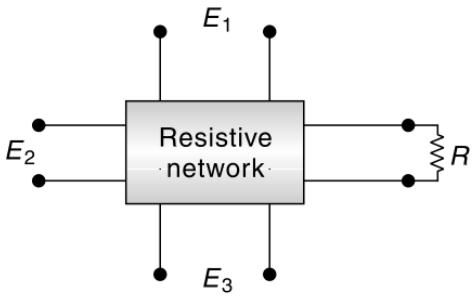
Phasor size convention. The book takes phasor magnitudes as rms values, so $P = |V_{th}|^2 / (4R_{th})$. If a source is given as a peak value, the power is half of that.

A35 · Problem 5.35

Power and superposition **new idea**

QUESTION

In the network shown, the power dissipated in R when E_1, E_2 or E_3 acting alone is (a) 20 W, 80 W and 5 W respectively; (b) 30 W, 270 W and 120 W respectively. Calculate the maximum power that R can dissipate due to the simultaneous action of all the sources. Calculate both for (a) and (b). What will be the minimum power dissipated in R when all the sources are acting simultaneously?



TRY IT FIRST

Power does not add. What does add? Turn each power back into a current, and remember you don't know each current's sign.

HINT: WHAT'S NEW

- **Superposition adds currents, not powers.** Each source alone gives $|i_k| = \sqrt{(P_k/R)}$; the directions are unknown, so the total is $\pm i_1 \pm i_2 \pm i_3$.
- $P = i^2R = (\pm\sqrt{(P_1)} \pm \sqrt{(P_2)} \pm \sqrt{(P_3)})^2$. Largest: all signs the same. Smallest: the sign choice that makes the sum closest to zero.

$$i = \pm\sqrt{\left(\frac{P_1}{R}\right)} \pm \sqrt{\left(\frac{P_2}{R}\right)} \pm \sqrt{\left(\frac{P_3}{R}\right)} \quad \Rightarrow \quad P = i^2R = (\pm\sqrt{(P_1)} \pm \sqrt{(P_2)} \pm \sqrt{(P_3)})^2$$

(a) $\sqrt{(20)} = 2\sqrt{(5)}$, $\sqrt{(80)} = 4\sqrt{(5)}$, $\sqrt{(5)}$:

$$P_{max} = (2 + 4 + 1)^2 \times 5 = 49 \times 5 = 245 \text{ W}, \qquad P_{min} = (4 - 2 - 1)^2 \times 5 = 5 \text{ W}$$

(b) $\sqrt{(30)}$, $\sqrt{(270)} = 3\sqrt{(30)}$, $\sqrt{(120)} = 2\sqrt{(30)}$:

$$P_{max} = (1 + 3 + 2)^2 \times 30 = 36 \times 30 = 1080 \text{ W}, \qquad P_{min} = (3 - 1 - 2)^2 \times 30 = 0 \text{ W}$$

ANSWER

(a) P_{max} = 245 W, P_{min} = 5 W. (b) P_{max} = 1080 W, P_{min} = 0 W (= book)

SIDE NOTE

Adding the powers would give 105 W and 420 W, which is wrong in both cases. The book's working line for (b) P_{max} misprints the bracket as $[4\sqrt{(5)} - 3\sqrt{(5)}]$; its answer 1080 W is right.

A36 · Problem 5.36

Maximum power · DC **teach**

QUESTION

Find the value of R in the circuit of the figure such that maximum power transfer takes place. What is the amount of this power?

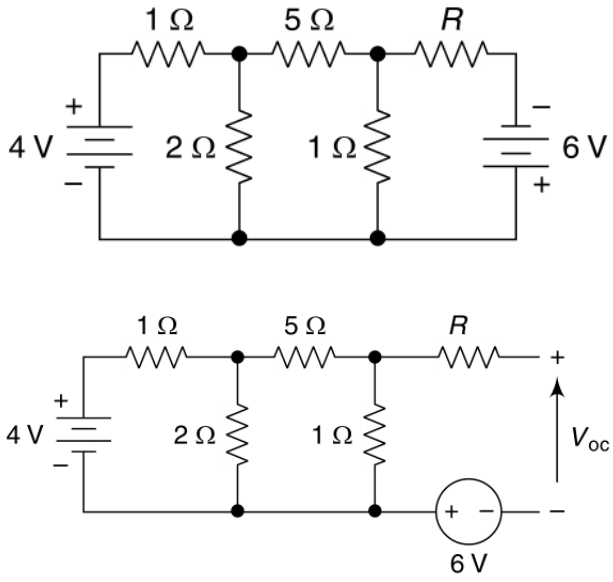
TRY IT FIRST

For each circuit: remove R, find V_{th} across its terminals and R_{th} looking back in. Then $R = R_{th}$ and $P_{max} = V_{th}^2/(4R_{th})$. In (b), look for an element that changes nothing. In (c), R is not connected to ground.

HINT: WHAT'S NEW

- **Maximum power transfer (DC):** $R = R_{th}$, $P_{max} = \dfrac{V_{th}^2}{4R_{th}}$. All the work is in finding V_{th} and R_{th}.
- **(b)** the 2 Ω directly across the 5 V source changes no other current (5.1's shortcut); then source transformations reduce the rest.
- **(c)** R sits between two non-ground nodes, so $V_{th} = V_P - V_Q$.

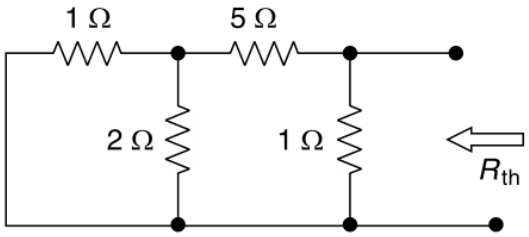
(a)



R removed; the 6 V source (+ at the bottom) is in the path to the terminal

V_{th}. Mesh currents with R removed: $3i_1 - 2i_2 = 4$, $-2i_1 + 8i_2 = 0 \Rightarrow i_1 = 4i_2$, so $10i_2 = 4$:

$$i_2 = \frac{2}{5}\ \text{A}, \qquad V_{th} = 1 \times i_2 + 6 = \frac{32}{5} = 6.4\ \text{V}$$



Sources shorted, looking back from R

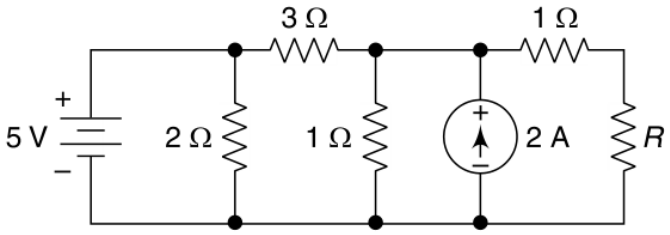
$$R_{th} = [(1 \parallel 2) + 5] \parallel 1 = \frac{17}{3} \parallel 1 = \frac{17}{20} = 0.85\ \Omega$$

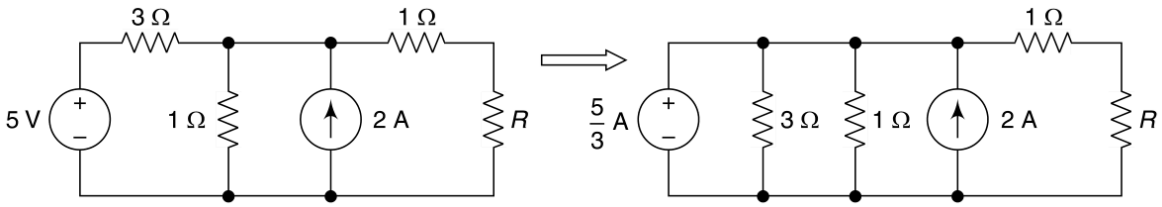
$$P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{6.4^2}{4 \times 0.85} = \frac{40.96}{3.4} = 12.05\ \text{W}$$

ANSWER

(a) R = 0.85 Ω, P_{max} = 12.05 W (= book, which rounds to 12 W)

(b)

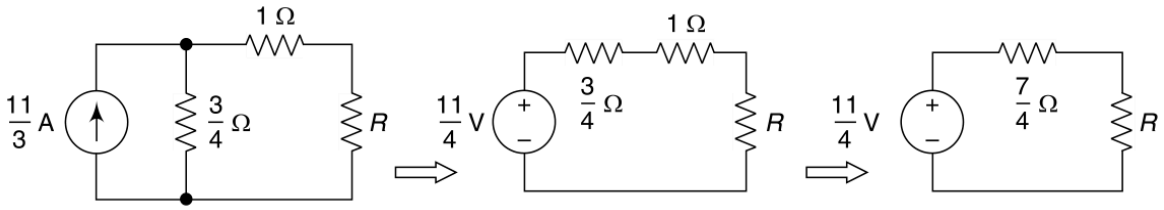




The 2 Ω across the 5 V is dropped; 5 V with 3 Ω → $\frac{5}{3}$ A ∥ 3 Ω

Combine the two current sources and the two resistors, then transform back:

$$\frac{5}{3} + 2 = \frac{11}{3}\ \text{A}, \qquad 3 \parallel 1 = \frac{3}{4}\ \Omega \qquad \Rightarrow \qquad \frac{11}{3} \times \frac{3}{4} = \frac{11}{4}\ \text{V in series with } \frac{3}{4}\ \Omega$$



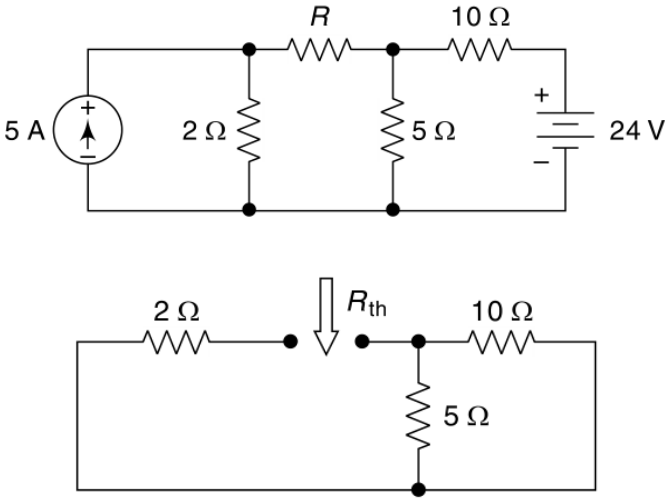
Successive source transformations

$$V_{th} = \frac{11}{4}\ \text{V}, \qquad R_{th} = \frac{3}{4} + 1 = \frac{7}{4}\ \Omega, \qquad P_{max} = \frac{(11/4)^2}{4 \times 7/4} = \frac{121}{112} = 1.08\ \text{W}$$

ANSWER

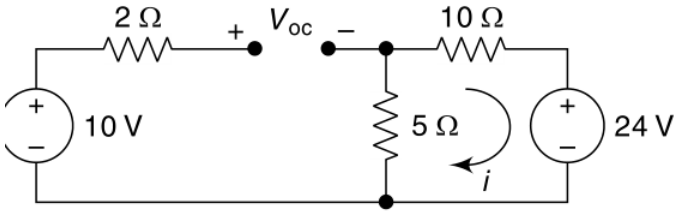
(b) R = 7/4 = 1.75 Ω, P_{max} = 1.08 W (= book)

(c)



R_{th}: 5 A opened, 24 V shorted

$$R_{th} = 2 + (10 \parallel 5) = 2 + \frac{10}{3} = 5.33\ \Omega$$



V_{oc}: 5 A ∥ 2 Ω → 10 V in series with 2 Ω

V_{th}. With R removed, no current flows in the 2 Ω, so V_P = 10 V. The right loop gives $V_Q = 24 \times \frac{5}{15} = 8$ V (the book writes this as i = −24/15 = −1.6 A, V_{oc} = 5i + 10).

$$V_{th} = V_P - V_Q = 10 - 8 = 2\ \text{V}, \qquad P_{max} = \frac{2^2}{4 \times 16/3} = \frac{3}{16} = 0.1875\ \text{W}$$

ANSWER

(c) R = 5.33 Ω, P_{max} = 0.1875 W (= book, 0.188 W)

CHECK

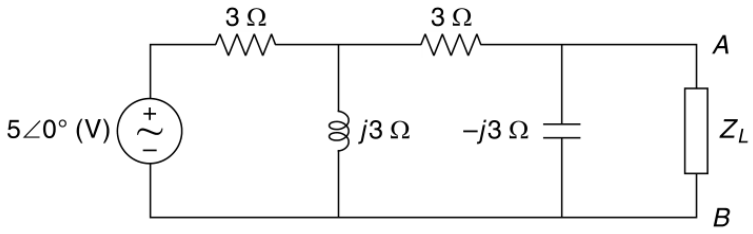
For each part, the full circuit was solved with R swept from 0 to 50 Ω; the power peaks at the R and P_{max} above.

A37 · Problem 5.37

Maximum power · AC conjugate match **new idea**

QUESTION

In the network shown, find the value of Z_L to which the maximum power can be delivered. Hence, find the value of the maximum power.



TRY IT FIRST

Find V_{th} and Z_{th} at A–B as in any AC Thevenin problem. Then choose Z_L so that the total impedance in the loop is purely resistive.

HINT: WHAT'S NEW

- **AC load free in R and X:** $Z_L = Z_{th}^*$. The reactances cancel ($X_L = -X_{th}$), leaving the DC case $R_L = R_{th}$, so $P_{max} = \frac{|V_{th}|^2}{4R_{th}}$.

V_{th}. Impedance seen by the source: 3 + j3 ∥ (3 − j3) = 3 + (3 + j3) = 6 + j3 Ω.

$$I = \frac{5}{6 + j3}, \qquad I_{right} = I \times \frac{j3}{j3 + 3 - j3} = jI$$

$$V_{th} = I_{right}(-j3) = 3I = \frac{15}{6 + j3} = 2 - j1 = 2.236\angle -26.57°\ \text{V}$$

$\mathbf{Z_{th}}$ (source shorted):

$$3 \parallel j3 = 1.5 + j1.5, \qquad (1.5 + j1.5 + 3) \parallel (-j3) = \frac{(4.5 + j1.5)(-j3)}{4.5 - j1.5} = 1.8 - j2.4\ \Omega$$

$$Z_L = Z_{th}^{*} = 1.8 + j2.4\ \Omega, \qquad I = \frac{V_{th}}{2 \times 1.8} = 0.621\angle -26.57^\circ\ \text{A}$$

$$P_{max} = \frac{2.236^2}{4 \times 1.8} = 0.694\ \text{W}$$

ANSWER

$Z_L = 1.8 + j2.4\ \Omega$, $P_{max} = 0.694$ W (= book)

A38 · Problem 5.38

Maximum power · AC load impedance **practice**

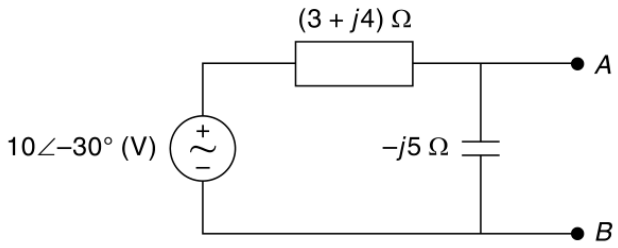
QUESTION

A loudspeaker is connected across terminals A and B of the network. What should be its impedance to obtain maximum power dissipation in it?

TRY IT FIRST

Only Z_{th} is needed: switch the sources off and look back from A–B.

(a)

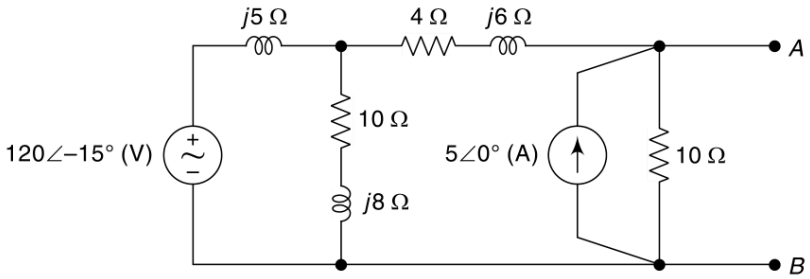


$$Z_{th} = \frac{(3 + j4)(-j5)}{3 + j4 - j5} = \frac{20 - j15}{3 - j1} = 7.5 - j2.5 = 7.91\angle -18.43^\circ\ \Omega$$

ANSWER

(a) $Z_L = Z_{th}^{*} = 7.5 + j2.5\ \Omega$ (the book prints 7.5 + j2.4, a typo; its own Z_{th} is 7.5 − j2.5)

(b)



Voltage source shorted and current source opened for Z_{th}.

$$(10 + j8) \parallel j5 = \frac{-40 + j50}{10 + j13} = 0.929 + j3.792\ \Omega$$

$$+ (4 + j6) = 4.929 + j9.792\ \Omega, \qquad \parallel 10\ \Omega \qquad \Rightarrow \qquad Z_{th} = 5.316 + j3.072 = 6.14\angle 30°\ \Omega$$

ANSWER

(b) $Z_L = Z_{th}^* = 5.316 - j3.072 = 6.14\angle -30°\ \Omega$ (= book)

A39 · Problem 5.39

Maximum power · resistive load on an AC source **new idea**

QUESTION

Two inductors each of 1 Ω reactance and negligible resistance are connected in series across a 2 V a.c. source. Find the value of resistance which should be connected across one of the inductors for maximum power dissipation. Also, find the maximum power.

TRY IT FIRST

Draw it: 2 V source, j1, then j1 ∥ R. Write the power in R as a function of R and maximise it, or find the Thevenin equivalent seen by R first.

HINT: WHAT'S NEW

- **Only a resistance is allowed**, so the reactance cannot be cancelled and $Z_L = Z_{th}^*$ is impossible. For a purely resistive load the best value is $R = |Z_{th}|$.

Route 1: calculus (as the book):

$$Z = \frac{jR}{R + j1} + j1 = \frac{-1 + j2R}{R + j1}, \qquad I_R = \frac{2(R + j1)}{-1 + j2R} \times \frac{j1}{R + j1} = \frac{j2}{-1 + j2R}$$

$$P = |I_R|^2 R = \frac{4R}{1 + 4R^2}, \qquad \frac{dP}{dR} = \frac{4(1 + 4R^2) - 4R(8R)}{(1 + 4R^2)^2} = 0 \quad \Rightarrow \quad 4 - 16R^2 = 0 \quad \Rightarrow$$

$$R = 0.5\ \Omega$$

$$P_{max} = \frac{4 \times 0.5}{1 + 4 \times 0.25} = 1\ \text{W}$$

Route 2: Thevenin. Seen by R: $V_{th} = 2 \times \frac{j1}{j1 + j1} = 1$ V, $Z_{th} = j1 \parallel j1 = j0.5\ \Omega$. Pure resistance $\Rightarrow R = |Z_{th}| = 0.5\ \Omega$, and $|I|^2 = \frac{1}{|0.5 + j0.5|^2} = 2$, so $P = 2 \times 0.5 = 1$ W ✓.

ANSWER

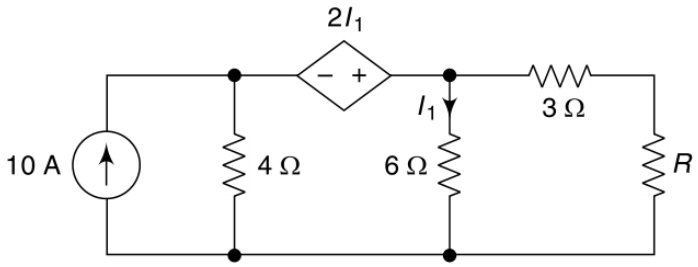
R = 0.5 Ω, P_{max} = 1 W (= book; its last line prints 4 × 0.4, a typo for 4 × 0.5)

A40 · Problem 5.40

Maximum power · dependent source **practice**

QUESTION

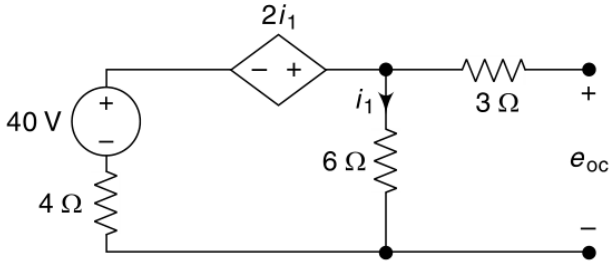
In the network shown, calculate the maximum power that may be dissipated in the external resistor R.



$2i_1$ is a current-controlled voltage source; i_1 flows down the 6 Ω.

TRY IT FIRST

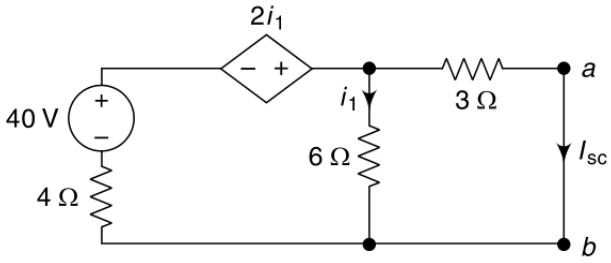
The dependent source rules out "switch off and combine", so find V_{oc} and I_{sc}. Start by turning 10 A ∥ 4 Ω into a voltage source.



10 A ∥ 4 Ω → 40 V in series with 4 Ω; terminals open

V_{oc} (no current in the 3 Ω, so the loop current is i_1):

$$-40 + 4i_1 - 2i_1 + 6i_1 = 0 \quad \Rightarrow \quad i_1 = 5\ \text{A}, \quad V_{oc} = 6i_1 = 30\ \text{V}$$



Terminals shorted

I_{sc}. Let V_M be the node above the 6 Ω: $i_1 = V_M/6$, $I_{sc} = V_M/3$, source current = $V_M/6 + V_M/3 = V_M/2$. Left loop:

$$40 - 4 \times \frac{V_M}{2} + 2 \times \frac{V_M}{6} - V_M = 0 \quad \Rightarrow \quad \frac{8}{3}V_M = 40$$

$$V_M = 15\ \text{V}, \quad I_{sc} = \frac{15}{3} = 5\ \text{A}$$

$$R_{th} = \frac{30}{5} = 6\ \Omega, \quad P_{max} = \frac{30^2}{4 \times 6} = 37.5\ \text{W}$$

ANSWER

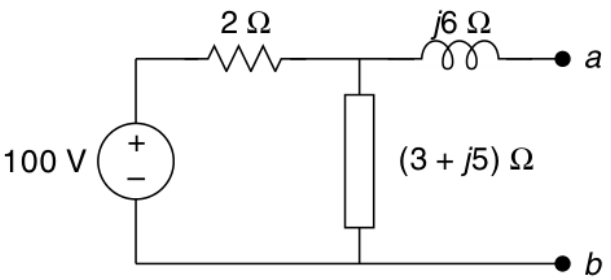
R = 6 Ω, P_{max} = 37.5 W (= book)

A41 · Problem 5.41

Maximum power · AC **practice**

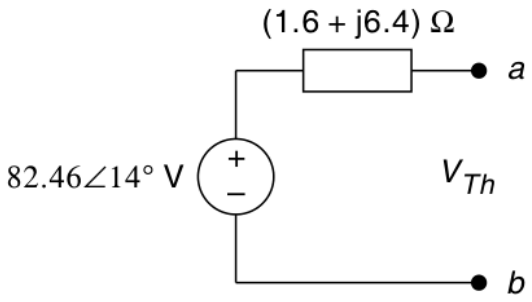
QUESTION

Find the Thevenin's equivalent between the points a and b for the circuit given in the figure. What should be the value of impedance connected between a and b for maximum power to be transferred from the sources? Obtain the amount of the maximum power.



$$I = \frac{100}{2 + 3 + j5} = 10 - j10\ \text{A}, \quad V_{th} = I(3 + j5) = 80 + j20 = 82.46\angle 14.04°\ \text{V}$$

$$Z_{th} = j6 + \frac{2(3 + j5)}{5 + j5} = j6 + 1.6 + j0.4 = 1.6 + j6.4\ \Omega$$



Thevenin equivalent

$$Z_L = Z_{th}^{*} = 1.6 - j6.4\ \Omega, \qquad P_{max} = \frac{|V_{th}|^2}{4R_{th}} = \frac{6800}{4 \times 1.6} = 1062.5\ \text{W}$$

ANSWER

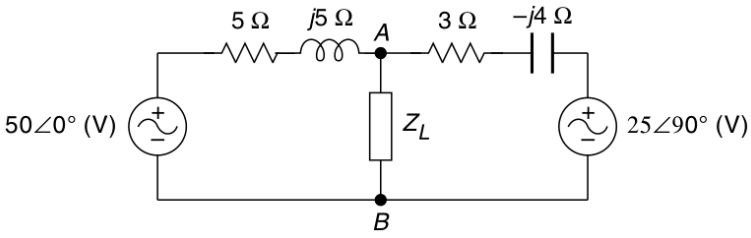
$V_{th} = 82.46\angle 14.04°$ V, $Z_{th} = 1.6 + j6.4\ \Omega$; $Z_L = 1.6 - j6.4\ \Omega$, $P_{max} = 1062.5$ W (= book)

A42 · Problem 5.42

Maximum power · AC, two sources **practice**

QUESTION

In the network in the figure, two voltage sources act on the load impedance connected to the terminal A and B. If the load is variable in both reactance and resistance, for what load Z_L will receive maximum power? What is the value of maximum power?



V_{th} (Z_L removed; one loop current from the 50 V towards the j25 V):

$$I = \frac{50 - j25}{(5 + j5) + (3 - j4)} = \frac{50 - j25}{8 + j1}$$

$$V_{th} = 50 - I(5 + j5) = \frac{25 - j75}{8 + j1} = 1.923 - j9.615 = 9.806\angle -78.69°\ \text{V}$$

$$Z_{th} = \frac{(5 + j5)(3 - j4)}{8 + j1} = \frac{35 - j5}{8 + j1} = 4.231 - j1.154\ \Omega$$

$$Z_L = Z_{th}^{*} = 4.231 + j1.154\ \Omega, \qquad P_{max} = \frac{9.806^2}{4 \times 4.231} = \frac{96.15}{16.92} = 5.68\ W$$

ANSWER

$Z_L = 4.23 + j1.154\ \Omega$, $P_{max} = 5.68$ W (= book to rounding: it squares 9.8 and prints 5.676 W)

A43 · Problem 5.43

Maximum power · equivalent from measurements **new idea**

QUESTION

A network has two output terminals. The open circuit voltage at these terminals is 260 V. The current flowing through the terminals is 20 A when the terminals are short circuited. Also, the current is 13 A when a coil of 11 ohm reactance and negligible resistance is connected across the terminals. Find the impedance components of the equivalent circuit feeding the terminals. What value of load impedance will give maximum power transfer and what is the value of this power?

TRY IT FIRST

Let $Z_{th} = R + jX$. The two current readings give two equations for the magnitudes $|R + jX|$ and $|R + j(X + 11)|$.

HINT: WHAT'S NEW

- **Z_{th} from terminal measurements:** $|Z_{th}| = V_{oc}/I_{sc}$ gives $R^2 + X^2$; a known reactance added in series gives a second equation. Subtracting them isolates X.

$$|Z_{th}| = \frac{260}{20} = 13\ \Omega \quad \Rightarrow \quad R^2 + X^2 = 169 \quad \ldots(i)$$

$$\frac{260}{|R + j(X + 11)|} = 13 \quad \Rightarrow \quad R^2 + (X + 11)^2 = 400 \quad \ldots(ii)$$

$$\text{(ii) } - \text{ (i):} \quad 22X + 121 = 231 \quad \Rightarrow \quad X = 5\ \Omega, \quad R = \sqrt{(169 - 25)} = 12\ \Omega$$

$$Z_L = Z_{th}^{*} = 12 - j5\ \Omega, \qquad P_{max} = \frac{260^2}{4 \times 12} = 1408.3\ W$$

ANSWER

Z_{th} = 12 + j5 Ω; Z_L = 12 − j5 Ω, P_{max} = 1408.3 W (= book)

CHECK

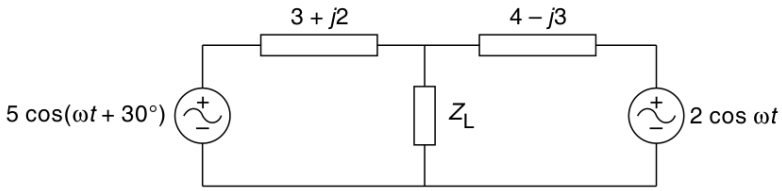
260/|12 + j5| = 260/13 = 20 A ✓; 260/|12 + j16| = 260/20 = 13 A ✓.

A44 · Problem 5.44

Maximum power · AC, two sources **practice**

QUESTION

What should be the value of Z_L for maximum power to be delivered in the circuit shown in the figure?



Impedances are given in ohms; both sources have the same ω.

$$Z_{th} = (3 + j2) \parallel (4 - j3) = \frac{(3 + j2)(4 - j3)}{7 - j1} = \frac{18 - j1}{7 - j1} = \frac{127 + j11}{50} = 2.54 + j0.22\ \Omega$$

ANSWER

$Z_L = Z_{th}^* = 2.54 - j0.22 = 2.55\angle -4.95°\ \Omega$ (= book)

SIDE NOTE

The source values do not affect Z_L; they only set V_{th} and hence P_{max}. Thevenin's equivalent works here because both sources share one frequency ω.

Part B · Sadiku (unsolved problems)

Alexander & Sadiku, *Fundamentals of Electric Circuits* 5e, ch. 4 "Circuit Theorems": practice problems, then end-of-chapter problems

Part A: every method, solved (done) → Part B: same theorems, your turn → **Superposition §4.3** → Source transformation §4.4 → Thevenin / Norton §4.5–4.6 → Max power §4.8

Where this sits. Part A taught every method on solved problems. Sadiku's problems are unsolved, so they are practice: the solutions here are compact, and an orange hint appears only when a problem adds something Part A did not show.

Answers. Practice problems print their answer beside the question ("= book"). End-of-chapter problems have answers only for odd numbers, in Appendix D ("= App. D"). Every answer here, odd or even, was checked by solving the full circuit with all sources on, and every superposition part was checked to add up to it.

A · Superposition (§4.3)

B1 · Practice Problem 4.3

Superposition · DC **practice**

QUESTION

Using the superposition theorem, find v_o in the circuit of Fig. 4.8.

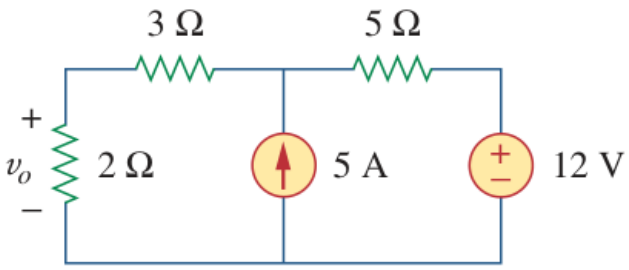


Fig. 4.8

5 A alone (12 V shorted). From the middle node the 5 A sees 3 + 2 = 5 Ω and 5 Ω, so it splits 2.5 A each way:

$$v_o' = 2.5 \times 2 = 5 \text{ V}$$

12 V alone (5 A opened). 5 Ω, 3 Ω and 2 Ω in series:

$$i = \frac{12}{5 + 3 + 2} = 1.2 \text{ A} \quad \Rightarrow \quad v_o'' = 1.2 \times 2 = 2.4 \text{ V}$$

ANSWER

v_o = 5 + 2.4 = 7.4 V (= book)

B2 · Practice Problem 4.4

Superposition · dependent current source **practice**

QUESTION

Use superposition to find v_x in the circuit of Fig. 4.11.

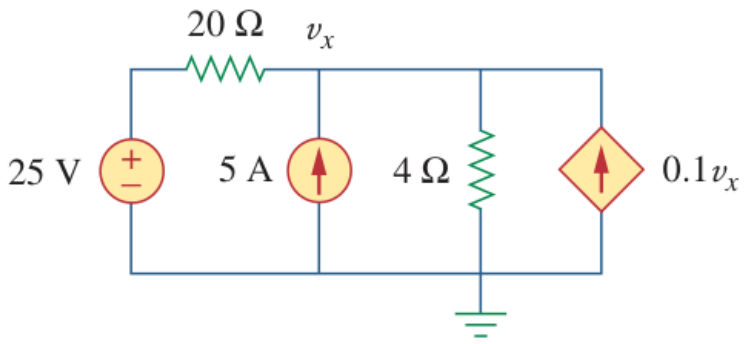


Fig. 4.11; $0.1v_x$ is a voltage-controlled current source.

TRY IT FIRST

The dependent source stays in both cases. One KCL at the v_x node per case.

25 V alone (5 A opened):

$$\frac{25 - v_x'}{20} + 0.1v_x' = \frac{v_x'}{4} \quad \Rightarrow \quad 1.25 = 0.2v_x' \quad \Rightarrow \quad v_x' = 6.25 \text{ V}$$

5 A alone (25 V shorted):

$$5 + 0.1v_x'' = \frac{v_x''}{4} + \frac{v_x''}{20} \quad \Rightarrow \quad 5 = 0.2v_x'' \quad \Rightarrow \quad v_x'' = 25 \text{ V}$$

ANSWER

v_x = 6.25 + 25 = 31.25 V (= book)

B3 · Practice Problem 4.5

Superposition · three sources **practice**

QUESTION

Find I in the circuit of Fig. 4.14 using the superposition principle.

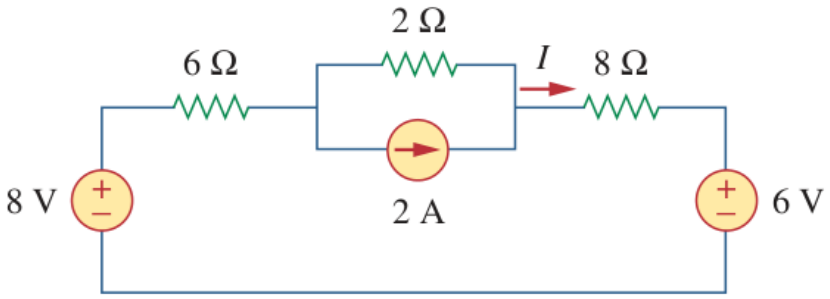


Fig. 4.14

With the other sources off, the loop is always 6 + 2 + 8 = 16 Ω.

8 V alone: $I' = \frac{8}{16} = 0.5$ A 6 V alone (it opposes I): $I'' = -\frac{6}{16} = -0.375$ A

2 A alone (2 A ∥ 2 Ω → 4 V in series with 2 Ω, pushing to the right): $I''' = \frac{4}{16} = 0.25$ A

ANSWER

I = 0.5 − 0.375 + 0.25 = 0.375 A = 375 mA (= book)

B4 · Problem 4.8

Superposition · one-node circuit **new idea**

QUESTION

Using superposition, find V_o in the circuit of Fig. 4.76. Check with PSpice or MultiSim.

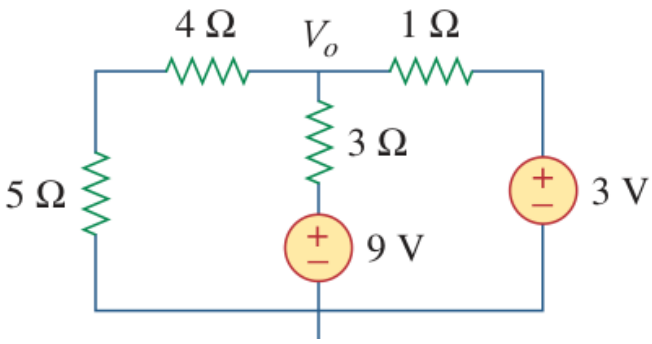


Fig. 4.76

HINT: WHAT'S NEW

- **One-node shortcut.** When every branch meets at one node, write one KCL at that node in each case instead of chaining dividers. Here the branches from V_o are 4 + 5 = 9 Ω, 3 Ω (+ 9 V) and 1 Ω (+ 3 V).

9 V alone (3 V shorted):

$$\frac{V_o'}{9} + \frac{V_o' - 9}{3} + \frac{V_o'}{1} = 0 \quad \Rightarrow \quad \frac{13}{9}V_o' = 3 \quad \Rightarrow \quad V_o' = \frac{27}{13}\ \text{V}$$

3 V alone (9 V shorted):

$$\frac{V_o''}{9} + \frac{V_o''}{3} + \frac{V_o'' - 3}{1} = 0 \quad \Rightarrow \quad \frac{13}{9}V_o'' = 3 \quad \Rightarrow \quad V_o'' = \frac{27}{13}\ \text{V}$$

ANSWER

$\mathbf{V_o = \frac{54}{13} = 4.154\ V}$ (even problem; full node solve agrees. The book's PSpice/MultiSim check is replaced by that solve.)

B5 · Problem 4.9

Superposition + linearity **new idea**

QUESTION

Given that I = 4 amps when V_s = 40 volts and I_s = 4 amps and I = 1 amp when V_s = 20 volts and I_s = 0, use superposition and linearity to determine the value of I when V_s = 60 volts and I_s = −2 amps.

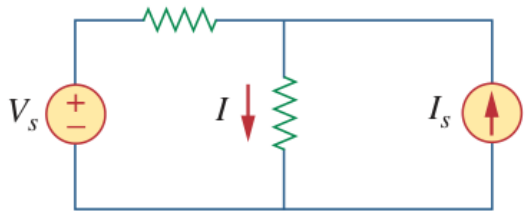


Fig. 4.77 (no resistor values given)

HINT: WHAT'S NEW

- **Superposition without the circuit values.** In a linear circuit any response is a weighted sum of the sources, $I = aV_s + bI_s$. The two measurements give two equations for a and b.

$$I_s = 0: \quad 1 = 20a \quad \Rightarrow \quad a = 0.05$$

$$4 = 40(0.05) + 4b \quad \Rightarrow \quad b = 0.5$$

$$I = 0.05 \times 60 + 0.5 \times (-2) = 3 - 1 = 2\ \text{A}$$

ANSWER

I = 2 A (= App. D)

B6 · Problem 4.10

Superposition · design problem **new idea**

QUESTION

Using Fig. 4.78, design a problem to help other students better understand superposition. Note, the letter k is a gain you can specify to make the problem easier to solve but must not be zero.

HINT: WHAT'S NEW

- **Design problem:** there is no single answer. Choose values, then solve your own problem. Below is one complete design; any other non-zero choice is equally valid.

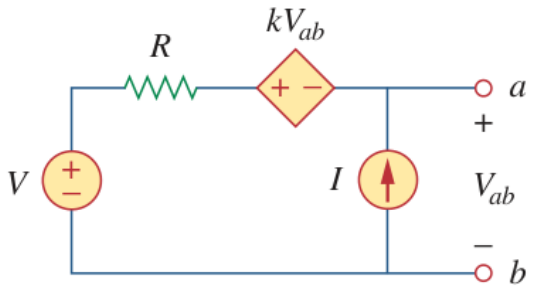


Fig. 4.78

One design. In Fig. 4.78 (V in series with R and the dependent source kV_{ab}, current source I across a–b), take **V = 12 V, R = 4 Ω, I = 3 A, k = 2**, and ask: *use superposition to find V_{ab}.*

Solution. Terminals a–b are open, so the current I must return through R and the source V: the series current from V towards a is −I. KVL around the loop: $V - R(-I) - kV_{ab} = V_{ab}$, so in general $V_{ab} = \dfrac{V + RI}{1 + k}$.

12 V alone (I = 0): $V_{ab}' = \dfrac{12}{1 + 2} = 4\ \text{V}$ 3 A alone (V = 0): $V_{ab}'' = \dfrac{4 \times 3}{1 + 2} = 4\ \text{V}$

ANSWER

(this design) $V_{ab} = 4 + 4 = 8$ V (design problem; the book lists "multiple answers")

B7 · Problem 4.11

Superposition · dependent source **practice**

QUESTION

Use the superposition principle to find i_o and v_o in the circuit of Fig. 4.79.

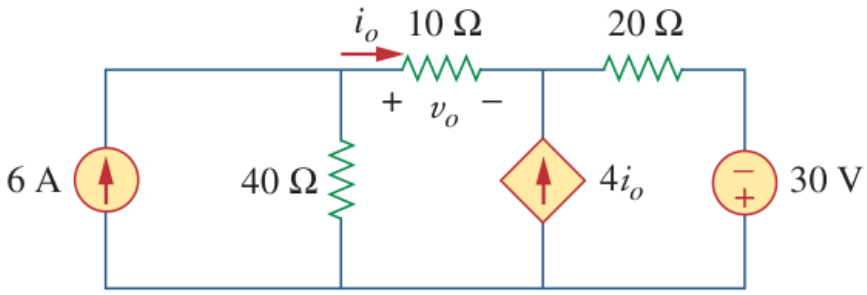


Fig. 4.79; $4i_o$ is a current-controlled current source.

Let the middle node be M and the left node A. KCL at M in either case: $5i_o$ flows on through the 20 Ω.

6 A alone (30 V shorted): $5i_o = V_M/20 \Rightarrow V_M = 100i_o$, $V_A = V_M + 10i_o = 110i_o$.

$$\text{KCL at A:} \quad 6 = \frac{110i_o}{40} + i_o = 3.75i_o \quad \Rightarrow \quad i_o' = 1.6\ \text{A}, \quad v_o' = 16\ \text{V}$$

30 V alone (6 A opened; the right end of the 20 Ω is at −30 V): $V_M = 100i_o - 30$, $V_A = 110i_o - 30$.

$$0 = \frac{110i_o - 30}{40} + i_o \quad \Rightarrow \quad 3.75i_o = 0.75 \quad \Rightarrow \quad i_o'' = 0.2\ \text{A}, \quad v_o'' = 2\ \text{V}$$

ANSWER

i_o = 1.8 A, v_o = 18 V (= App. D to rounding: 1.799 A, 17.99 V)

B8 · Problem 4.12

Superposition · three sources **practice**

QUESTION

Determine v_o in the circuit of Fig. 4.80 using the superposition principle.

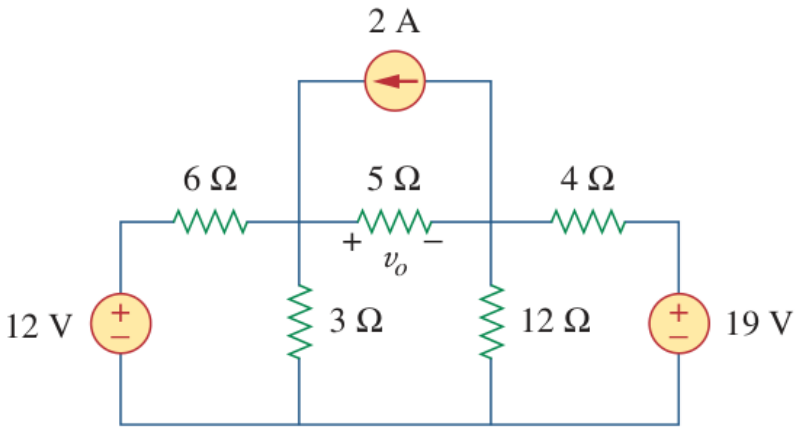


Fig. 4.80; the 2 A pushes current from the right node B into the left node A.

12 V alone. B sees $12 \parallel 4 = 3\ \Omega$; A sees $3 \parallel (5 + 3) = \frac{24}{11}\ \Omega$.

$$V_A = 12 \times \frac{24/11}{6 + 24/11} = 3.2\ \text{V}, \qquad V_B = 3.2 \times \frac{3}{8} = 1.2\ \text{V}, \qquad v_o' = 2\ \text{V}$$

19 V alone. A sees $6 \parallel 3 = 2\ \Omega$; B sees $12 \parallel (5 + 2) = \frac{84}{19}\ \Omega$.

$$V_B = 19 \times \frac{84/19}{4 + 84/19} = 9.975\ \text{V}, \qquad V_A = 9.975 \times \frac{2}{7} = 2.85\ \text{V}, \qquad v_o'' = -7.125\ \text{V}$$

2 A alone. A has 2 Ω to ground, B has 3 Ω; the 2 A splits equally between the 5 Ω and the 2 + 3 = 5 Ω path, so 1 A flows A → B in the 5 Ω:

$$v_o''' = 5 \times 1 = 5\ \text{V}$$

ANSWER

$v_o = 2 - 7.125 + 5 = -0.125$ V (even problem; full node solve agrees)

B9 · Problem 4.13

Superposition · three sources **practice**

QUESTION

Use superposition to find v_o in the circuit of Fig. 4.81.

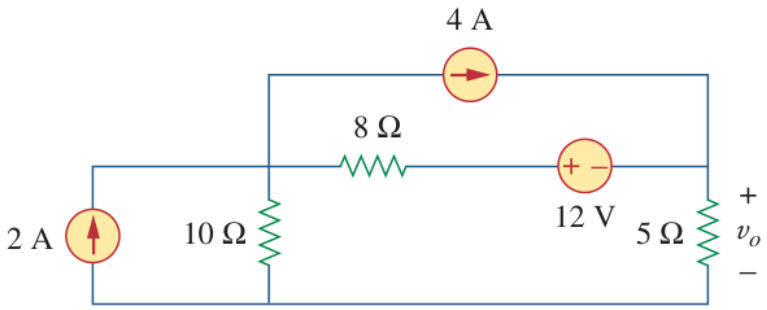


Fig. 4.81

2 A alone. It splits between 10 Ω and 8 + 5 = 13 Ω:

$$i_{5\Omega} = 2 \times \frac{10}{23} = \frac{20}{23}\ \text{A}, \qquad v_o' = 5 \times \frac{20}{23} = \frac{100}{23}\ \text{V}$$

12 V alone. One loop of 10 + 8 + 5 = 23 Ω; clockwise current (L → 8 Ω → source → 5 Ω):

$$i = -\frac{12}{23}\ \text{A}, \qquad v_o'' = 5i = -\frac{60}{23}\ \text{V}$$

4 A alone. It enters the right node and returns to the left node through 8 Ω or through 5 + 10 = 15 Ω:

$$i_{5\Omega} = 4 \times \frac{8}{23} = \frac{32}{23}\ \text{A}, \qquad v_o''' = \frac{160}{23}\ \text{V}$$

ANSWER

$\mathbf{v_o = \frac{100 - 60 + 160}{23} = \frac{200}{23} = 8.696\ V}$ (= App. D)

B10 · Problem 4.14

Superposition · three sources **practice**

QUESTION

Apply the superposition principle to find v_o in the circuit of Fig. 4.82.

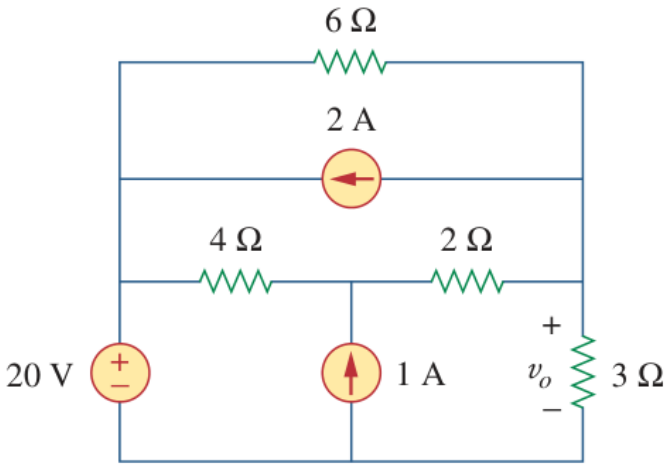


Fig. 4.82; the left rail is the top of the 20 V source, the right rail is the top of the 3 Ω.

20 V alone. Between the rails: 6 Ω ∥ (4 + 2) Ω = 3 Ω, then 3 Ω to ground: $v_o' = 20 \times \frac{3}{6} = 10$ V.

2 A alone (it pulls 2 A out of the right rail). The right rail sees 3 ∥ 6 ∥ 6 = 1.5 Ω: $v_o'' = -2 \times 1.5 = -3$ V.

1 A alone. From the middle node: 4 Ω to ground, or 2 Ω + (3 ∥ 6 = 2 Ω). Current towards the right rail = $1 \times \frac{4}{4+4} = 0.5$ A: $v_o''' = 0.5 \times 2 = 1$ V.

ANSWER

$\mathbf{v_o = 10 - 3 + 1 = 8\ V}$ (even problem; full node solve agrees)

B11 · Problem 4.15

Superposition · power **practice**

QUESTION

For the circuit in Fig. 4.83, use superposition to find i. Calculate the power delivered to the 3-Ω resistor.

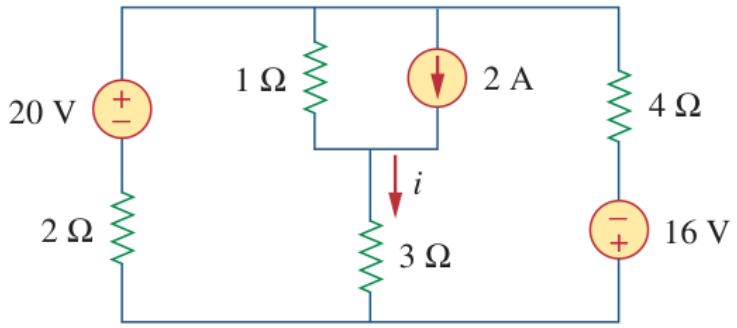


Fig. 4.83; the 16 V has its + at the bottom.

20 V alone. The top node sees 4 Ω (1 + 3) and 4 Ω (right): 2 Ω. $V_T = 20 \times \frac{2}{2+2} = 10$ V, $i' = \frac{10}{4} = 2.5$ A.

2 A alone. The 2 A enters the 3 Ω node and returns to the top node either through 1 Ω or through 3 Ω + (2 ‖ 4 = $\frac{4}{3}$ Ω): $i'' = 2 \times \frac{1}{1 + 13/3} = 0.375$ A.

16 V alone. The bottom of the right 4 Ω sits at −16 V; the top node sees 2 ‖ 4 = $\frac{4}{3}$ Ω: $V_T = -16 \times \frac{4/3}{4/3 + 4} =$ −4 V, $i''' = -1$ A.

$$i = 2.5 + 0.375 - 1 = 1.875 \text{ A}, \qquad P_{3\Omega} = i^2 \times 3 = 10.55 \text{ W}$$

ANSWER

i = 1.875 A, P = 10.55 W (= App. D)

POWER DOES NOT SUPERPOSE

Adding the three partial powers (18.75 + 0.42 + 3 W) gives a wrong answer. Add the currents first, then square.

B12 · Problem 4.16

Superposition · three sources **practice**

QUESTION

Given the circuit in Fig. 4.84, use superposition to obtain i_0.

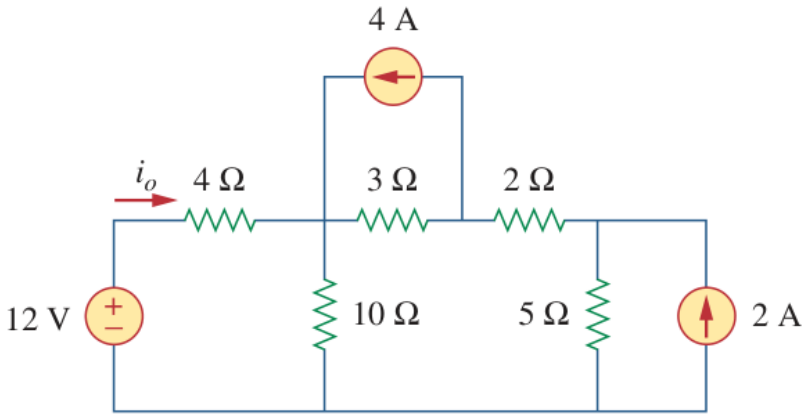


Fig. 4.84; the 4 A pushes current from the node between 3 Ω and 2 Ω into the node above the 10 Ω.

12 V alone. Seen from the 4 Ω: 10 ‖ (3 + 2 + 5) = 5 Ω, so $i_0' = \frac{12}{4+5} = \frac{4}{3}$ A.

2 A alone. From its node: 5 Ω to ground, or 2 + 3 + (4 ‖ 10 = $\frac{20}{7}$) Ω. Current towards the 10 Ω node = 2 × $\frac{5}{5+5+20/7} = \frac{7}{9}$ A, so $V_A = \frac{7}{9} \times \frac{20}{7} = \frac{20}{9}$ V and $i_0'' = -\frac{V_A}{4} = -\frac{5}{9}$ A.

4 A alone. It returns from A to B through 3 Ω, or through $\frac{20}{7}$ + 5 + 2 = $\frac{69}{7}$ Ω. Current down the ground path = 4 × $\frac{3}{3+69/7} = \frac{14}{15}$ A, so $V_A = \frac{14}{15} \times \frac{20}{7} = \frac{8}{3}$ V and $i_0''' = -\frac{2}{3}$ A.

ANSWER

$\mathbf{i_0 = \frac{4}{3} - \frac{5}{9} - \frac{2}{3} = \frac{1}{9} = 0.111\ A}$ (even problem; full node solve agrees)

B13 · Problem 4.17

Superposition · three sources **practice**

QUESTION

Use superposition to obtain v_x in the circuit of Fig. 4.85. Check your result using PSpice or MultiSim.

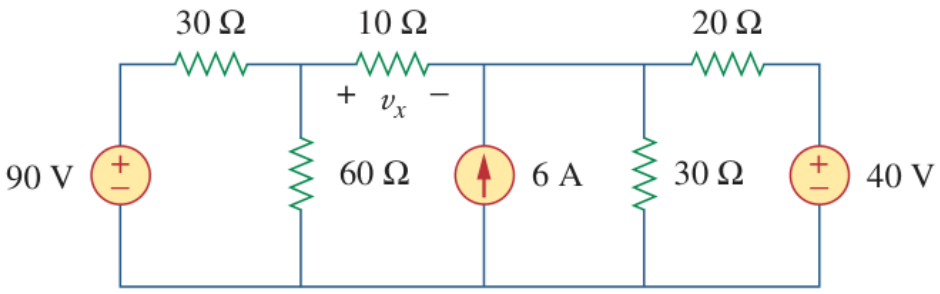


Fig. 4.85; A is the node after the 30 Ω, B the node after the 10 Ω.

90 V alone. B sees 30 ‖ 20 = 12 Ω; A sees 60 ‖ (10 + 12) = 16.10 Ω.

$$V_A = 90 \times \frac{16.10}{30 + 16.10} = 31.43\ \text{V}, \qquad v_x' = V_A \times \frac{10}{22} = 14.29\ \text{V}$$

6 A alone. B sees $12\ \Omega \parallel (10 + 30 \parallel 60) = 12 \parallel 30 = \frac{60}{7}\ \Omega$.

$$V_B = 6 \times \frac{60}{7} = 51.43\ \text{V}, \qquad V_A = V_B \times \frac{20}{30} = 34.29\ \text{V}, \qquad v_x'' = -17.14\ \text{V}$$

40 V alone. B sees $30 \parallel (10 + 20) = 15\ \Omega$.

$$V_B = 40 \times \frac{15}{20 + 15} = 17.14\ \text{V}, \qquad V_A = V_B \times \frac{20}{30} = 11.43\ \text{V}, \qquad v_x''' = -5.71\ \text{V}$$

ANSWER

v_x = 14.29 − 17.14 − 5.71 = −8.571 V (= App. D)

B14 · Problem 4.18

Superposition · dependent current source **practice**

QUESTION

Use superposition to find V_o in the circuit of Fig. 4.86.

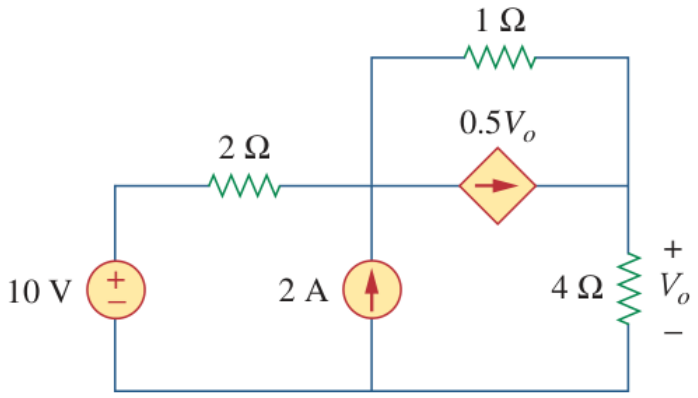


Fig. 4.86; $0.5V_o$ is a voltage-controlled current source from node A to the output node.

KCL at the output node (true in every case): $(V_A - V_o) + 0.5V_o = \frac{V_o}{4} \Rightarrow V_A = 0.75V_o$.

10 V alone (2 A opened), KCL at A:

$$\frac{10 - V_A}{2} = (V_A - V_o) + 0.5V_o \qquad \Rightarrow \qquad 5 = 1.5V_A - 0.5V_o$$

$$5 = 1.125V_o - 0.5V_o = 0.625V_o \quad \Rightarrow \quad V_o' = 8\ \text{V}$$

2 A alone (10 V shorted), KCL at A:

$$2 - \frac{V_A}{2} = V_A - 0.5V_o \quad \Rightarrow \quad 2 = 0.625V_o \quad \Rightarrow \quad V_o'' = 3.2\ \text{V}$$

ANSWER

V_o = 8 + 3.2 = 11.2 V (even problem; full node solve agrees)

B15 · Problem 4.19

Superposition · floating dependent source **new idea**

QUESTION

Use superposition to solve for v_x in the circuit of Fig. 4.87.

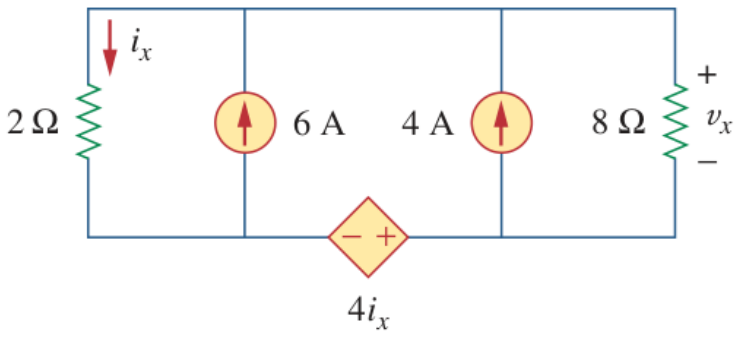


Fig. 4.87; $4i_x$ sits in the bottom wire, + on the right.

HINT: WHAT'S NEW

- **A dependent source in the return wire.** The bottom wire is split into two nodes that differ by $4i_x$. Take the left bottom node as ground; then the right bottom node is at $4i_x$, and $v_x = V_{top} - 4i_x$.

With $i_x = V_{top}/2$: $v_x = V_{top} - 2V_{top} = -V_{top}$. KCL at the top node (sources entering = currents leaving):

$$I_{in} = \frac{V_{top}}{2} + \frac{v_x}{8} = \frac{V_{top}}{2} - \frac{V_{top}}{8} = \frac{3}{8}V_{top}$$

$$\text{6 A alone:} \quad 6 = \frac{3}{8}V_{top} \quad \Rightarrow \quad V_{top} = 16\ \text{V}, \quad v_x' = -16\ \text{V}$$

$$\text{4 A alone:} \quad 4 = \frac{3}{8}V_{top} \quad \Rightarrow \quad V_{top} = 10.67\ \text{V}, \quad v_x'' = -10.67\ \text{V}$$

ANSWER

$v_x = -16 - 10.67 = -26.67$ V (= App. D)

B · Source transformation (§4.4)

Superposition (done) → **Source transformation: swap V + series R ↔ I ∥ R** → Thevenin / Norton (next)

Where this sits. Part A used source transformation inside superposition cases (5.9, 5.10). Here it is the whole method: keep swapping sources until the circuit is one loop or one node.

The rule. A voltage source V_s **in series** with R is equivalent to a current source $i_s = V_s/R$ **in parallel** with the same R, and back again. The current source's arrow points towards the voltage source's + terminal.

- It works for dependent sources too, as long as the controlling voltage or current is not hidden inside the swap.
- A source with **no** series R (voltage) or **no** parallel R (current) cannot be transformed.
- Parallel current sources add; series voltage sources add; resistors combine as usual between swaps.

B16 · Practice Problem 4.6

Source transformation **practice**

QUESTION

Find i_o in the circuit of Fig. 4.19 using source transformation.

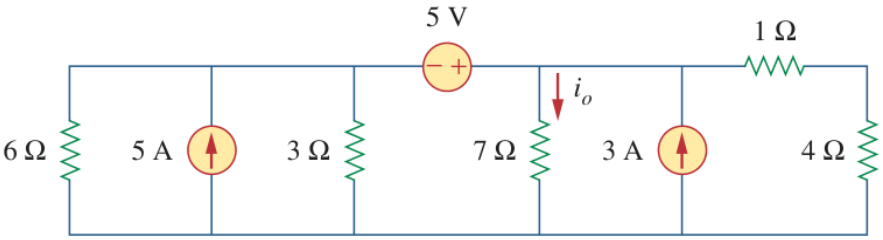


Fig. 4.19; the 5 V has + on the right.

TRY IT FIRST

Merge everything left of the 5 V into one source with one resistor, then add the 5 V.

$$6 \parallel 3 = 2\ \Omega \quad \text{with 5 A} \quad \Rightarrow \quad \text{10 V behind 2 } \Omega; \quad + \text{ 5 V in series} \quad \Rightarrow \quad \text{15 V behind 2 } \Omega$$

At the 7 Ω node (call it M), the 3 A enters and 1 + 4 = 5 Ω goes to ground:

$$\frac{15 - V_M}{2} + 3 = \frac{V_M}{7} + \frac{V_M}{5} \quad \Rightarrow \quad 10.5 = 0.8429 V_M \quad \Rightarrow \quad V_M = 12.46\ \text{V}$$

ANSWER

$\mathbf{i_o = \frac{12.46}{7} = 1.78\ A}$ (= book)

B17 · Practice Problem 4.7

Source transformation · dependent source **practice**

QUESTION

Use source transformation to find i_x in the circuit shown in Fig. 4.22.

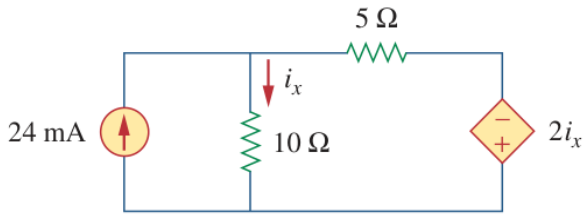


Fig. 4.22; $2i_x$ has − at the top.

TRY IT FIRST

Transform the dependent source with its 5 Ω, but keep the 10 Ω (it carries i_x).

The $2i_x$ source (− at top) in series with 5 Ω becomes a current source $\frac{2i_x}{5} = 0.4i_x$ pulling current **out of** the top node, in parallel with 5 Ω. With $V_{top} = 10i_x$:

$$0.024 - 0.4i_x = \frac{10i_x}{10} + \frac{10i_x}{5} = 3i_x \quad \Rightarrow \quad 3.4i_x = 0.024$$

ANSWER

i_x = 7.059 mA (= book)

B18 · Problem 4.20

Source transformation · reduce to one source **practice**

QUESTION

Use source transformation to reduce the circuit in Fig. 4.88 to a single voltage source in series with a single resistor.

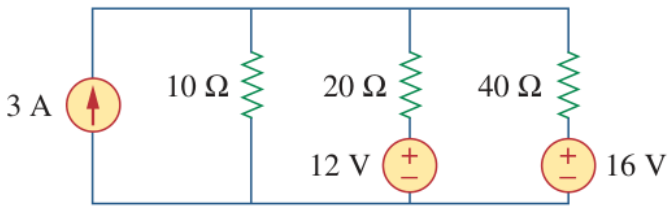


Fig. 4.88

Every branch runs between the two rails, so turn the voltage branches into current sources and add:

$$\frac{12}{20} = 0.6\ \text{A}, \qquad \frac{16}{40} = 0.4\ \text{A}, \qquad 3 + 0.6 + 0.4 = 4\ \text{A} \qquad \text{in parallel with}$$

$$10 \parallel 20 \parallel 40 = \frac{40}{7}\ \Omega$$

ANSWER

$V = 4 \times \frac{40}{7} = \frac{160}{7} = 22.86$ V (+ at the top rail) in series with $\frac{40}{7} = 5.714\ \Omega$ (even problem; open-circuit and short-circuit solves agree)

B19 · Problem 4.21

Source transformation · design problem **practice**

QUESTION

Using Fig. 4.89, design a problem to help other students better understand source transformation.

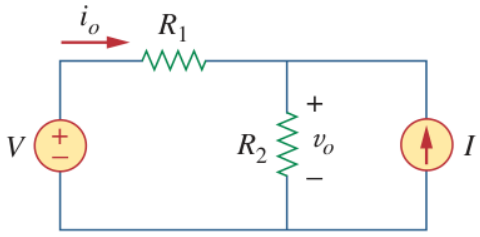


Fig. 4.89

One design. Take **$V = 20$ V, $R_1 = 5\ \Omega$, $R_2 = 20\ \Omega$, $I = 2$ A** and ask: *use source transformation to find v_o and i_o.*

$$20\ \text{V behind}\ 5\ \Omega \quad \Rightarrow \quad 4\ \text{A} \parallel 5\ \Omega; \quad 4 + 2 = 6\ \text{A} \quad \text{into} \quad 5 \parallel 20 = 4\ \Omega \quad \Rightarrow \quad v_o = 24\ \text{V}$$

$$i_o = \frac{V - v_o}{R_1} = \frac{20 - 24}{5} = -0.8\ \text{A}$$

ANSWER

(this design) $v_o = 24$ V, $i_o = -0.8$ A (design problem; the book lists "multiple answers")

B20 · Problem 4.22

Source transformation **practice**

QUESTION

For the circuit in Fig. 4.90, use source transformation to find i.

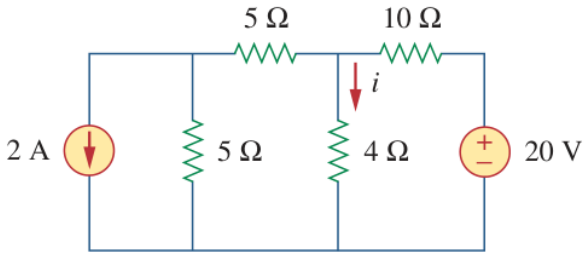


Fig. 4.90; the 2 A points down.

2 A (down) ∥ 5 Ω ⇒ −10 V behind 5 Ω; + 5 Ω ⇒ −10 V behind 10 Ω

At the middle node M, with 20 V behind 10 Ω on the right:

$$\frac{-10 - V_M}{10} + \frac{20 - V_M}{10} = \frac{V_M}{4} \quad \Rightarrow \quad 1 = 0.45V_M \quad \Rightarrow \quad V_M = 2.222\ \text{V}$$

ANSWER

i = $\frac{V_M}{4} = \frac{5}{9}$ = 0.556 A (even problem; full node solve agrees)

B21 · Problem 4.23

Source transformation · power **practice**

QUESTION

Referring to Fig. 4.91, use source transformation to determine the current and power absorbed by the 8-Ω resistor.

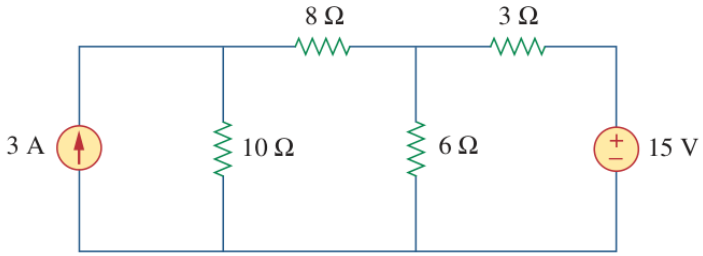


Fig. 4.91

Left: 3 A ∥ 10 Ω ⇒ 30 V behind 10 Ω Right: 15 V behind 3 Ω ⇒
5 A ∥ 3 Ω; ∥ 6 Ω ⇒ 5 A ∥ 2 Ω ⇒ 10 V behind 2 Ω

$$i_{8\Omega} = \frac{30 - 10}{10 + 8 + 2} = 1\ \text{A}, \qquad P = 1^2 \times 8 = 8\ \text{W}$$

ANSWER

i = 1 A, P = 8 W (= App. D)

B22 · Problem 4.24

Source transformation · dependent source **practice**

QUESTION

Use source transformation to find the voltage V_x in the circuit of Fig. 4.92.

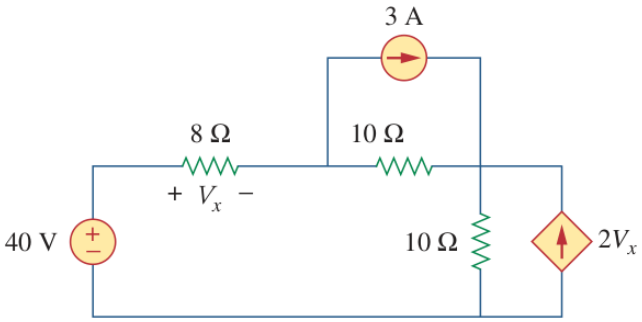


Fig. 4.92; $2V_x$ is a voltage-controlled current source.

TRY IT FIRST

Both current sources have a resistor in parallel. Turn the circuit into a single loop, keeping the 8 Ω (it carries V_x).

- 3 A ‖ 10 Ω (top) → 30 V in series with 10 Ω, + on the right.
- $2V_x$ ‖ 10 Ω (right) → $20V_x$ in series with 10 Ω, + at the top.

One loop, clockwise current i, with $V_x = 8i$:

$$40 = 8i + 10i - 30 + 10i + 20V_x = 28i - 30 + 160i \quad \Rightarrow \quad 188i = 70 \quad \Rightarrow$$

$$i = 0.3723 \text{ A}$$

ANSWER

$V_x = 8i = 2.979$ V (even problem; full node solve agrees)

B23 · Problem 4.25

Source transformation · three current sources **new idea**

QUESTION

Obtain v_o in the circuit of Fig. 4.93 using source transformation. Check your result using PSpice or MultiSim.

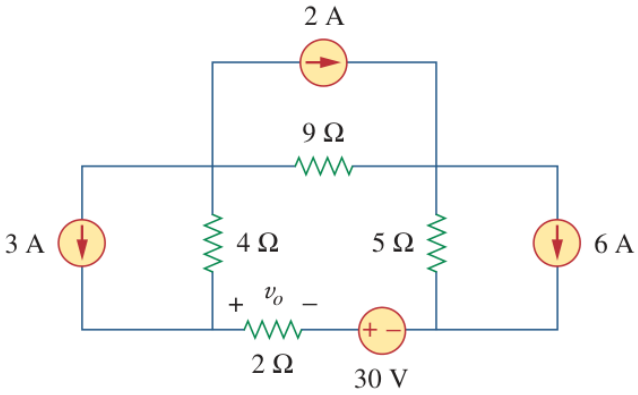


Fig. 4.93

HINT: WHAT'S NEW

- **Every current source here has a resistor in parallel**, so all three transform: the whole circuit collapses into **one loop**. Keep track of each new + sign: it sits where the arrow pointed.

- 3 A (down) ‖ 4 Ω → 12 V in series with 4 Ω, + at the bottom.
- 2 A (left → right) ‖ 9 Ω → 18 V in series with 9 Ω, + on the right.
- 6 A (down) ‖ 5 Ω → 30 V in series with 5 Ω, + at the bottom.

Clockwise loop current i (top: left → right; bottom: right → left through the 30 V and the 2 Ω):

$$18 + 30 + 30 - 12 = (9 + 5 + 2 + 4)i \quad \Rightarrow \quad 66 = 20i \quad \Rightarrow \quad i = 3.3 \text{ A}$$

The current runs right → left in the 2 Ω, against the + of v_o:

ANSWER

$v_o = -2 \times 3.3 = -6.6$ V (= App. D)

B24 · Problem 4.26

Source transformation **practice**

QUESTION

Use source transformation to find i_o in the circuit of Fig. 4.94.

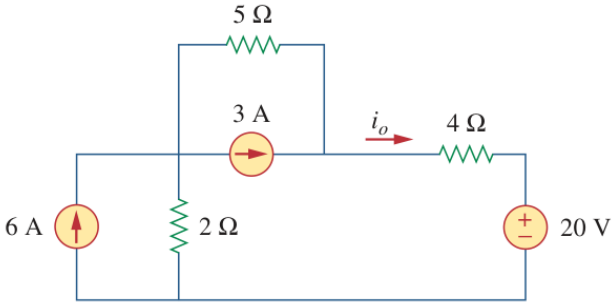


Fig. 4.94

6 A ‖ 2 Ω ⇒ 12 V behind 2 Ω; 3 A ‖ 5 Ω ⇒ 15 V behind 5 Ω (+ on the right)

$$i_o = \frac{12 + 15 - 20}{2 + 5 + 4} = \frac{7}{11}\ \text{A}$$

ANSWER

i_o = 0.636 A (even problem; full node solve agrees)

B25 · Problem 4.27

Source transformation **practice**

QUESTION

Apply source transformation to find v_x in the circuit of Fig. 4.95.

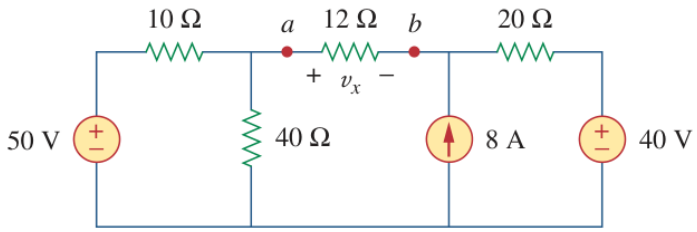


Fig. 4.95

Left: 50 V/10 Ω ⇒ 5 A ‖ 10 Ω; ‖ 40 Ω ⇒ 5 A ‖ 8 Ω ⇒ 40 V behind 8 Ω

Right: 40 V/20 Ω ⇒ 2 A ‖ 20 Ω; + 8 A ⇒ 10 A ‖ 20 Ω ⇒ 200 V behind 20 Ω

$$i_{a\to b} = \frac{40 - 200}{8 + 12 + 20} = -4\ \text{A}$$

ANSWER

$v_x = 12 \times (-4) = -48$ V (= App. D)

B26 · Problem 4.28

Source transformation · dependent source **practice**

QUESTION

Use source transformation to find I_o in Fig. 4.96.

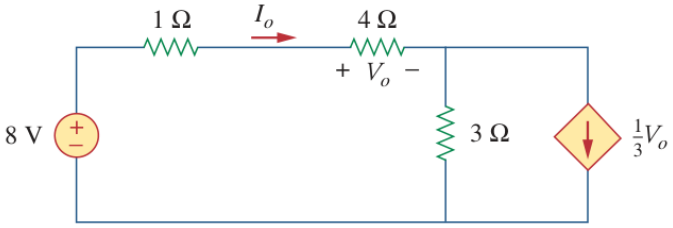


Fig. 4.96; $\frac{1}{3}V_o$ points down.

$\frac{1}{3}V_o$ (down) ∥ 3 Ω → a voltage V_o in series with 3 Ω, + at the bottom. One loop, with $V_o = 4I_o$:

$$8 = 1\cdot I_o + 4I_o + 3I_o - V_o = 8I_o - 4I_o = 4I_o$$

ANSWER

I_o = 2 A (even problem; full node solve agrees)

B27 · Problem 4.29

Source transformation · dependent source **new idea**

QUESTION

Use source transformation to find v_o in the circuit of Fig. 4.97.

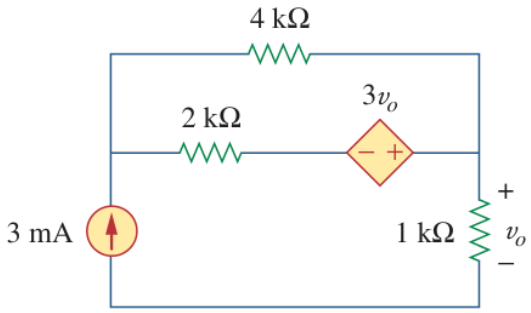


Fig. 4.97

HINT: WHAT'S NEW

- **Look for the only return path first.** Every path out of the 3 mA source ends in the 1 kΩ, so all 3 mA flows through it, whatever the 4 kΩ, 2 kΩ and $3v_o$ do.

$$v_o = 3\text{ mA} \times 1\text{ k}\Omega = 3\text{ V}$$

ANSWER

v_o = 3 V (= App. D)

CHECK

With $v_o = 3$ V the dependent source is 9 V, and the 3 mA splits between the 4 kΩ and the 2 kΩ + $3v_o$ branches; a full node solve confirms $v_o = 3$ V.

B28 · Problem 4.30

Source transformation · dependent source **practice**

QUESTION

Use source transformation on the circuit shown in Fig. 4.98 to find i_x.

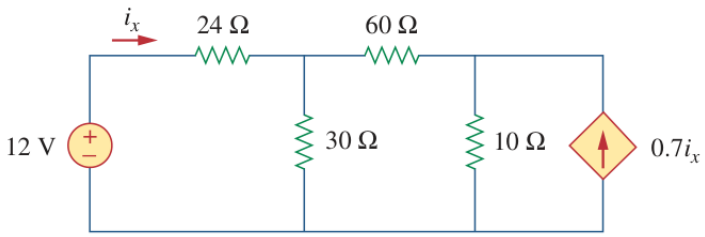


Fig. 4.98

$0.7i_x \parallel 10\ \Omega \rightarrow 7i_x$ in series with 10 Ω, + at the top. From node A (after the 24 Ω) the right side is then 60 + 10 = 70 Ω in series with $7i_x$. With $V_A = 12 - 24i_x$, KCL at A:

$$i_x = \frac{V_A}{30} + \frac{V_A - 7i_x}{70} \quad \Rightarrow \quad 210i_x = 7(12 - 24i_x) + 3(12 - 31i_x) \quad \Rightarrow$$

$$471i_x = 120$$

ANSWER

$i_x = 0.2548$ A (even problem; full node solve agrees)

B29 · Problem 4.31

Source transformation · dependent source **practice**

QUESTION

Determine v_x in the circuit of Fig. 4.99 using source transformation.

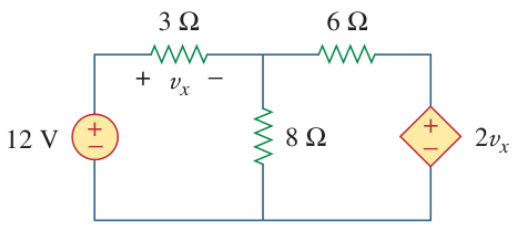


Fig. 4.99

Both voltage sources have a series resistor: 12 V/3 Ω → 4 A ‖ 3 Ω, $2v_x/6\ \Omega \rightarrow \frac{v_x}{3}$ ‖ 6 Ω. At the middle node ($V_M = 12 - v_x$):

$$4 + \frac{v_x}{3} = V_M\left(\frac{1}{3} + \frac{1}{8} + \frac{1}{6}\right) = 0.625(12 - v_x) \quad \Rightarrow \quad \frac{23}{24}v_x = 3.5$$

ANSWER

$\mathbf{v_x = \frac{84}{23} = 3.652\ V}$ (= App. D)

B30 · Problem 4.32

Source transformation · dependent source **practice**

QUESTION

Use source transformation to find i_x in the circuit of Fig. 4.100.

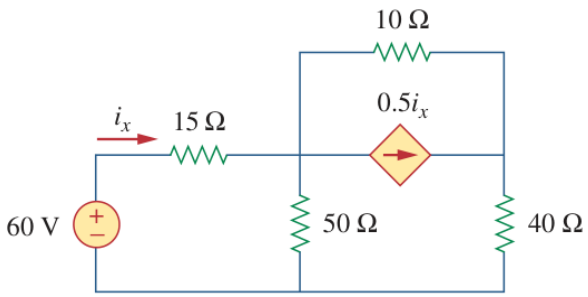


Fig. 4.100

$0.5i_x$ ‖ 10 Ω (between A and B) → $5i_x$ in series with 10 Ω, + at B. The A–B branch now meets only the 40 Ω, so it carries $\frac{V_A + 5i_x}{50}$. With $V_A = 60 - 15i_x$, KCL at A:

$$i_x = \frac{V_A}{50} + \frac{V_A + 5i_x}{50} \quad \Rightarrow \quad 50i_x = 2(60 - 15i_x) + 5i_x \quad \Rightarrow \quad 75i_x = 120$$

ANSWER

$\mathbf{i_x = 1.6\ A}$ (even problem; full node solve agrees)